FOUNDATIONS OF
ENVIRONMENTAL STUDIES
PRINCIPLES, CHALLENGES AND SUSTAINABLE PRACTICES



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Foundations of Environmental Studies: Principles, Challenges and Sustainable Practices

First Edition

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Preface

This book, *Foundations of Environmental Studies*, serves as an essential guide for understanding the intricate and multidimensional interactions between humans and their environment. It presents a comprehensive overview of environmental science, focusing on the interplay between natural resources, ecosystems, and sustainable practices. Aimed at learners and professionals, this text bridges theoretical knowledge with real-world applications, fostering environmental awareness and responsibility. The content emphasizes key environmental challenges, from resource management to ecological balance, underscoring the necessity for a global perspective and individual action in addressing contemporary issues. Through case studies, illustrations, and clear explanations, this book aspires to equip readers with the knowledge and motivation needed to contribute to a sustainable future.

Acknowledgement

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Unit-1

The Interdisciplinary Approach of Environmental Studies

1.1 Introduction

The term "environment" comes from the French word environner, which means to encircle or surround. According to the Environmental Protection Act (1986), the environment refers to the combination of water, air, and land, along with the interrelationships among these elements and their connection to humans, other living organisms, and property. Environmental studies are the systematic examination of human interactions with the environment.

1.2 Inter-disciplinary Characteristics of Environmental Studies

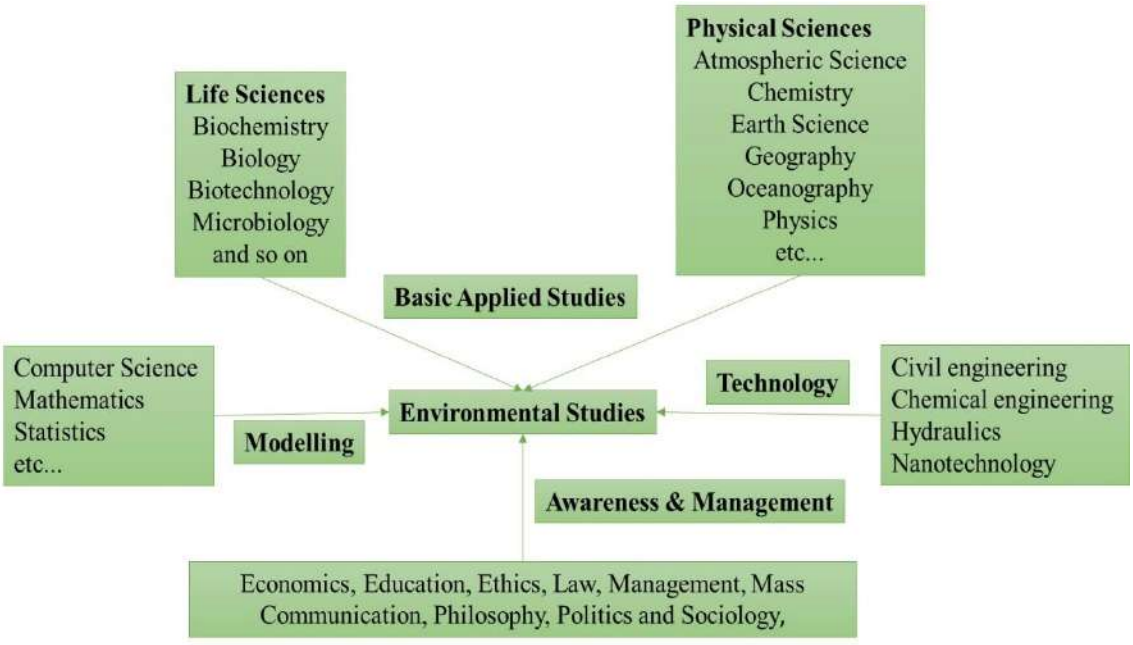


Figure 1: Inter-disciplinary characteristics of environmental studies

1.3 Scope

Environmental studies have a wide scope. The following are some elements of environmental studies' scope. Investigating how biotic and abiotic elements interact to create a sustainable human ecosystem, conducting environmental audits and impact analyses for additional Disastrous actions. Addressing and controlling pollution from both emerging and established industries, and preventing the use of nuclear and biological weapons to wipe out humanity, handling unforeseen disasters, and so forth. Major issues that will impact humanity as a whole include global warming, ozone layer depletion, declining Forests and energy supplies, decline in global biodiversity and more so we must think globally.

1.4 Importance

Environmental studies focus on everyday interactions with the environment, to which humans are closely related. These are linked to many scientific fields and Focused on the significance of wildlife concerned with the importance of wild life and how to preserve it. Environmental studies explain the role that biodiversity plays in preserving ecological balance. They provide information about population growth, population explosion, and the consequences of population growth. They also provide information on water conservation, watershed management, and the values of water. These are also offering information on water conservation, watershed management and the importance of water.

1.5 Need for Public Awareness

Public awareness is essential for environmental safety and pollution control for several reasons:

- Public awareness is key to fostering a culture of responsibility, where individuals and societies understand their impact on the environment and take action to ensure a sustainable future.

- Understanding environmental issues and their effects requires a high level of public awareness. Public knowledge about air, water, and soil contamination helps in mitigating health risks and preventing pollution-related diseases.
- Laws alone cannot eradicate environmental pollution; therefore, appropriate implementation—and particularly public participation—are crucial. For instance, Government initiatives to prohibit polythene littering will not be effective until the general public is aware of the negative effects they have on the environment.
- Only when people are aware of ecological and environmental issues can they participate in them. Awareness helps people Embrace environmentally friendly practices, like recycling, saving energy, and minimizing waste.
- The public needs to be made aware that we are hurting ourselves when we damage our environment. Awareness encourages participation in activities like tree planting, wildlife protection, and conservation projects.

1.6 Sustainability

Sustainability aims to meet present needs without compromising the ability of future generations to meet their own. It focuses on balancing environmental, social, and economic factors to ensure the long-term well-being and prosperity of both the planet and humanity. Preserving ecological balance, encouraging resource conservation, and cultivating behaviours that result in long-term social and economic prosperity are the corner stones of sustainability.

Public awareness and action must support international policy-making and agreements, according to the lessons learnt from the Stockholm and Rio summits. In order for people to understand their part in global sustainability, awareness must be raised. These summits' interconnected approach shows that a sustainable future can only be achieved when public awareness and business, government, and scientific efforts are in sync. This creates a cycle in which

knowledgeable citizens support and assist in putting the changes required for a sustainable world into action.

1.6.1 Stockholm Conference

From June 5 to 16, 1972, Stockholm, Sweden, hosted the United Nations Conference on the Human Environment. During the conference, a resolution was adopted, along with an action plan containing 109 recommendations, and a declaration with 26 environmental and development-related principles.

The Stockholm Declaration's guiding principles include the need to protect natural resources:

- The planet's capacity to produce renewable resources must be maintained.
- Wildlife needs to be protected.
- Resources that are not renewable must be shared and not depleted.
- The ability of the environment to purify itself must not be surpassed by pollution.
- Pollution of the oceans must be avoided.
- The environment must be improved through development. Planning for integrated development is required.
- Development and environmental conflicts should be resolved by rational planning.
- Planning human settlements is necessary to solve environmental issues.
- Governments ought to develop suitable population policies on their own.
- The environment must be improved through the application of science and technology.
- Education about the environment is crucial.
- Every country needs to set its own standards.
- We must eradicate weapons of mass destruction.

1.6.2 Rio-Summit/ Earth Summit

From June 3 to June 14, 1992, Rio de Janeiro hosted the United Nations Conference on The Environment and Development Conference (UNCED), commonly known as the Rio de Janeiro Earth Summit, Rio Summit, Rio Conference, or Earth Summit, saw 116 of the 172 participating governments sending their heads of state or government. In addition to the 17,000 participants at the concurrent NGO "Global Forum" (also referred to as Forum Global), which had Consultative Status, around 2,400 representatives from non-governmental organizations (NGOs) were also present.

The central theme of the summit is "Environment and Development.

The following documents were created as a result of the Earth Summit:

- Rio Declaration on Environment and Development
- Agenda 21
- Convention on Biological Diversity
- Forest Principles
- United Nations Framework Convention on Climate Change (UNFCCC)

Questions

1. Why are environmental studies considered as a multidisciplinary Subject?
2. What is the scope of Environmental Studies?
3. Explain the significance of Environmental Studies.
4. How can environmental awareness contribute to the protection of our Environment?
5. Why public awareness requires saving environment.
6. Define Environment.
7. Discuss about Sustainability.
8. Discuss about Rio summit and Stockholm conference.
9. Discuss about guidelines suggested in Stockholm conference.
10. How can you create awareness in your community?

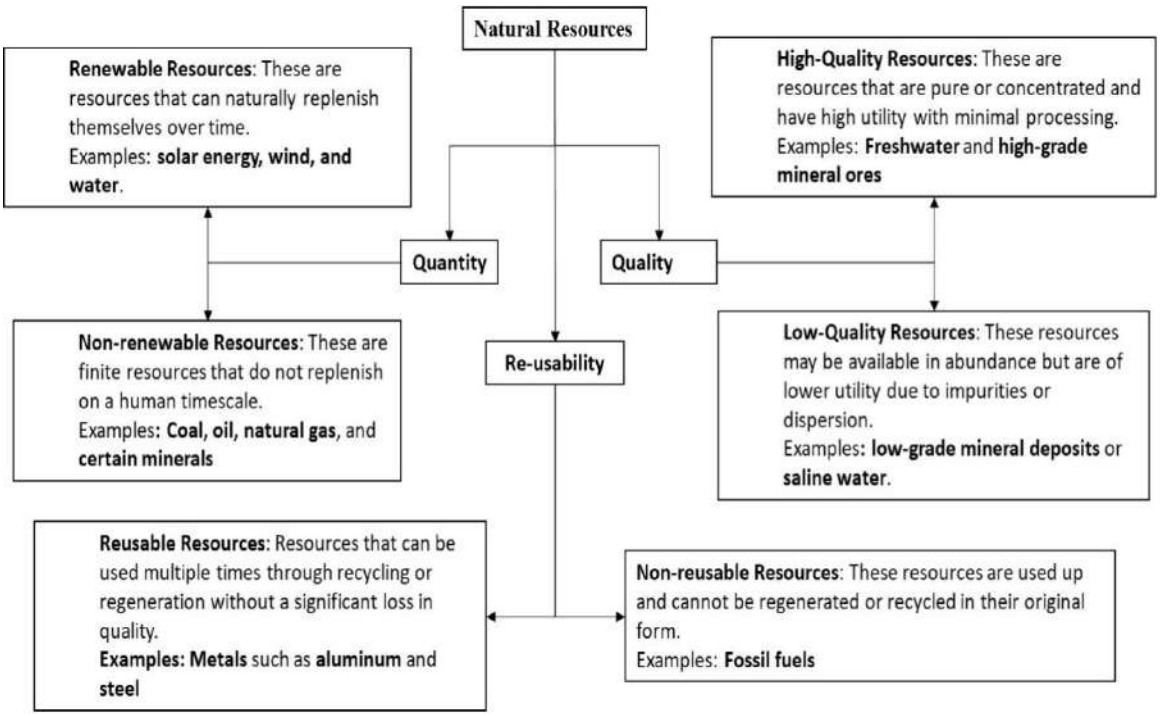
Unit-2

Natural Resources

2.1 Introduction

In general, a resource is a supply or support source that can be accessed when required. All materials and components found in nature that are necessary for human survival and economic activity are included in the context of natural resources. Examples of natural resources include water, air, soil, minerals, coal, forests, crops, and wildlife. From providing industrial raw materials to maintaining biodiversity and ecological balance, each of these resources is essential to life support and a variety of human needs.

Classification of Natural Resources



Major Natural Resources

- Forest resources
- Water resources
- Mineral resources
- Food resources
- Energy resources
- Land resources

2.2 Forest Resources

Forests primarily composed of trees, shrubs, and other woody plants, are among the Earth's most valuable natural resources. The earth's forests cover about one-third of its total land area.

2.2.1 Uses of Forests

The uses of forests can be broadly classified as follows

- **Commercial Uses**
  - Numerous commercial products, such as lumber, fuel wood, wood for paper production, edible products, gums, resins, inedible oils, Latex, bamboo, medications, etc., are produced by forests.
  - Cooking fuel is made from half of the annual amount of timber that is cut.
  - One-third of the wood harvest is utilised to make building supplies like hardwood, plywood, etc.
  - The paper industry uses one-sixth of the wood harvest to make pulp.

- **Ecological Uses**
  - **O_2 Production:** Through photosynthesis, trees generate O_2.

- **Mitigating Global Warming:** The forest absorbs CO_2 as a precursor to photosynthesis. These forests lessen the issue of greenhouse gas-induced global warming.
- **Habitat for Wildlife:** Forests alone provide a home to millions of wild plants and animals.
- **Conservation of Soil:** Forests keep soil erosion at bay by firmly binding soil particles in their roots.
- **Moderators of Pollution:** Forests have the ability to absorb a variety of harmful gases and maintain clean, fresh air. Apart from the aforementioned, forests shield people from radiation, floods, drought, etc.

2.2.2 Over Exploitation of Forests

Forests are classified as renewable resources, as they are replenished through natural cycles. The highest rate at which forest resources can be used indefinitely is known as the sustainable yield. This yield ensures that the resource supply is not diminished. However, if the rate of utilization exceeds the natural replenishment rate, the available supply begins to decline, leading to rapid environmental degradation.

Reasons for Over Exploitation

- For short-term financial gain without considering the long-term effects.
- Overexploitation of forest resources reduces the supply of wood and other products, which raises their price. This results in even more overuse of forests.
- People are compelled to overuse forest resources in order to meet their basic needs.
  Overexploitation is also caused by ignorance and a lack of awareness.
- Forests are cleared as a result of mining operations.
- For food, medicine, shelter, wood, and fuel, man is largely dependent on forests.
- Large-scale forest destruction is a result of hydropower projects.

2.2.3 Deforestation and causes of Deforestation

The removal of trees from forests is known as deforestation. Tree removal has negative effects because it exposes the soil, which causes erosion, rapid water runoff, and wildlife loss. Forest ecosystems are very healthy and contain a lot of water. Approximately 80% of the planet's original forests have already been destroyed. According to the National Forest Policy, satellite data only shows 20% of the forest area, or 63 million hectares.

Causes of Deforestation

The primary causes of deforestation are listed below:

- Development projects
- Shifting cultivation
- Fuel needs, including cutting and burning
- Construction of dams
- Increasing demand for food
- Forest fires
- Over grazing

Disadvantages of Deforestation

- The availability of food items such as coffee, tea, spices, nuts, fruits etc will be reduced.
- The number of different species may decrease, leading to an imbalance in the forest ecosystem.
- Soil erosion will increase significantly.
- Rainfall significantly decreases.
- Weather conditions are changed.
- Wildlife populations decline.
- Historical significance is lost.
- Bio diversity is lost.
- Hydrological cycle gets affected.

2.2.4 Case Studies on Deforestation

2.2.4.1 Deforestation in the Amazon Rainforest

Location: Brazil and other parts of South America.

Background: The Amazon Rainforest, often called the **"lungs of the Earth,"** covers approximately 5.5 million square kilometres and The Amazon plays a vital role in regulating the global climate and maintaining biodiversity. In recent decades, it has faced significant deforestation caused by various human activities.

Causes

- **Agricultural Expansion:** Large-scale cattle ranching and soy farming are primary drivers of deforestation, with significant land being cleared for pastures and crops.
- **Logging:** Both legal and illegal timber harvesting contribute to the destruction of forests.
- **Infrastructure Development:** The construction of roads and dams accelerates deforestation by opening up previously inaccessible areas.

Impacts

- **Climate Change:** Deforestation releases vast amounts of carbon dioxide stored in trees, contributing to global warming.
- **Biodiversity Loss:** The Amazon is home to an estimated 10% of all known species. Habitat loss threatens countless plant and animal species with extinction.
- **Indigenous Communities:** Indigenous tribes that rely on the forest for their way of life face displacement and cultural erosion.

Mitigation Efforts

- **Protected Areas:** Creation of national parks and reserves.

- **International Agreements:** Partnerships between countries to fund forest protection, such as the Amazon Fund.
- **Sustainable Practices:** Promotion of sustainable farming and logging practices.

2.2.4.2. Deforestation in India – The Western Ghats

Location: The Western Ghats, stretching across the western coast of India through states like Maharashtra, Karnataka, Kerala, Tamil Nadu, and Goa, are a UNESCO World Heritage Site and one of the most biodiverse regions in the world.

Background: The Western Ghats are home to a diverse range of plant and animal species, many of which are unique and found only in this region. This mountain range is essential for regulating the local climate and sustaining millions of people who rely on its resources for their livelihoods However, deforestation has significantly impacted this area due to human activities.

Causes of Deforestation

1. **Agricultural Expansion**
   - **Plantation Agriculture:** Large tracts of land have been cleared for plantations, particularly for crops like tea, coffee, and rubber.
   - **Slash-and-Burn Cultivation:** Practiced in some tribal areas, this traditional form of agriculture clears patches of forest, affecting regeneration.
2. **Urbanization and Infrastructure Development**
   - **Road Construction:** Expansion of roads and highways cuts through forests, fragmenting habitats and increasing accessibility for further exploitation.
   - **Urban Sprawl:** Population growth and the expansion of cities have led to deforestation for housing and industrial projects.

3. **Mining**
    - **Extraction of Minerals:** Mining for bauxite, iron ore, and other minerals has led to significant deforestation and environmental deterioration in the Western Ghats.
    - **Quarrying:** Stone and sand mining disrupts local ecosystems and accelerates habitat loss.

4. **Timber Extraction and Logging**
    - **Commercial Logging:** Legal and illegal logging for timber and firewood contributes to forest loss.
    - **Forest Fires:** Both intentional and accidental fires cause significant damage and prevent forest regeneration.

Impacts of Deforestation

1. **Biodiversity Loss**
    - **Endangered Species:** The Western Ghats are Habitat for species such as the lion-tailed macaque, Nilgiri tahr, and Malabar civet, which face severe threats from habitat loss.

2. **Habitat Fragmentation:** Forest fragmentation leads to isolated Wildlife populations decline, leading to reduced genetic diversity and heightened risk of extinction.

3. **Climate Change**
    - **Disruption of Rainfall Patterns:** The Western Ghats play a key role in monsoon regulation. Deforestation affects local and regional climate, altering rainfall and contributing to droughts and floods.
    - **Increased Carbon Emissions:** The loss of trees reduces the capacity of forests to absorb carbon dioxide, contributing to greenhouse gas emissions.

4. Water Resources

- **Reduced Water Availability:** Forests in the Western Ghats are critical for the recharge of rivers and groundwater. Deforestation impacts the water cycle, leading to water shortages and affecting agriculture and drinking water supplies.
- **Soil Erosion:** The removal of tree cover leads to increased soil erosion, which affects soil fertility and leads to sedimentation in rivers and dams.

Mitigation and Conservation Efforts

1. Forest Conservation Laws

- The Indian government has enacted laws such as the Forest Conservation Act (1980) and the Wildlife Protection Act (1972) to regulate deforestation and protect wildlife.
- **Protected Areas:** Establishment of national parks, wildlife sanctuaries, and biosphere reserves in the Western Ghats has helped conserve significant portions of the forest.

2. Afforestation and Reforestation

- **Government Initiatives:** Programs like the National Afforestation Programme (NAP) aim to restore degraded forest lands and promote sustainable forest management.
- **Community Participation:** Local community involvement in tree-planting and conservation projects has been encouraged to restore ecosystems.

3. Sustainable Practices

- **Agro Forestry:** Integrating trees with crops and livestock to maintain a balance between forest preservation and agricultural productivity.
- **Eco-Tourism:** Promoting responsible tourism in the Western Ghats to generate revenue for local communities while raising awareness about the importance of conservation.

4. International Collaboration

- **UNESCO:** The Western Ghats' designation as a World Heritage Site has brought international attention to its conservation.
- **Partnerships with NGOs:** Organizations like WWF-India and local NGOs work with governments and communities to promote sustainable land use and conservation projects.

2.2.5 The Impact of Dams on Forests and Indigenous Communities

Long-term ecological issues and ecosystem destruction result from the construction of massive dams in sensitive areas and forests. There are advantages and disadvantages to building massive dams and reservoirs. Reservable reservoirs can be created by building dams across rivers to store excess water that would otherwise enter the sea. At the desired rate, the stored water can be used for a number of things, including irrigation, water supply, and the production of hydroelectric power.

Effects on Tribal People

- The largest social cost of large dams is the mass eviction of local residents; it is estimated that up to 20 million people have been impacted directly or indirectly by all irrigation projects in India over the previous 50 years.
- More than 20,000 people living in 250 villages have been displaced by the Hirakud dam, one of the biggest dams built in the 1950s.

Effects on Forests

- The region's natural ecological balance has been upset by the clearing of thousands of hectares of forest to build river valley projects. In these places, landslides and floods are more common.

- For instance, 3.5 lakh hectares of the best forest, which includes rich teak and bamboo forests, have been submerged by the Narmada Sagar project alone.
- According to a study conducted by the Botanical Survey of India, the Tehri dam flooded 1000 hectares of forest, impacting roughly 430 plant species.

2.3 Water Resources

A vital component of all living things is water. Water covers almost 80 percent of the earth's surface.

2.2.1 Uses of Water

- Water's unique properties make it versatile and essential for various functions in all living organisms.
- Water is absolutely essential for life.
- The majority of life processes occur in the water present within the body.
- Water facilitates the absorption of nutrients, their distribution throughout the body, temperature regulation, and the removal of waste.
- Human beings depend on water for almost every developmental activity.
- Water is essential for drinking, irrigation, transportation, industrial cleaning, waste disposal, and serves as a coolant in thermal power plants.
- Water has shaped the Earth's surface and plays a key role in regulating our climate.

2.3.2 Hydrological Cycle

Evaporation: Water moving from the ground or bodies of water into the surrounding atmosphere undergoes evaporation, which is the

change from liquid to gas phases. Solar radiation is the main energy source for evaporation. Although they are specifically referred to as evapotranspiration, transpiration from plants is frequently implicitly included in evaporation.

Rainfall: Condensed water vapor that falls to the Earth's surface is known as precipitation. Besides rain, it can also occur as snow, hail, fog drip, graupel, or sleet. Condensation is the process by which water vapour turns into liquid water droplets in the atmosphere, forming fog and clouds. Water vapour is released into the atmosphere by plants and soil through transpiration. One invisible gas is water vapour.

Snowmelt: The runoff that results from snow melting.

Runoff: The different ways that water travels across the land are referred to as runoff. This covers both channel and surface runoff. Water can be stored in lakes or reservoirs, evaporate into the atmosphere, seep into the ground, or be sourced for use in agriculture or other human endeavours.

Infiltration: Infiltration water moving into the ground from the ground surface. Once inside, the water turns into groundwater or soil moisture.

Flow beneath the Surface: The underground water flow found in aquifers. Subsurface water may eventually seep into the oceans or return to the surface (for example, by pumping or springing). Under the influence of gravity or pressures caused by gravity, water flows back to the land surface at a lower elevation than where it entered. Groundwater can stay in aquifers for thousands of years because it tends to move and replenish slowly.

Sublimation: Water vapour is the direct result of the state transition from solid water (ice or snow).

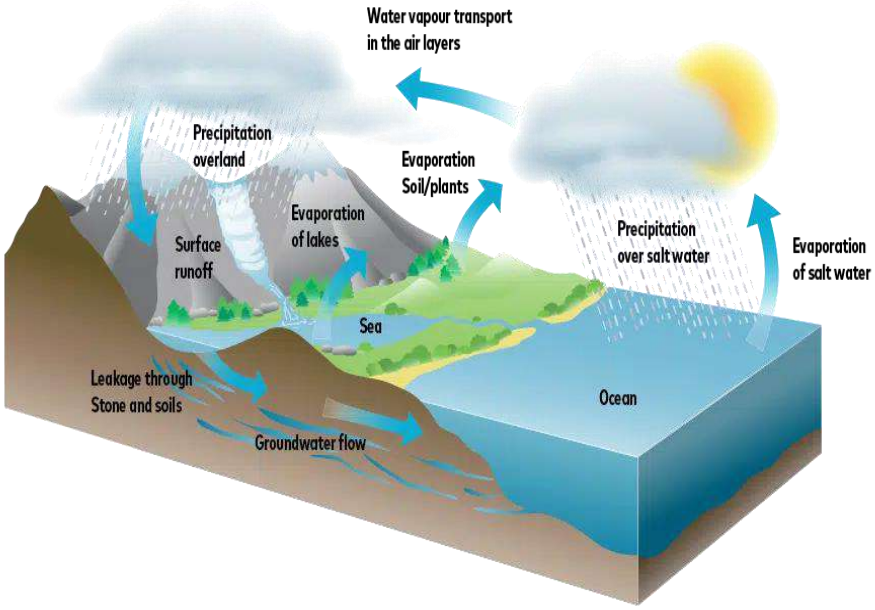


Figure 2.1: Water Cycle

2.3.3 Ground Water

About 9.68% of the total fresh water resources are in the form of ground water.

Aquifer: An aquifer is a highly permeable sedimentary rock layer that holds water. While clay and crystalline rock are poor aquifers, layers of sand and gravel are good ones.

There are two types of aquifers:

1. **An Aquifer without Confines:** Water seeping down from above in the form of rain and snowmelt replenishes unconfined aquifers that are covered by permeable earth materials.

2. **A Small Aquifer:** Constrained aquifers are only replenished where they meet the land surface; they are positioned between two permeable layers of rock or sediments.

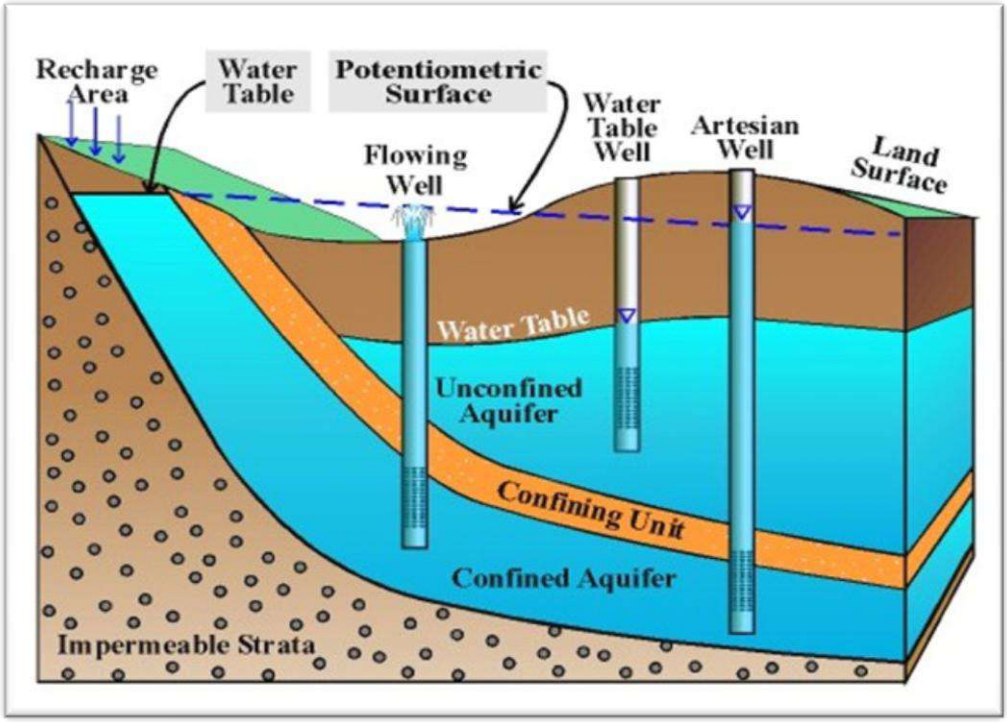


Figure 2.2: The Ground Water System.

2.3.4 The Over-Exploitation of Groundwater can lead to Several Serious Consequences

1. **Depletion of Water Tables:** Continuous overuse can cause groundwater levels to drop, making it more difficult and expensive to access water.

2. **Land Subsidence:** Excessive groundwater extraction can cause the ground to sink or collapse, leading to infrastructure damage and changes in the landscape.

3. **Reduced Water Quality:** Over-exploitation can lead to the contamination of groundwater by pollutants, as there may be insufficient recharge to dilute or flush contaminants.

4. **Saltwater Intrusion:** In coastal regions, over-extraction of groundwater can cause saltwater to intrude into freshwater aquifers, rendering the water undrinkable.

5. **Decreased Agricultural Productivity:** As groundwater levels drop, irrigation for crops may become less reliable, leading to reduced crop yields and food shortages.

6. **Ecosystem Disruption:** The depletion of groundwater affects wetlands, rivers, and other ecosystems that rely on a consistent flow of fresh water, disrupting habitats and biodiversity.

7. **Increased Costs for Water Supply:** As groundwater levels decrease, it becomes more expensive and energy-intensive to extract water, which can strain resources and increase water costs for communities.

8. **Drying of Wells:** Overuse can cause wells to dry up, impacting households, agriculture, and industries that rely on groundwater.

These effects underline the need for sustainable groundwater management to avoid long-term negative consequences.

2.3.5. Floods and Drought

- Floods are the overflow of water into normally dry land, often caused by heavy rainfall, rapid snowmelt, or the blockage of water flow. Flooding can occur in rivers, lakes, or coastal areas, and may also result from man-made factors like dam failures.
- Droughts are prolonged periods of abnormally low rainfall or dry conditions that result in a shortage of water. Droughts can occur due to natural climatic variations or human factors such as overuse of water resources and land degradation.

Causes of Floods

- **Heavy Rainfall:** Intense or prolonged rainfall can overwhelm drainage systems and cause rivers and streams to overflow, leading to flooding.
- **Snowmelt:** Rapid melting of snow and ice, often in spring, can cause rivers and lakes to rise and overflow.

- **Storm Surges:** Coastal floods can be triggered by strong storms, hurricanes, or typhoons, which push seawater onto the land, inundating coastal areas.
- **River Overflow:** When rivers receive more water than they can carry, such as from heavy rain or upstream snowmelt, they may overflow their banks and flood surrounding areas.
- **Dam or Levee Failure:** The failure or breach of dams, levees, or other water control structures can lead to catastrophic flooding downstream.
- **Urbanization:** Increased urban development, particularly in floodplains, reduces the natural absorption of rainwater by soil, causing surface runoff to flow into drains and rivers more quickly, leading to flooding.
- **Deforestation:** The removal of trees and vegetation can reduce the ability of the land to absorb rainfall, increasing runoff and the likelihood of flooding.
- **Coastal Erosion:** The degradation of natural coastal barriers, such as sand dunes or wetlands, can make coastal areas more vulnerable to flooding during storms.
- **Tsunamis:** Underwater earthquakes can generate massive ocean waves, leading to flooding in coastal regions.
- **Poor Drainage Systems:** Inadequate or clogged drainage systems can prevent rainwater from being properly channeled away, causing localized flooding, especially in cities.

Causes of Droughts

- **Low or Prolonged Rainfall:** The most common cause of drought is an extended period of below-average rainfall, which reduces the amount of water available in rivers, lakes, and groundwater.
- **Climate Change:** Global warming is altering weather patterns, leading to more frequent and severe droughts. Changes in

temperature can increase evaporation rates and reduce precipitation in certain areas.

- **High Temperatures:** Prolonged heatwaves can increase evaporation from soil, rivers, and lakes, exacerbating water shortages and contributing to drought conditions.
- **Deforestation:** The removal of forests reduces the ability of the land to retain moisture and can disrupt the water cycle, leading to drier conditions.
- **Soil Degradation:** Overuse of land, particularly in agriculture, can lead to soil erosion and loss of its ability to retain moisture, making it more susceptible to drought conditions.
- **Water Mismanagement:** Over-extraction of groundwater, excessive irrigation, and poor water management can deplete available water sources, contributing to drought-like conditions even when rainfall is adequate.
- **Changes in Atmospheric Circulation:** Shifts in atmospheric pressure systems can alter the distribution of rainfall, leading to droughts in regions that typically receive regular precipitation.
- **Overgrazing and Agriculture:** Excessive grazing by livestock or unsustainable agricultural practices can degrade the land, making it less able to absorb and retain water, contributing to drought.
- **Volcanic Activity:** Large volcanic eruptions can release ash and aerosols into the atmosphere, potentially blocking sunlight and disrupting rainfall patterns, which can result in regional droughts.

Preventive Measures of Floods

- Floodplain Management
- Improved Drainage Systems
- River and Coastal Protection
- Flood Forecasting and Early Warning Systems
- Reforestation and Vegetation Protection
- Wetland Restoration

- Sustainable Land Use Planning
- Education and Community Awareness

Preventive Measures of Droughts

- Water Conservation
- Rainwater Harvesting
- Sustainable Water Management
- Drought-Resistant Crops
- Soil Conservation
- Desalination and Water Recycling
- Reforestation and Afforestation
- Early Warning Systems
- Land Use Planning
- Public Awareness and Education

2.3.6 Conflicts over Water

The essential nature of water and its unequal distribution has frequently sparked inter-state and international conflicts. Disputes over the sharing of river water have particularly impacted farmers and created tensions within governments. Numerous countries are involved in intense rivalries over this vital resource.

National Water Conflicts

India faces several inter-state water disputes that have intensified in recent years:

- **Krishna River Water Dispute:** States like Karnataka, Telangana, Andhra Pradesh, and Maharashtra have been in conflict over the equitable distribution of Krishna River waters. The Krishna Water Disputes Tribunal II, established in 2004, continues to address these issues.
- **Mahanadi River Water Dispute:** Odisha and Chhattisgarh have been at odds over the sharing of Mahanadi River waters. The Mahanadi Water Disputes Tribunal was constituted in 2018 to resolve this conflict.

- **Mahadayi River Water Dispute:** Goa, Karnataka, and Maharashtra are engaged in a dispute over the sharing of Mahadayi River waters. The Mahadayi Water Disputes Tribunal was established in 2010 to adjudicate this matter.
- **Cauvery River Water Dispute:** The longstanding conflict between Karnataka and Tamil Nadu over the sharing of Cauvery River waters has seen various legal interventions, including the establishment of the Cauvery Water Disputes Tribunal.

International Water Conflicts

- Water scarcity and disputes over water resources are escalating globally, leading to conflicts and instability. The United Nations World Water Development Report 2024 highlights that tensions over water are exacerbating conflicts worldwide.
- In **Asia,** the Brahmaputra River, which flows through China, India, and Bangladesh, is a source of contention. China's plans to construct dams and divert water for hydroelectric projects have raised concerns in India and Bangladesh about reduced water availability for agriculture and daily use.
- The **Pacific** Institute's Water Conflict Chronology has identified over 350 new instances of conflicts associated with water resources and systems, reflecting a global increase in water-related disputes.
- In the **Middle East**, the Grand Ethiopian Renaissance Dam (GERD) on the Nile River has intensified tensions between Ethiopia, Egypt, and Sudan. Ethiopia's move to fill the dam's reservoir could reduce Nile flows by as much as 25%, potentially devastating Egyptian farmlands.

2.3.7 Big-Dams –Benefits and Problems

Big dams offer several significant benefits, including:

- **Water Supply:** Dams store large amounts of water that can be used for irrigation, drinking, and industrial purposes, especially in areas prone to droughts.

- **Hydroelectric Power Generation:** Dams are a major source of renewable energy. The water stored in reservoirs is used to generate hydroelectric power, contributing to a sustainable energy supply.
- **Flood Control:** By controlling the flow of rivers, dams help prevent floods during heavy rainfall, protecting communities, farmland, and infrastructure.
- **Irrigation:** Dams provide a reliable source of water for irrigation, which is crucial for agriculture, especially in arid regions.
- **Recreation and Tourism:** Reservoirs created by dams often become popular sites for recreational activities such as boating, fishing, and water sports, contributing to local economies.
- **Navigation:** Dams can improve waterway navigation by regulating water levels, allowing for easier transportation of goods by boat.
- **Ecosystem Services:** In some cases, dams can help maintain local ecosystems by regulating water flow and supporting aquatic habitats, though this can also have negative environmental impacts.
- **Economic Growth:** By providing water for agriculture, industry, and power generation, dams can stimulate economic development and improve living standards.

Despite these benefits, it's important to consider the environmental and social impacts of big dams, such as displacement of communities and the disruption of natural ecosystems.

Problems

The impacts of big dams can be upstream as well as downstream levels.

Upstream Problems Include the Following

- Water Quality Degradation
- Sediment Accumulation
- Ecosystem Disruption
- Displacement of local communities
- Loss of Agricultural Land
- Altered Water Flow
- Cultural Impacts
- Increased Erosion
- Changes in River Ecosystems
- Increased Water Demand

Downstream Problems Include the Following

Downstream problems of dams can have significant impacts on the environment, communities, and economies located downstream of a dam site. Some of the key downstream issues include:

- Reduced Water Flow
- Altered River Ecosystems
- Sediment deposition
- Loss of Fertile Floodplains
- Water Quality Issues
- Erosion and Bank Instability
- Flooding Risks
- Disruption of Fisheries
- Impact on Local Communities
- Reduced Hydroelectric Power Generation
- Decline in Wetland Areas
- Changes in Water Temperature

2.4 Mineral Resources

Minerals are naturally occurring, inorganic substances with a definite chemical composition and a crystalline structure. They are the building blocks of rocks and are formed through geological processes.

Key Characteristics of Minerals

1. **Naturally Occurring:** Minerals are formed by natural geological processes without human intervention.
2. **Inorganic:** They are not derived from living organisms.
3. **Solid State:** Minerals exist in a solid state under normal Earth conditions.
4. **Definite Chemical Composition:** Each mineral has a specific chemical formula, though slight variations can occur.
5. **Crystalline Structure:** The atoms in minerals are arranged in an orderly and repeating pattern, forming crystals.

Examples of Minerals

- Quartz (SiO_2)
- Feldspar
- Mica
- Calcite ($CaCO_3$)
- Pyrite (FeS_2)

2.4.1 Uses of Minerals

Minerals have a wide range of uses across various industries and aspects of daily life. Here's an overview of their applications:

1. **Industrial Uses**
   - **Construction Materials:** Minerals like limestone, granite, and gypsum are used in building materials (cement, plaster, concrete, and bricks).
   - **Metallurgy:** Ores such as iron, copper, aluminum, and zinc are refined to produce metals used in tools, machinery, and vehicles.
   - **Glass and Ceramics:** Minerals like quartz, feldspar, and kaolin are used in making glass, ceramics, and tiles.

2. **Energy Production**
   - **Coal:** Used as a fuel for electricity generation and industrial processes.
   - **Uranium:** Fuels nuclear power plants.
   - **Oil and Gas:** Fossil fuels derived from mineral-rich geological formations.

3. **Electronics and Technology**
   - **Rare Earth Elements:** Essential for manufacturing smart-phones, computers, and renewable energy technologies like wind turbines.
   - **Silicon:** Used in semiconductors and photovoltaic cells for solar panels.

4. **Agriculture**
   - **Fertilizers:** Minerals like phosphate and potash are used to enhance soil fertility and crop yields.
   - **Lime:** Helps to neutralize soil acidity.

5. **Health and Nutrition**
   - **Dietary Supplements:** Minerals such as calcium, iron, and magnesium are critical for human health.
   - **Salt:** Used as a preservative and seasoning in food.

6. **Jewellery and Ornamentation**
   - **Precious Stones:** Diamonds, emeralds, and rubies are used in Jewellery.
   - **Gold and Silver:** Used in ornaments and coins.

7. **Environmental Applications**
   - **Water Purification:** Minerals like zeolite are used in water filtration systems.

- **Carbon Sequestration:** Certain minerals, like olivine, can trap carbon dioxide from the atmosphere.

8. **Transportation**
   - **Road Construction:** Minerals like gravel, sand, and asphalt are essential in paving roads.
   - **Vehicles:** Steel (from iron ore) and aluminum are used in vehicle manufacturing.

9. **Cultural and Artistic Uses**
   - **Sculptures and Monuments:** Marble and granite are used in art and architecture.
   - **Pigments:** Minerals like hematite and malachite are used in paints and dyes.

2.4.2 Environmental Impacts of Mineral Extraction

Here are some major mines in India along with their extracted minerals and associated environmental and social impacts:

1. **Jharia Coalfield (Jharkhand)**
   - **Minerals Extracted:** Coal.
   - **Impacts**
     - **Underground Fires:** Persistent coal seam fires release toxic gases, impacting air quality and contributing to greenhouse gas emissions.
     - **Land Subsidence:** Mining weakens the land, causing sinkholes and displacing communities.
     - **Health Issues:** Air pollution leads to respiratory diseases among workers and nearby residents.

2. **Kudremukh Iron Ore Mine (Karnataka)**
   - **Minerals Extracted:** Iron ore.

- **Impacts**
  - **Deforestation:** Mining in the Western Ghats has led to loss of biodiversity in this UNESCO World Heritage Site.
  - **Soil Erosion:** Mining operations have destabilized the slopes, increasing erosion.
  - **Water Pollution:** Mining tailings contaminate local rivers, affecting aquatic ecosystems and downstream water users.

3. **Bauxite Mines (Odisha)**
   - **Minerals Extracted:** Bauxite (used for aluminum production).
   - **Impacts**
     - **Deforestation:** Mining operations in areas like the Niyamgiri Hills have led to large-scale forest loss.
     - **Loss of Tribal Land:** Indigenous communities, such as the Dongria Kondh, face displacement and loss of cultural heritage.
     - **Soil and Water Contamination:** Toxic residues from bauxite processing seep into the environment.

4. **Singareni Collieries (Telangana)**
   - **Minerals Extracted:** Coal.
   - **Impacts**
     - **Air Pollution:** Dust and emissions from mining operations affect local air quality.
     - **Water Scarcity:** Mining depletes and contaminates local water sources.
     - **Community Displacement:** Mining projects have led to relocation of local populations.

5. **Hutti Gold Mines (Karnataka)**
   - **Minerals Extracted:** Gold.
   - **Impacts**
     - **Water Pollution:** Cyanide and other chemicals used in gold extraction contaminate water bodies.
     - **Health Risks:** Workers are exposed to hazardous substances, leading to long-term health issues.
     - **Land Degradation:** Mining leaves barren landscapes unsuitable for agriculture.

6. **Ratnagiri Bauxite Mines (Maharashtra)**
   - **Minerals Extracted:** Bauxite.
   - **Impacts**
     - **Ecosystem Disruption:** Mining in this ecologically sensitive area harms local flora and fauna.
     - **Water Pollution:** Runoff from mining sites affects nearby streams and agricultural lands.

7. **Jaduguda Uranium Mines (Jharkhand)**
   - **Minerals Extracted:** Uranium.
   - **Impacts**
     - **Radiation Risks:** Mining waste poses serious health risks to workers and nearby residents.
     - **Water Contamination:** Radioactive materials leach into local water sources.
     - **Community Health Issues:** Increased cases of cancer and birth defects reported in the region.

8. Korba Coalfield (Chhattisgarh)

- **Minerals Extracted:** Coal.
- **Impacts**
  - **Air Pollution:** High levels of particulate matter and emissions from mining affect local air quality.
  - **Deforestation:** Mining has cleared large tracts of forest land.
  - **Water Depletion:** Heavy water use for mining operations reduces availability for agriculture and local needs.

9. Zawar Mines (Rajasthan)

- **Minerals Extracted:** Zinc and lead.
- **Impacts**
  - **Soil Contamination:** Lead and zinc residues pollute the soil, affecting agriculture.
  - **Water Pollution:** Toxic chemicals from processing seep into water bodies.
  - **Health Risks:** Local populations face risks from heavy metal exposure.

10. Sukinda Chromite Mines (Odisha)

- **Minerals Extracted:** Chromite (used in stainless steel production).
- **Impacts**
  - **Water Pollution:** Hexavalent chromium, a carcinogenic substance, contaminates water bodies.
  - **Health Issues:** Local populations face higher rates of cancer and respiratory diseases.
  - **Soil Degradation:** Toxic waste from chromite mining renders land unsuitable for agriculture.

By adopting sustainable mining practices and enforcing stricter environmental regulations, India can work toward mitigating the negative impacts of these mining operations.

2.4.3 Impacts of Mining

The extraction of minerals from major mines has profound environmental, social, and economic impacts. These impacts often depend on the scale, location, and method of extraction, but some common major impacts are as follows:

- **Environmental Impacts**

1. Habitat and Biodiversity Loss

- **Deforestation and Land Use Change:** Large-scale mining operations often clear forests and vegetation, leading to habitat destruction.
- **Endangered Species Risk:** Mining disrupts ecosystems, threatening local flora and fauna.

2. Water Pollution

- **Acid Mine Drainage (AMD):** Mining of sulfide minerals releases acidic water that contaminates rivers, streams, and groundwater.
- **Toxic Wastewater:** Heavy metals such as mercury, arsenic, and lead often leach into water bodies.
- **Sedimentation:** Increased sediment from mining can clog rivers and harm aquatic life.

3. Air Pollution

- **Dust Emissions:** Blasting and transporting minerals produce fine particulate matter harmful to human and animal health.
- **Toxic Gas Emissions:** Smelting and refining release sulfur dioxide and other harmful gases.

4. **Soil Erosion and Degradation**
    - **Loss of Fertile Land:** Mining removes topsoil, often making the land unsuitable for agriculture or reforestation.
    - **Soil Contamination:** Toxic chemicals from mining seep into the soil, affecting plant growth.

5. **Climate Change Contribution**
    - **Greenhouse Gas Emissions:** Mining operations and associated industries are significant sources of carbon dioxide.
    - **Deforestation Effects:** Loss of forests reduces carbon sequestration capabilities.

➢ **Social Impacts**

1. **Community Displacement**
    - Large mining projects often force local populations to relocate, disrupting livelihoods and cultural ties.

2. **Health Hazards**
    - Workers and nearby communities may suffer from exposure to toxic substances, leading to respiratory problems, skin diseases, and other health issues.

3. **Water Scarcity**
    - High water demand for mining operations often competes with local agricultural and drinking water needs, causing shortages.

4. **Loss of livelihoods**
    - Communities dependent on farming, fishing, or forestry may lose their income sources due to environmental degradation caused by mining.

- **Economic Impacts**

1. Positive Contributions

- **Job Creation:** Mines often provide employment opportunities for local populations.
- **Economic Growth:** Mining contributes to national GDP and foreign exchange earnings.

2. Negative Economic Impacts

- **Overdependence:** Reliance on mining can lead to economic instability if mineral prices drop.
- **Resource Depletion:** Unsustainable mining exhausts non-renewable resources, leaving regions economically vulnerable.

2.4.4 Mitigation Strategies

- **Sustainable Mining Practices:** Reduce environmental damage through better technology and management.
- **Land Rehabilitation:** Restore ecosystems after mining operations.
- **Community Engagement:** Include local populations in decision-making and ensure fair compensation.
- **Regulatory Compliance:** Enforce stricter environmental and safety regulations.

Balancing the benefits of mining with the need to minimize its impacts is crucial for sustainable development.

2.5 Food Resources

2.5.1 World Food Problems

World Food Problems refer to the challenges related to food production, distribution, and consumption that contribute to global

hunger, malnutrition, and food insecurity. These problems are complex and multifaceted, often intertwined with economic, environmental, political, and social factors. Here's an in-depth look at some of the major food problems facing the world today:

1. **Hunger and Food Insecurity**
   - Global Hunger
   - Food Insecurity

2. **Malnutrition**
   - Under nutrition
   - Over nutrition
   - Micronutrient Deficiencies

3. **Poverty and Access to Food**
   - Poverty
   - Unequal Distribution

4. **Climate Change and Environmental Impact**
   - Extreme Weather Events
   - Changing Weather Patterns
   - Soil Degradation

5. **Food Waste**
   - Global Food Waste
   - Environmental Impact

6. **Global Population Growth**
   - Increasing Demand
   - Urbanization

Solutions and Approaches to Address Food Problems

1. Improved Agricultural Practices
2. Better Food Distribution
3. Policy and Trade Reform
4. Nutrition Education

5. Technology and Innovation
6. Climate Action

2.5.2 Impacts of Overgrazing and Agriculture

Overgrazing

Overgrazing refers to the excessive grazing of vegetation by livestock, which can lead to several detrimental environmental, ecological, and socio-economic impacts.

Below are the primary effects of overgrazing

Impact of Overgrazing

1. **Soil Degradation**
   - Erosion
   - Soil Compaction
   - Reduced Soil Fertility

2. **Loss of Vegetation and Biodiversity**
   - Decreased Plant Cover
   - Increased Invasive Species
   - Disrupted Ecosystems

3. **Desertification**
   - Land Desertification
   - Declining Productivity

4. **Water Quality Issues**
   - Water Pollution
   - Reduced Water Availability

5. **Carbon Emissions and Climate Change**
   - Increased Carbon Release
   - Loss of Carbon Sequestration

Agriculture

Traditional Agriculture and its Impacts

1. Positive Impacts of Traditional Agriculture

- Sustainability
- Biodiversity

2. Negative Impacts of Traditional Agriculture

- Lower Productivity
- Limited Technology
- Vulnerability to Climate Change
- Soil Degradation

Modern Agriculture and its Effects

- Modern agriculture refers to farming practices that rely on advanced technologies, machinery, synthetic chemicals, and high-yield crops to increase productivity. While it has led to higher food production, it also has several positive and negative effects on the environment, society, and economy.

- **Positive Effects of Modern Agriculture**
  - Increased Food Production
  - Efficiency
  - Reduced Labor Requirements
  - Technological Innovation
  - Improved Crop Resistance

- **Negative Effects of Modern Agriculture**
  - Environmental Pollution
  - Loss of Biodiversity
  - Soil Degradation
  - Climate Change
  - Water Overuse
  - Health Risks
  - Loss of Traditional Farming

- **Balancing the Impacts**

  While modern agriculture has made significant advancements in feeding the global population, there is a growing need to find sustainable solutions. Some approaches include:

- **Agro Ecology:** Incorporating ecological principles into farming to make it more sustainable.
- **Sustainable Intensification:** Increasing food production without harming the environment, often by using less water, fewer chemicals, and improving soil health.
- **Organic Farming:** Focusing on natural methods for pest control and soil management.
  - It makes use of hybrid seeds of selected and single crop variety.
  - High-tech equipments, lots of energy subsidies in the form of fertilizers and, pesticides
  - Irrigation water

2.6 Energy Resources

Energy is the capacity to do work or produce change. In scientific terms, it is a property of objects or systems that enables them to perform work or cause other changes in the system, like moving an object, heating it, or creating light.

- Energy exists in various forms, including **kinetic energy** (motion), **potential energy** (stored energy due to position), **thermal energy** (heat), **chemical energy** (stored in bonds of molecules), electrical energy (from the flow of electrons), and **nuclear energy** (stored in atomic nuclei).
- The **Law of Conservation of Energy** states that energy cannot be created or destroyed, only transformed from one form to another.

- In simpler terms, energy is what allows things to happen—whether that's moving, growing, changing temperature, or even powering our daily activities

2.6.1 Growing Energy Needs

Here are some key factors contributing to growing energy needs:
1. Population Growth
2. Economic Development
3. Urbanization
4. Technological Advancements
5. Increased Standards of Living
6. Electrification of Transportation
7. Climate Change and Weather Patterns
8. Industrialization and Automation

Implications of Growing Energy Needs

1. Environmental Impact
2. Energy Security
3. Renewable Energy Transition
4. Innovation in Energy Efficiency
5. Energy Storage and Grid Modernization

2.6.2 Renewable Energy Resources

A **renewable energy resource** is any source of energy that is naturally replenished on a human timescale and can be used continuously without running out. These resources are considered sustainable because they are not depleted when used, unlike fossil fuels, which can take millions of years to form.

Common types of renewable energy resources include:

1. **Solar Energy:** Energy from the sun, harnessed through solar panels or solar thermal systems.

2. **Wind Energy:** Energy derived from the movement of air, captured by wind turbines to generate electricity.
3. **Hydropower (Water Energy):** Energy from the flow of water, usually harnessed through dams or water turbines in rivers and streams.
4. **Geothermal Energy:** Energy from the heat stored beneath the Earth's surface, typically used for electricity generation or heating.
5. **Biomass Energy:** Energy from organic materials (wood, agricultural waste, or dedicated crops) that can be burned or converted into biofuels like ethanol or biodiesel.
6. **Tidal and Wave Energy:** Energy from the movement of the ocean's tides and waves, captured by specialized turbines or other technologies.

These resources are considered environmentally friendly because they produce little to no greenhouse gas emissions, making them key to combating climate change and achieving sustainability goals.

2.6.3 Solar Energy

Solar energy is the energy we get from the sun. It can be captured and used to generate electricity or heat, making it a renewable and clean source of energy.

Methods of Harvesting Solar Energy

1. Solar Cells (or) Photovoltaic Cells (or) PV Cells

Solar cells, also called **photovoltaic (PV) cell**s, are devices that convert **sunlight directly into electricity** using a process called the **photovoltaic effect.** The principle behind how they work is based on the behavior of semiconductor materials when exposed to light.

Here's a breakdown of how solar cells work:

1. **Structure of a Solar Cell:** A solar cell is typically made up of **two layers** of semiconductor materials, usually **silicon:**
   - **N-type Layer (Negative):** This layer is doped (treated) with materials that give it extra electrons, making it negatively charged.
   - **P-type Layer (Positive):** This layer is doped with materials that create "holes" (spaces where electrons can go), giving it a positive charge.

     The **junction** between these two layers is called the **P-N junction,** which is key to how solar cells generate electricity.

2. **The Photovoltaic Effect:** The **photovoltaic effect** describes the process by which light (photons) is absorbed by a solar cell and generates electricity. The steps are as follows:
   - **Absorption of Light**
     - When sunlight (which is made of photons) hits the surface of the solar cell, the **photons** are absorbed by the semiconductor material.
     - This energy from the photons excites the electrons in the **N-type layer**, causing them to break free from their atoms.
   - **Generation of Electron-Hole Pairs**
     - The freed electrons leave behind **holes** (empty spaces), which are essentially places where electrons used to be. This creates **electron-hole pairs** in the semiconductor material.
   - **Movement of Electrons**
     - The **electric field** at the P-N junction (the boundary between the N-type and P-type layers) forces the electrons to move towards the **N-type side** and the holes towards the P**-type side.**
     - This movement of electrons creates a **flow of electric current.**

- **Electric Current**
  - This flow of electrons through the external circuit (connected to the solar cell) generates **electricity.**
  - The electricity can then be used to power devices, stored in batteries, or sent to the grid for use elsewhere.

3. Output and Connection

- A **single solar cell** generates a small voltage (about 0.5 to 0.6 volts). To produce usable power, **multiple solar cells** are connected together to form a **solar panel.**
- The panels generate **DC (direct current)** electricity, which can be used directly, stored in batteries, or converted to **AC (alternating current)** electricity using an **inverter** for use in homes and businesses.

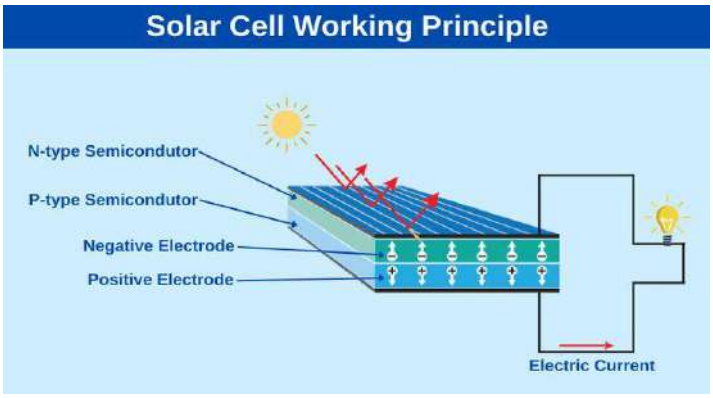


Figure 2.3: Solar energy working scheme

Uses

- Solar cells convert sunlight into electricity, making them ideal for powering small devices like calculators and streetlights. They are also used in larger applications, such as rooftop solar panels for homes and businesses, to generate renewable energy.

Solar Battery

A solar battery stores the extra energy generated by solar panels during the day so you can use it later, like at night or on cloudy days.

It helps make the most of solar power and reduces reliance on the grid.

Solar Heat Collectors

Solar heat collectors capture the sun's heat and use it to warm water or air. They're often used in homes and buildings for heating or hot water.

Solar Water Heater

- A solar water heater is a device that uses sunlight to heat water for household or commercial use.
- It is an environmentally friendly and cost-effective alternative to traditional water heaters that rely on electricity or gas.
- The system consists of three main parts: solar collectors, a storage tank, and a circulation system. The solar collectors, usually installed on the roof, absorb sunlight and convert it into heat.
- This heat warms up the water that flows through pipes inside the collectors.
- There are two common types of solar collectors: **flat-plate collectors** and **evacuated tube collectors.** Both are designed to capture and retain heat effectively.
- The heated water is stored in an insulated tank, ready for use when needed.
- The circulation system ensures that water moves between the collectors and the tank. Some systems use pumps for this process, while others rely on natural circulation.
- Solar water heaters can be classified into **active systems**, which use pumps, and passive systems, which do not. **Passive systems** are simpler and more affordable, while active systems are more efficient.

- These heaters work best in sunny areas but can still function in colder climates with the help of backup systems or antifreeze. They help reduce energy bills and lower greenhouse gas emissions, making them a sustainable choice for heating water.

By using a solar water heater, you can save money and help protect the planet.

2.6.4 Wind Energy

Wind energy is the process of converting the kinetic energy of wind into mechanical or electrical energy using wind turbines. It is a clean, renewable energy source that helps generate electricity without producing greenhouse gas emissions.

Harvesting of Wind Energy

Wind Mills

A **windmill** is a machine that uses the energy of the wind to do work, like grinding grain or pumping water. It has large blades that turn when the wind blows, powering its mechanism.

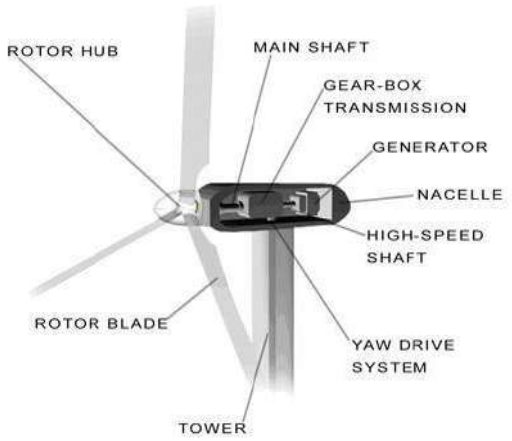


Figure 2.4: Wind Mill

Wind Farms

A **wind farm** is a group of wind turbines located in the same area, working together to generate electricity from wind energy. These farms can be on land or offshore in bodies of water.

Advantages of Wind Farms

1. **Renewable Energy:** Wind is an endless resource, making wind farms a sustainable way to produce electricity.

2. **Environmentally Friendly:** They produce no greenhouse gas emissions or air pollution during operation.

3. **Cost-Effective:** After installation, the operational costs are low since wind is free.

4. **Energy Independence:** Wind farms reduce reliance on fossil fuels and imported energy.

5. **Land Use Efficiency:** Onshore wind farms can coexist with agricultural activities, allowing dual use of land.

6. **Job Creation:** They create jobs in manufacturing, installation, and maintenance.

2.6.5 Tidal Energy (or) Tidal Power

- Tidal energy is the power generated by the movement of ocean tides.
- It uses the rise and fall of seawater caused by the gravitational pull of the moon and sun. Special devices, like tidal turbines or barrages, capture this energy and convert it into electricity.
- Tidal energy is renewable and predictable since tides follow regular cycles.

- It produces no pollution and helps reduce reliance on fossil fuels.
- However, tidal energy systems can be costly to build and may impact marine ecosystems.
- Despite these challenges, tidal energy is a reliable and sustainable source of clean electricity for coastal areas.

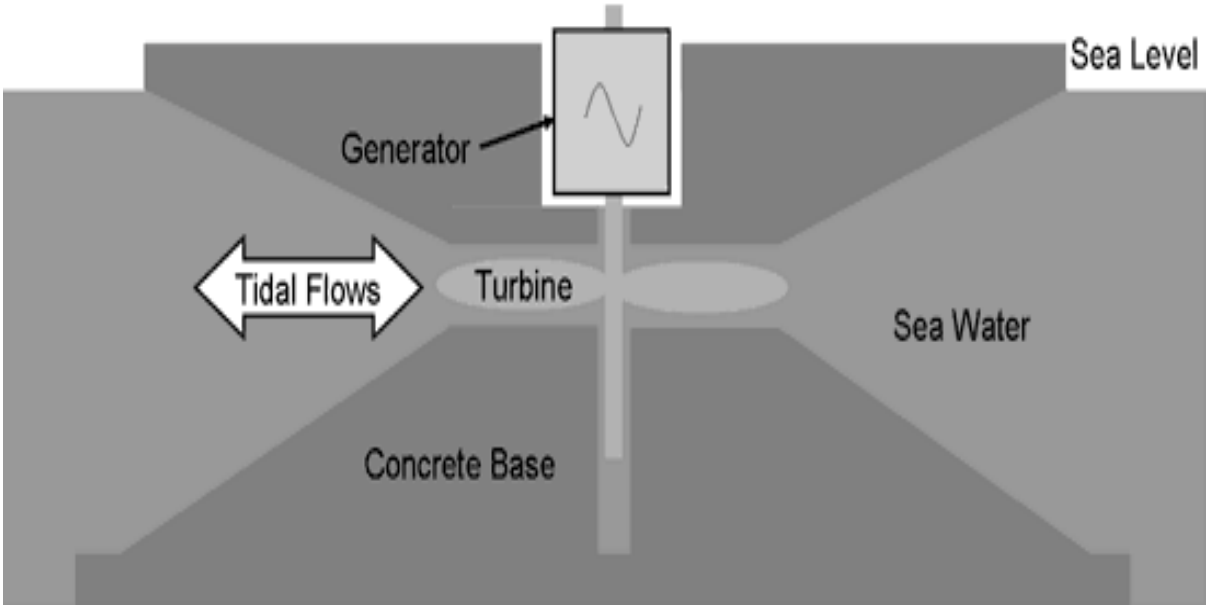


Figure 2.5: Water Flows in to the reservoir to turn the turbine

2.6.6 Ocean Energy

It can be generated by following ways.

1. **Ocean Thermal Energy (OTE)**
   - Ocean thermal energy is the power generated using the temperature difference between the warm surface water and the cold deep water of the ocean.
   - This process, called Ocean Thermal Energy Conversion (OTEC), uses the warm water to produce steam, which drives a turbine to generate electricity.
   - The cold water helps cool the system and condense the steam back into water.

- Ocean thermal energy is renewable, available 24/7, and eco-friendly since it doesn't produce pollution.
- However, it requires specific conditions, like tropical waters, and the technology can be expensive to build and maintain.
- It's a promising clean energy source for coastal regions.

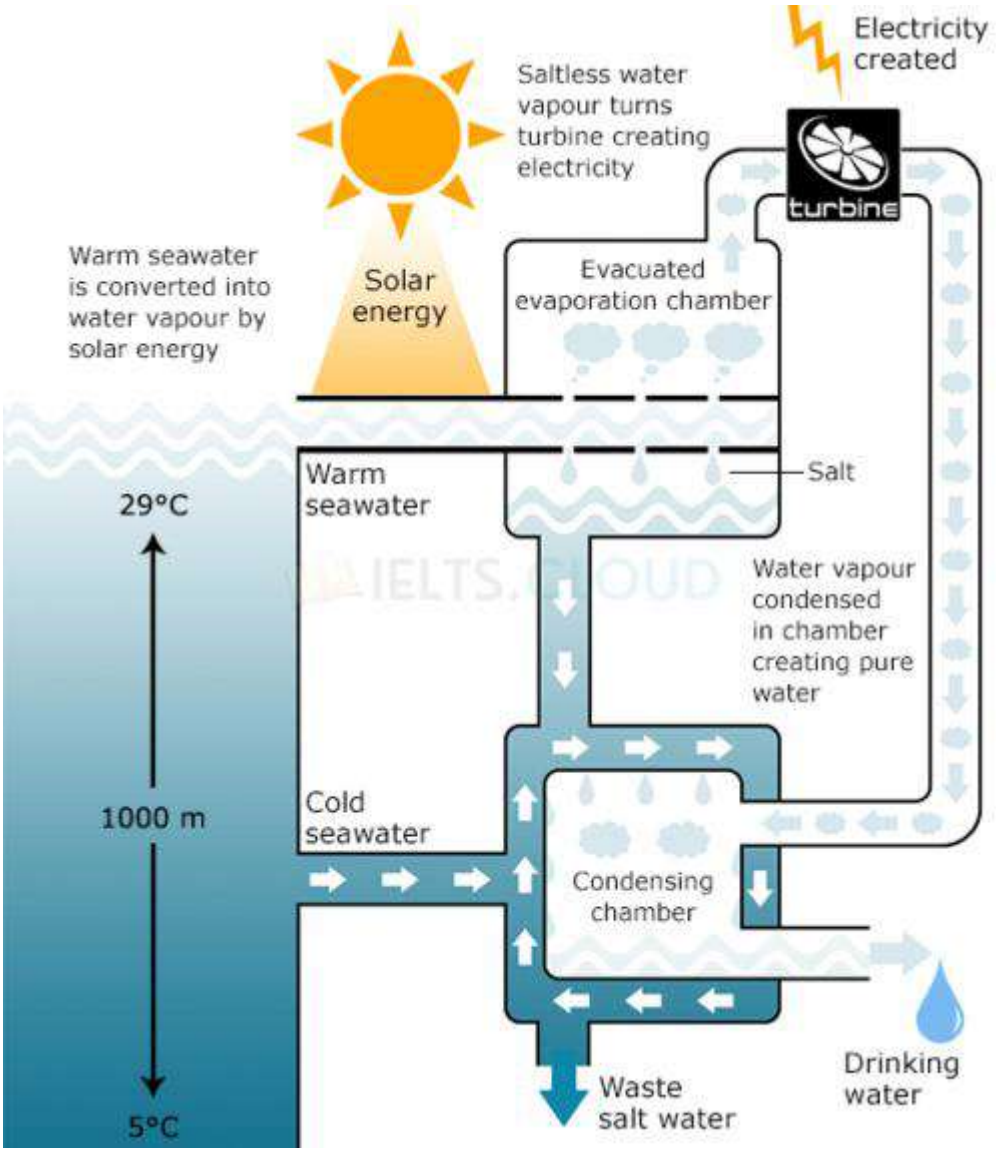


Figure 2.6: Ocean Thermal Energy Plant (Image source: www.ielts.clod).

2.6.7 Geo-thermal Energy

1. Geothermal energy is heat that comes from inside the Earth. This heat can be found in hot rocks, steam, or hot water beneath the Earth's surface.
2. Geothermal power plants use this heat to generate electricity or to heat buildings.
3. It is a clean, renewable energy source because it doesn't produce harmful emissions.
4. Geothermal energy is available all the time, making it a reliable energy source, especially in areas close to volcanic or tectonic activity.
5. It is a sustainable way to generate power and reduce our reliance on fossil fuels.
6. Temperature of the earth increases at a rate of 20-750C per km, when we move down the earth surface.
7. High temperature and high-pressure steam fields exists below the earth’s surface in many places.
8. The energy harnessed from the high temperature present inside the earth is called geothermal energy.

1. **Natural Geysers:** Natural geysers are hot springs that periodically release bursts of steam and hot water into the air. This happens when underground water is heated by the Earth's heat and builds up pressure until it erupts.

2. **Artificial Geysers:** Artificial geysers are man-made systems that imitate natural geysers. They use pressure and heat to force water or steam to shoot out of a pipe, creating a burst of hot water or steam similar to a natural geyser. These are often created for display purposes in spas or amusement parks.

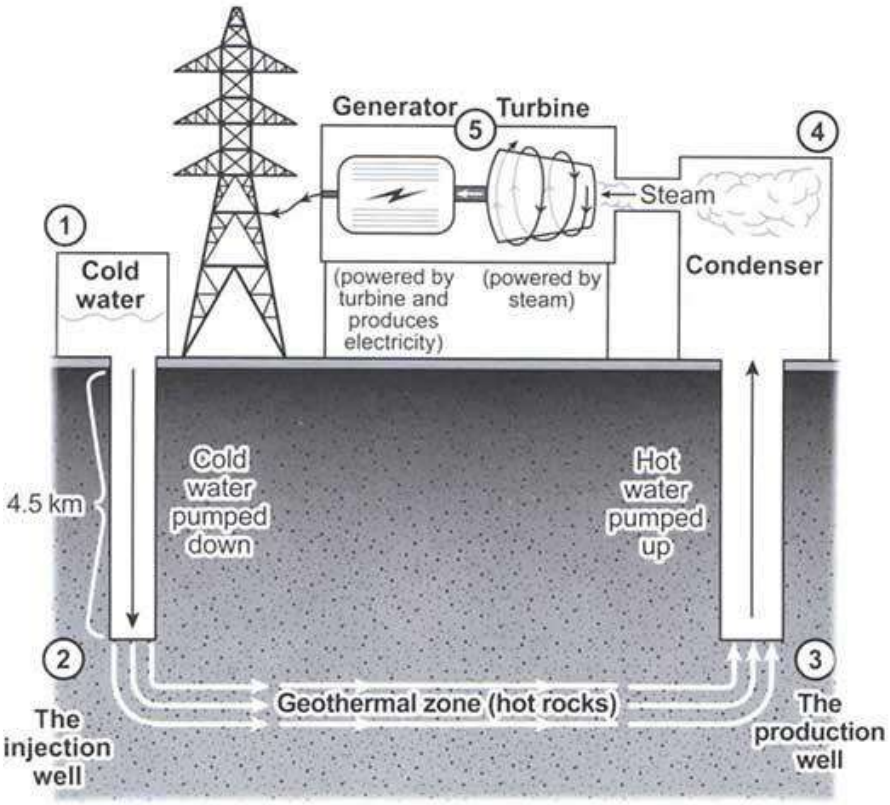


Figure 2.7: Geothermal energy power plant.

2.6.8 Biomass Energy

Biomass energy is energy produced from organic materials like plants, wood, or animal waste. When these materials are burned or processed, they release energy that can be used for heating, electricity, or fuel.

Example: Burning wood in a fireplace or using crops like corn to make biofuels such as ethanol, Cattle dung

Biogas

- Biogas is a type of energy produced from the breakdown of organic waste, like food scraps, animal manure, or plant materials, by bacteria in an oxygen-free environment.
- It mostly consists of methane, which can be used for cooking, heating, or generating electricity.

- Biogas is a renewable and eco-friendly energy source.
- Bio gas plants used in our country are basically of two types:

1. Floating Gas Holder Type Biogas Plant

- A floating gas holder type biogas plant is a type of biogas system where the gas is stored in a floating container (called a gas holder) that rises and falls with the amount of gas produced.
- In this system:
- Organic waste (like food scraps or animal manure) is placed in a digester where bacteria break it down and produce biogas.
- The biogas rises and is collected in the floating gas holder, which floats on top of the digester.
- As more gas is produced, the gas holder rises. When the gas is used, the holder lowers.
- This type of plant is simple, cost-effective, and commonly used for small-scale biogas production, especially in rural areas. It helps generate energy from waste while also producing useful by-products like compost.

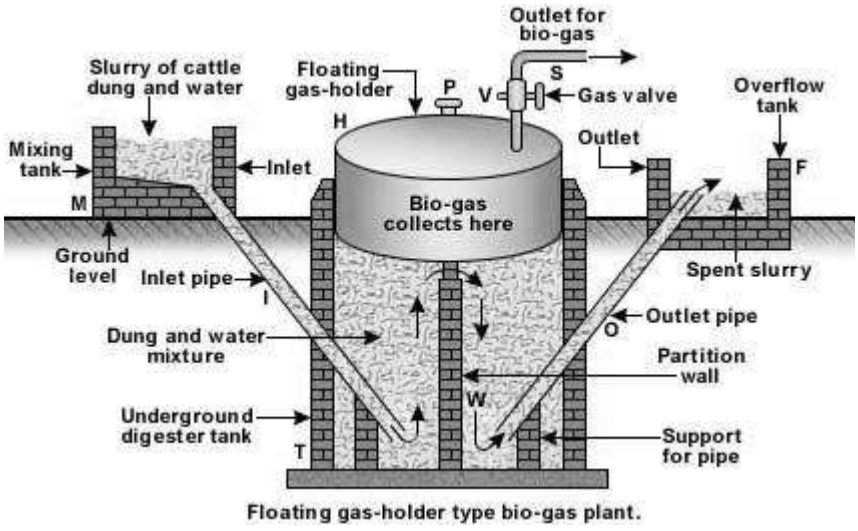


Figure 2.8: Bio-Gas Plant (Floating)

2. **Fixed dome type biogas plant: A fixed dome type biogas plant** is a common design used for producing biogas from organic waste. In this system:

1. Organic materials, such as food scraps, animal manure, or plant waste, are placed in a digester (a large underground chamber).
2. Bacteria break down the waste in the absence of oxygen, producing biogas (mostly methane).
3. The biogas collects in a fixed dome at the top of the digester, which does not move.
4. The gas is then stored and can be used for cooking, heating, or generating electricity.

The main feature of a fixed dome biogas plant is the dome-shaped structure that remains in place. It is a durable, efficient, and low-maintenance system, especially suitable for small to medium-scale biogas production.

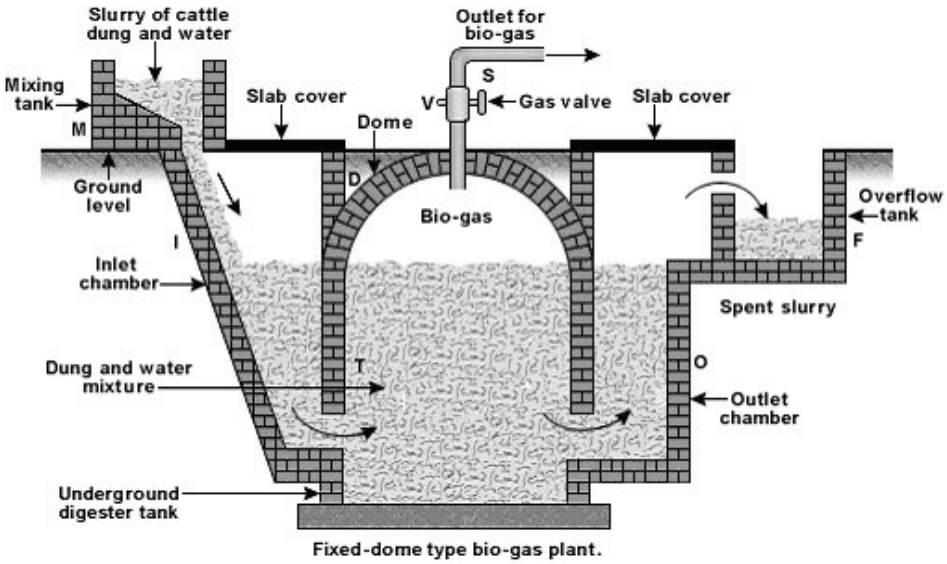


Figure 2.9: Bio-Gas Plant (Fixed Dome)

2.6.9 Bio Fuels

Biofuels are fuels made from organic materials like plants, algae, or animal waste. They can be used for transportation, heating, or generating electricity and are considered a renewable energy source.

Examples include ethanol (from crops like corn) and biodiesel (from vegetable oils or animal fats).

Ethanol: Ethanol can be easily produced from sugarcane. Its calorific value is lower than that of petrol, and it generates significantly less heat when compared to petrol.

Methanol: Methanol can be easily derived from ethanol or plants containing sugar. Its calorific value is also much lower compared to gasoline and diesel.

1. Gasohol

Gasohol is a mixture of **ethanol + gasoline.**

- Gasohol is a mixture of gasoline and ethanol, typically containing 10-20% ethanol and the rest gasoline.
- It is used as an alternative fuel for internal combustion engines, such as those in cars and motorcycles.
- Gasohol is considered a cleaner fuel because ethanol is derived from renewable resources like corn, sugarcane, or other plants, and it reduces the reliance on fossil fuels.
- Using gasohol helps lower carbon emissions and improves air quality.
- It also helps reduce the environmental impact of gasoline. However, gasohol has a lower energy content compared to pure gasoline, which can affect engine performance.

2.6.10 Hydrogen Fuel

- Hydrogen fuel is energy stored in hydrogen gas, which can be used to power vehicles, generate electricity, or heat.
- When hydrogen reacts with oxygen, it produces energy, with water as the only by product, making it a clean and eco-friendly fuel.

- Hydrogen fuel is used in fuel cells, which convert the gas into electricity without harmful emissions.
- It is a renewable energy source, especially when produced using renewable methods like solar or wind energy, and can be an alternative to fossil fuels.

$2H_2+O_2 \rightarrow 2H_2O+150KJ$

Disadvantages of Hydrogen Fuel

1. Production Cost
2. Storage and Transport
3. Infrastructure
4. Energy Efficiency
5. Safety Concerns

2.6.11 Non-Renewable Energy Sources

Non-renewable energy resources are natural resources that cannot be replenished on a human timescale. Once they are used up, they are gone for good. These include:

1. **Fossil Fuels:** Such as coal, oil, and natural gas. They are formed from the remains of plants and animals over millions of years.

2. **Nuclear Energy:** Derived from uranium, which is a finite resource. While nuclear energy itself doesn't produce carbon emissions, uranium mining and waste disposal can have environmental impacts.

Coal

Coal is a non-renewable energy source that comes from the remains of plants that lived millions of years ago. It is burned to produce heat and electricity.

Various Stages of Coal

Coal forms in several stages over millions of years, as plant material is gradually transformed by heat and pressure. The stages are:

1. Peat
2. Lignite
3. Sub-bituminous coal
4. Bituminous coal
5. Anthracite

Disadvantages

- Environmental Pollution
- Greenhouse Gas Emissions
- Non-renewable
- Mining Impact
- Health Issues
- **Contribution to Climate Change:** The large amount of CO2 produced from burning coal is a major factor in global warming and climate change.

2.6.12 Petroleum

Petroleum or crude oil = hydrocarbons +small amount S, O, N.

Occurrence

Petroleum, also known as crude oil, is found deep beneath the Earth's surface, typically in sedimentary rock formations.

It is formed over millions of years from the remains of ancient plants and animals, which are buried and subjected to heat and pressure.

LPG (Liquefied Petroleum Gas)

LPG (Liquefied Petroleum Gas) is a type of fuel made from a mixture of gases, primarily used for cooking, heating, and even as

fuel for vehicles. It is produced by refining crude oil or processing natural gas.

Composition of LPG

- **Propane (C3H8):** The main component, typically making up about 60-70% of LPG. It is used for heating and cooking.
- **Butane (C4H10):** Makes up around 30-40% of LPG. It is often used in areas with milder weather or in portable gas stoves.

LPG is stored in liquid form under pressure, making it easy to transport and use in homes, industries, and vehicles.

Natural Gas

Natural gas is a fossil fuel made mostly of (50-90%) methane (CH_4), along with smaller amounts of other gases like ethane, propane, and butane. It is found deep underground, often alongside oil deposits, and is extracted through drilling.

Natural gases are two types. Those are

- **Dry Gas:** Dry gas is natural gas that is mostly made up of methane and contains little to no liquid hydrocarbons, such as oil or water.
- **Wet Gas:** Wet gas is natural gas that contains a higher amount of heavier hydrocarbons, such as propane, butane, and natural gasoline, along with methane. These liquids can be separated and processed for use in other products, like fuel or chemicals.

2.6.13 Nuclear Energy

Nuclear energy is the energy released from the nucleus (center) of atoms through processes like **nuclear fission** (splitting atoms) or **nuclear fusion** (combining atoms). In nuclear fission, when the nucleus of an atom, like uranium, splits, it releases a large amount of energy. This energy can be used to produce electricity in nuclear power plants.

Nuclear Fusion: Nuclear fusion is the process where two small atomic nuclei combine to release energy, like how hydrogen atoms fuse to form helium in the Sun. This reaction provides the energy that powers stars and could potentially be used for clean energy on Earth in the future.

$$_1H^2 +_1H^2 \rightarrow 2He^3 +_0n^1 + \text{energy}$$

Nuclear Fission: Nuclear fission is the process where a heavy atomic nucleus, like uranium, splits into smaller nuclei, releasing energy. An example is the reaction in nuclear power plants, where fission of uranium atoms produces heat to generate electricity.

When U235nucleus is hit by a thermal neutron, it undergoes the following reaction with the release of 3 neutrons.

$$_{92}U^{235} + n_0^1 \rightarrow {}_{56}Ba^{139} + {}_{36}Kr^{94} + n_0^1 + \text{Energy}$$

- Each of the three neutrons produced in the fission reaction strikes another U-235 nucleus, leading to (3x3) 9 additional reactions.
- These 9 reactions then trigger (3x9) 27 more reactions.
- This continuous multiplication of reactions at each fission step is known as a chain reaction.

Fission reaction of U^{235} is given below.

$$_{92}U^{235} + n_0^1 \rightarrow {}_{56}Ba^{139} + {}_{36}Kr^{94} + 3\ n_0^1 + \text{Energy}$$

Nuclear Power Plants in India

Some of the major nuclear power plants in India include:

- **Tarapur Atomic Power Station (TAPS):** Located in Maharashtra, it was the first nuclear power plant in India, consisting of two reactors.

- **Kakrapar Atomic Power Station (KAPS):** Located in Gujarat, it has multiple reactors.
- **Kalpakkam Nuclear Power Plant:** Located in Tamil Nadu, it operates reactors for electricity generation and research.
- **Rajasthan Atomic Power Station (RAPS):** Located in Rajasthan, this plant consists of several reactors.
- **Kudankulam Nuclear Power Plant:** Located in Tamil Nadu, it is one of the largest nuclear power plants in India, with reactors built in collaboration with Russia.

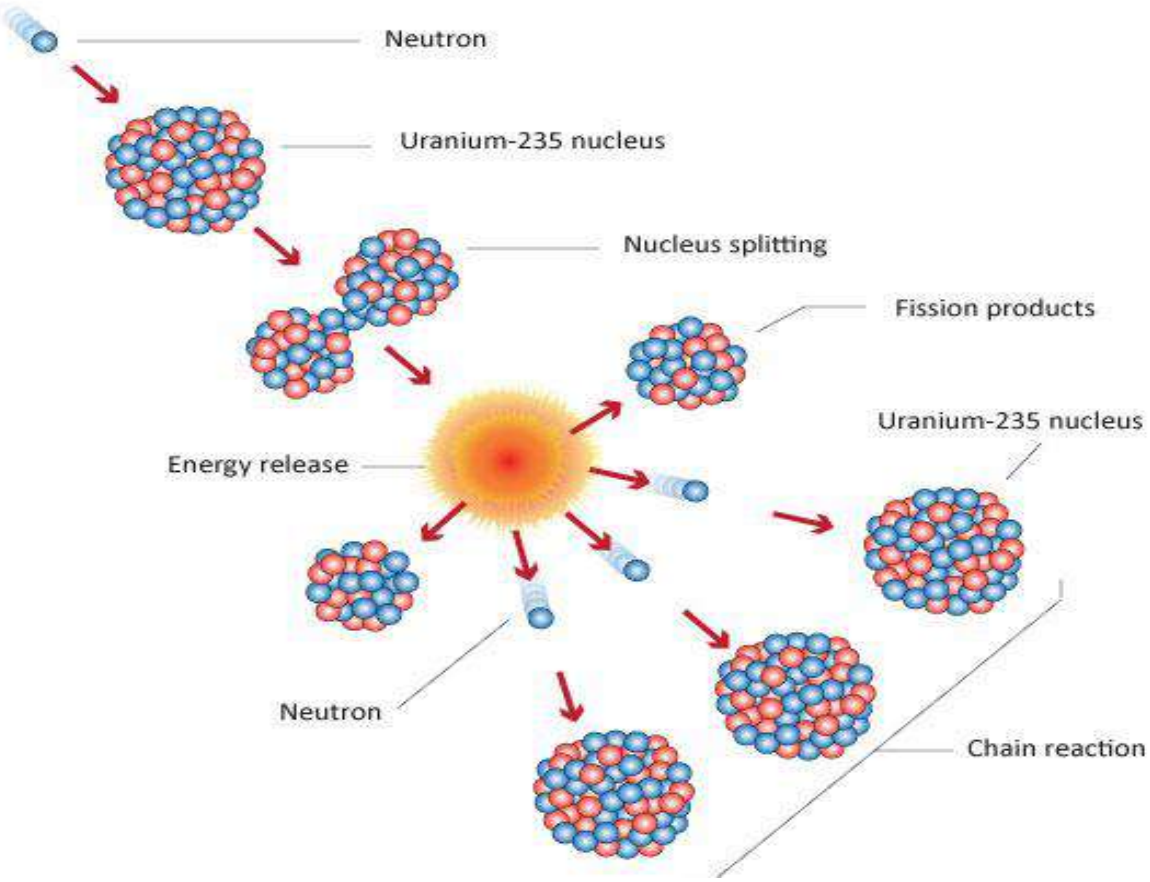


Figure 2.10: Nuclear Fission Reactions
(Image source: Pass My Exams)

2.7 Uses of Alternate (Renewable) Energy Sources

Alternate (renewable) energy sources have many uses, including:

1. **Electricity Generation:** Solar, wind, and hydroelectric power are commonly used to generate electricity for homes, businesses, and industries.
2. **Heating:** Solar energy can be used to heat water in homes and buildings through solar water heaters.
3. **Transportation:** Biofuels (like ethanol and biodiesel) and electricity from renewable sources power vehicles, reducing reliance on fossil fuels.
4. **Cooking:** Biogas is used for cooking in rural areas, replacing traditional fuels like wood or coal.
5. **Industrial Processes:** Renewable energy sources are used in various industries for processes like drying, heating, and powering machinery.
6. **Water Desalination:** Solar and wind energy can be used to power desalination plants to provide fresh water in areas with limited water resources.

These energy sources help reduce greenhouse gas emissions, improve air quality, and create sustainable, long-term solutions for energy needs.

Objectives

- To generate more energy to meet the growing needs of the increasing population.
- To decrease environmental pollution
- To minimize the safety and security risks linked to the use of nuclear energy.

2.8 Land Resources

Land resources refer to the natural resources found on the Earth's surface, including soil, water, forests, and minerals, which are essential for various human activities. These resources support agriculture, urban development, transportation, and the extraction of raw materials.

Land resources have several important uses, including:

1. **Agriculture:** Land is used for growing crops, raising livestock, and producing food, which is essential for human survival.
2. **Forestry:** Forests provide timber, fuelwood, and other forest products, as well as play a role in maintaining biodiversity and regulating the climate.
3. **Urban Development:** Land is used for building cities, towns, and infrastructure like roads, schools, and hospitals to support human populations.
4. **Mineral Extraction:** Land is used to mine valuable minerals, coal, and other raw materials used in various industries.
5. **Water Resources:** Land supports natural water systems like rivers, lakes, and groundwater, which are essential for drinking water, irrigation, and industry.
6. **Recreation and Tourism:** Parks, forests, and scenic areas are used for leisure activities like hiking, camping, and tourism, which contribute to the economy.
7. **Conservation:** Some land is set aside for conservation purposes to protect endangered species and maintain ecological balance.

2.8.1 Land Degradation

The process of soil degradation refers to the decline in soil quality or the loss of soil fertility.

Harmful Effects of Land Degradation

The most important harmful effects of land degradation include:

- **Reduced Agricultural Productivity:** Land degradation leads to a loss of soil fertility, making it harder to grow crops and resulting in lower food production.
- **Soil Erosion:** The removal of topsoil by wind or water reduces the soil's ability to retain moisture and nutrients, which further degrades the land.
- **Desertification:** Overuse of land, deforestation, and poor farming practices can transform fertile land into deserts, making it unproductive and unsuitable for agriculture.
- **Loss of Biodiversity:** Degraded land can lead to the destruction of habitats, causing a decline in plant and animal species.
- **Water Scarcity:** Soil degradation can reduce the land's ability to store water, leading to increased water runoff, lower groundwater levels, and water shortages.
- **Increased Flooding:** Degraded land is less able to absorb rainfall, leading to more surface runoff, which can cause flooding.

Soil Erosion

Soil erosion is the process by which the top layer of soil is worn away by natural forces like wind, water, and ice, or by human activities like farming, deforestation, and construction.

Types of Soil Erosion

1. Normal Erosion

Normal erosion refers to the natural process through which the Earth's surface is gradually worn away by forces like wind, water, and ice over time.

2. Accelerated Erosion

Accelerated erosion occurs when the natural rate of soil erosion is significantly increased due to human activities, such as deforestation, overgrazing, construction, and improper agricultural practices.

Causes of Soil Erosion

1. Water

- Affects soil erosion in the form of rain, run-off, rapid flow, wave action.
- **Sheet Erosion:** Sheet erosion is a type of soil erosion where thin layers of soil are removed evenly from large areas of land by the action of wind or water.
- **Rill Erosion:** Rill erosion is a type of soil erosion where small channels, or "rills," form on the surface of the soil as water flows over it.
- **Gully Erosion:** Gully erosion is a severe form of soil erosion where large channels, or "gullies," are formed in the landscape due to the concentrated flow of water.
- **Slip Erosion:** Slip erosion occurs when a section of soil, often along a slope, moves downhill due to gravity. This movement is typically caused by water saturation, which weakens the soil, making it unstable.
- **Stream Bank Erosion:** Stream bank erosion is the process where the banks of a stream or river are worn away by the movement of water.

2. Wind

- Loss of Topsoil
- Reduced Agricultural Productivity
- Desertification
- Airborne Dust
- Sedimentation in Water Bodies
- Damage to Vegetation

Control of soil Erosion (or) Soil Conservation Practices

1. Conservational Till Farming (or) No-Till-Farming

- **It** is an agricultural practice where the soil is left undisturbed and crops are planted without turning the soil over.
- In this method, farmers avoid plowing or tilling the land, which helps reduce soil erosion, improve soil structure, and retain moisture.
- By not disturbing the soil, the soil's natural layers and organic matter are preserved, which enhances its fertility.
- This practice also helps reduce fuel and labor costs.
- No-till farming is especially beneficial for preventing soil erosion and promoting sustainability in farming.

2. Contour Farming

- **Contour farming** is a method where crops are planted along the contours of a slope, rather than up and down. This helps reduce soil erosion by slowing down water flow and allowing it to soak into the soil.

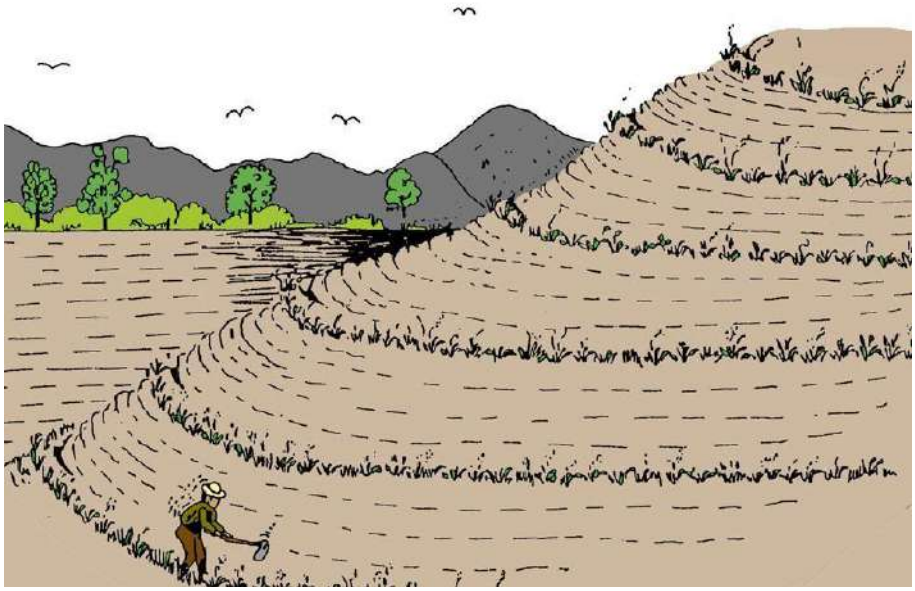


Figure 2.11: Contour Farming

3. Terracing

- It involves transforming steep slopes into wide terraces that run along the contours. This method helps retain water for crops and reduces soil erosion by controlling water runoff.



Figure 2.12: Terracing

4. Alley Cropping (or) Agro Forestry

- It involves planting crops in strips or alleys between rows of trees or shrubs that provide fruits and fuelwood.
- Even after harvesting the crops, soil erosion is prevented as the trees and shrubs remain, helping to anchor the soil particles.

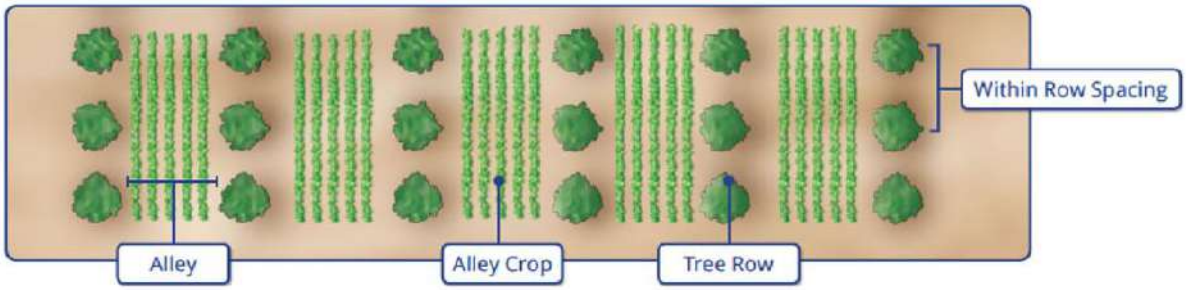


Figure 2.13: Alley cropping

5. **Wind Breaks or Shelterbelts**
    - Trees are planted in long rows along the edges of cultivated land to block the wind and reduce soil erosion.
    - Windbreaks also help retain soil moisture, provide wood for fuel, and offer habitats for birds.

2.8.2 Landslides

Landslides are the sudden movement of a large amount of soil, rocks, and debris down a slope or hill. They usually happen after heavy rainfall, earthquakes, or when the land becomes unstable, causing the earth to slide downhill.

Man Induced Landslides

Man-induced landslides are landslides caused by human activities, such as deforestation, construction, mining, and improper land use. These activities can make the land unstable, increasing the risk of soil, rocks, and debris sliding down slopes.

Causes of Landslides

- Heavy Rainfall
- Earthquakes
- Volcanic Activity
- Deforestation
- Mining and Construction
- Soil Erosion
- Weak or Fractured Rocks
- Rapid Snowmelt
- Overloading of Slopes

Harmful Effect of Landslides

- Loss of Life
- Damage to Property

- Soil Erosion
- Displacement of Communities
- Environmental Damage
- Transport Disruption
- Increased Risk of Flooding
- Economic Loss

2.8.3 Desertification

- **Desertification** is the process by which fertile land becomes increasingly arid and unproductive, often transforming into desert-like conditions.
- This occurs due to a combination of natural factors, such as drought, and human activities, like deforestation, overgrazing, and poor agricultural practices.

Harmful Effect of Desertification

- **Loss of Agricultural Productivity:** Soil becomes less fertile, making it difficult to grow crops and sustain livestock, leading to food insecurity.
- **Water Scarcity:** Desertification reduces the availability of water sources, worsening drought conditions and limiting access to clean water.
- **Loss of Biodiversity:** As the land becomes barren, plant and animal species that depend on fertile land for survival are threatened or displaced.
- **Soil Erosion:** Desertification leads to the removal of topsoil, making the land more vulnerable to erosion by wind and water, further degrading the environment.
- **Economic Loss:** Communities dependent on agriculture and livestock suffer financial losses due to decreased productivity, which can lead to poverty and migration.

- **Increased Flooding:** When desertification occurs, the land's ability to absorb water diminishes, leading to increased surface runoff, which can cause flooding in nearby areas.
- **Health Problems:** Dust storms and poor air quality from desertification can lead to respiratory issues and other health problems for local populations.
- **Social Displacement:** As land becomes uninhabitable, communities may be forced to migrate, causing social and economic challenges in new areas.

Causes of Desertification

1. **Deforestation**
   - Deforestation is the large-scale removal or destruction of forests, often due to human activities such as logging, agriculture, urban expansion, and mining.
   - Deforestation leads to soil erosion, loss of habitat for wildlife, and contributes to climate change by reducing the amount of carbon dioxide absorbed by trees.

2. **Overgrazing:** The growing cattle population overgrazes grasslands or forests, resulting in the depletion of the land. This degraded land becomes dry, loose, and highly vulnerable to soil erosion, eventually leading to desertification.

3. **Water Management:** Overuse of groundwater, especially in coastal areas, is causing saline water to intrude into aquifers, making the water unsuitable for irrigation.

4. **Mining and Quarrying:** These activities also contribute to the loss of vegetation and the degradation of large land areas, resulting in desertification.

5. **Climate Change:** Desert formation can also occur due to climate change, such as the failure of monsoons and frequent droughts.

6. **Pollution:** Excessive use of fertilizers and pesticides, along with the disposal of toxic water onto the land, also contributes to desertification.

2.9 Conservation of Natural Resources - Role of an Individual

Various natural resources such as forests, water, soil, food, minerals, and energy are crucial for a nation's development. Although conservation efforts are being made at both national and international levels, individual contributions to preserving natural resources can make a significant impact.

1. Conserve Water

Conserving water involves using water wisely and reducing waste to ensure its availability for future generations. Simple ways to conserve water include:

- **Fixing Leaks:** Repairing leaking taps, pipes, and toilets to prevent water wastage.
- **Shortening Showers:** Taking shorter showers and turning off the tap while brushing teeth or washing hands.
- **Water-Efficient Appliances:** Using water-saving appliances like low-flow showerheads, faucets, and dishwashers.
- **Collecting Rainwater:** Using rainwater harvesting systems to collect and store rainwater for irrigation or other non-drinking uses.
- **Watering Plants Wisely:** Watering plants early in the morning or late in the evening to minimize evaporation.
- **Reusing Water:** Reusing water from activities like washing vegetables for gardening or cleaning.

2. Conserve Energy

Conserving energy involves using less energy by being more efficient and reducing unnecessary consumption. Simple ways to conserve energy include:

- **Turn off Lights and Appliances:** Always switch off lights, fans, and electrical appliances when not in use.
- **Use Energy-Efficient Appliances:** Choose appliances with high energy efficiency ratings, such as LED bulbs, energy-efficient refrigerators, and washing machines.
- **Unplug Devices:** Unplug chargers, electronics, and appliances when they are not in use to avoid "phantom" energy consumption.
- **Use Natural Light:** Make the most of natural sunlight during the day, and minimize artificial lighting when possible.
- **Optimize Heating and Cooling:** Set thermostats to a comfortable, but not excessive temperature, and use fans or natural ventilation to cool spaces instead of air conditioning.
- **Carpool or Use Public Transport:** Reduce fuel consumption by sharing rides or using energy-efficient public transportation options.

3. Protect the Soil

Protecting the soil is essential for maintaining healthy ecosystems, agriculture, and water systems. Simple ways to protect the soil include:

- **Avoid Overgrazing:** Limit the number of animals grazing on land to prevent soil compaction and erosion.
- **Practice Crop Rotation:** Rotating crops helps maintain soil fertility and reduces soil degradation.
- **Use Mulching:** Applying a layer of mulch around plants helps conserve moisture, prevent erosion, and add nutrients to the soil.
- **Reduce Tilling:** Minimizing tilling prevents the disruption of soil structure and helps retain organic matter.
- **Plant Vegetation:** Planting trees, grass, and cover crops helps anchor the soil, preventing erosion and promoting soil health.

- **Control Water Runoff:** Installing terraces, dams, or trenches can slow down water runoff, preventing soil erosion and water loss.
- **Avoid Pollutants:** Reducing the use of harmful chemicals like pesticides and fertilizers helps maintain soil health.

2.10 Equitable use of Resources for Sustainable Life Style

- **Equitable use of resources for a sustainable lifestyle** means ensuring that natural resources are used fairly and responsibly, so they are available for future generations while meeting the needs of all people today. This involves:
- **Fair Distribution:** Ensuring that everyone, regardless of income or location, has access to essential resources like clean water, energy, and food.
- **Efficient Resource Use:** Using resources wisely and reducing waste by adopting sustainable practices such as recycling, reusing, and reducing consumption.
- **Supporting Renewable Resources:** Prioritizing renewable resources like solar, wind, and water, which can be replenished naturally, over finite resources like fossil fuels.
- **Conservation:** Protecting ecosystems, biodiversity, and natural habitats, while managing land, water, and energy resources to prevent depletion and environmental damage.
- **Promoting Equity:** Ensuring that vulnerable and marginalized communities have access to the resources they need to thrive while reducing inequalities in resource distribution.

Questions

1. Examine, through a case study, the impact of large dams on forests and tribal communities, focusing on the environmental and social consequences.
2. Provide a concise overview of non-renewable energy sources.

3. Describe the ways in which an individual can contribute to the conservation of natural resources.
4. Write about world food problems?
5. Explain the major environmental impacts of mineral extraction.
6. Provide a brief discussion on the occurrence and impacts of droughts and floods.
7. Discuss the merits and demerits of wind energy.
8. What is nuclear energy? Discuss its two types?
9. Discuss the different types of energy that can be harnessed from the oceans.
10. Discuss the primary causes and effects of deforestation.
11. Discuss briefly about renewable energy resources.
12. How to use resources to develop sustainable life style?
13. How to prevent land degradation?
14. Discuss the biogas preparation.

Unit-3

Ecosystem

An ecosystem is a dynamic complex of living organisms (plants, animals, and microorganisms) and their non-living environment (such as air, water, soil, and climate) interacting as a functional unit. These interactions result in a flow of energy and cycling of nutrients that sustain life. Ecosystems can be as small as a puddle or as large as an entire forest or ocean.

Ecosystem is the study of an interaction between living organisms and surroundings in one unit area.

3.1 Ecosystem Structure

The Structure of an ecosystem consists of:

1. **Biotic structure** means living organisms present in an Ecosystem.
   Eg: Flora, Fauna, Different tiny organisms.
   - **Producer (Autotrophs):** They can prepare their own food material through the photosynthesis process.
     Eg: Flora, Algae
   - **Consumer (Heterotrophs):** Organisms which can eat other organisms. Consumers are Classified into
     - **Herbivores:** Plant Eaters. They can eat only plants (Producers).
       Eg: Rabbits, Deer's, Insects.
     - **Carnivores:** Organisms which can eat only flesh.
       Eg: Lion, Tiger
     - **Omnivores:** Organisms which can eat both plants & animals.
       Eg: Human

- **Detritivores (saprotrophs):** Organisms which can eat dead parts of organisms.
  Eg: Beetles, Termites, Earthworms, Crabs, Ants, etc.

- **Decomposer: These** are organisms like bacteria and fungi that break down dead plants, animals, and waste materials into simpler substances. This process returns nutrients to the soil, making them available for plants to use and helping keep the ecosystem clean.

  Examples: Bacteria and fungi

2. **Abiotic structure** means non-living organisms present in an Ecosystem.
   Abiotic components classified into two types
   - Physical Factors Eg: Sunlight, Temperature, Rainfall…
   - Chemical Factors Eg: Carbohydrates, Proteins.

3.1.1 Functions of Eco-system

- Energy Flow
- Nutrient Cycling
- Oxygen Production
- Waste Breakdown
- Climate Regulation
- Habitat Provision
- Pollination
- Water Purification
- Soil Formation
- Support for Biodiversity

3.1.2 Food Chain

- A food chain is a simple way to show how energy and nutrients flow through an ecosystem.
- It starts with plants (producers) that make their own food, followed by animals (consumers) that eat plants or other

animals, and ends with decomposers that break down dead matter.
- Each step is called a trophic level.

For instances

1. Grass → Grasshopper → Frog → Snake →Eagle (grassland ecosystem)
2. Phyto-planktons→Zooplanktons→ Small Fish → Large Fish (pond ecosystem)
3. Plant leaf →Caterpillar →Sparrow →Hawk.

A grasshopper eats a Grass, a Frog eats the Grasshopper, a snake eats the Frog and an Eagle eats the Snake. When they all die, they are all consumed by microorganisms like bacteria (or) fungi which break down the organic matter and convert it into simple inorganic substances that can again be used by the plants.

3**.1.3 Food Web**

- A food web is a network of interconnected food chains in an ecosystem.
- It shows how different plants and animals are linked through various feeding relationships, with multiple paths for energy and nutrients to flow.
- Unlike a simple food chain, a food web highlights the complexity of these interactions.

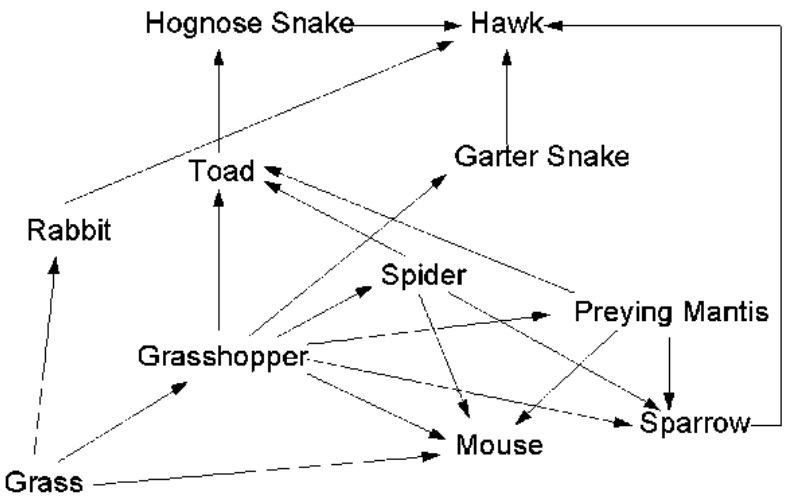


Figure 3.1: Food Web

Significance of Food Chain and Food Web

- **Energy Flow:** It shows how energy moves from one organism to another in a straight line.
- **Nutrient Cycling:** It helps explain how nutrients are passed between plants, animals, and decomposers.
- **Ecosystem Balance:** It maintains the balance by linking producers, consumers, and decomposers.
- **Survival Understanding:** It helps us see how every organism depends on others for food.

Bio Magnification / Biological Magnification

Numerous chemicals, including heavy metals and pesticides, are not naturally biodegradable. These substances continue to move from one tropic level to another and are not broken down by microorganisms. And their concentration continues to rise at each subsequent tropic level. Biomagnification is the term for this phenomenon.

3.1.4 Ecological Pyramids

Ecological pyramids are simple diagrams that show the relationship between different levels in a food chain or food web. They represent

the flow of energy, the number of organisms, or the biomass (total weight) at each level:

- **Pyramid of Energy:** Shows how energy decreases as it moves up the food chain.
- **Pyramid of Numbers:** Shows the number of organisms at each level.
- **Pyramid of Biomass:** Shows the total weight of living things at each level.

Pyramid of Numbers

The pyramid of numbers illustrates the relative number of individual organisms at each trophic level. It represents the relationship between producers, herbivores, and carnivores across successive trophic levels based on their population size.

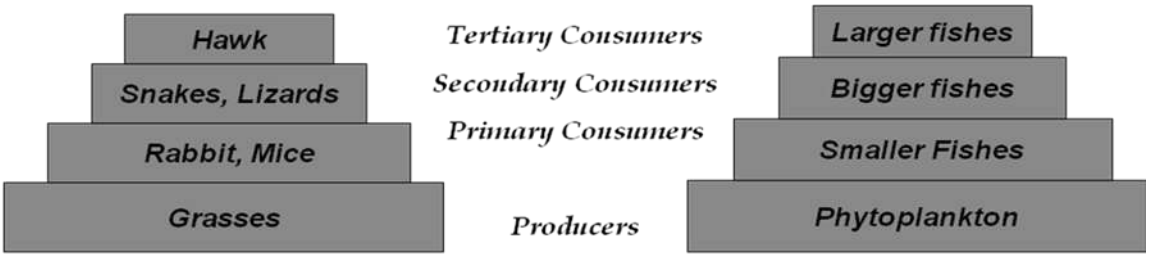


Figure 3.2: Grass and Pond Ecosystems

- In a grassland ecosystem the producers, which are mainly grasses, are always maximum in number.
- The primary consumers number is obviously less compared to producers.
- The secondary consumers, are less in number than the Primary Consumer number.
- In the top (tertiary) consumers are least in number. In this way the pyramid becomes upright.

In a pond ecosystem, also the pyramid is upright as under:

- The producers, primarily phytoplankton such as algae and bacteria, are the most numerous.
- The herbivores, including smaller fish and rotifers, are fewer in number compared to the producers.
- The secondary consumers (carnivores), such as small fish that prey on each other and water beetles, are fewer in number than the herbivores.
- Finally, the top (tertiary) consumers, the bigger fish are least in number.

3.1.5 Pyramid of Biomass

A **pyramid of biomass** is a diagram that shows the total mass of living organisms at each trophic level in an ecosystem. It represents the amount of biological material (biomass) available to each level, with producers at the base and higher trophic levels (herbivores, carnivores) above.

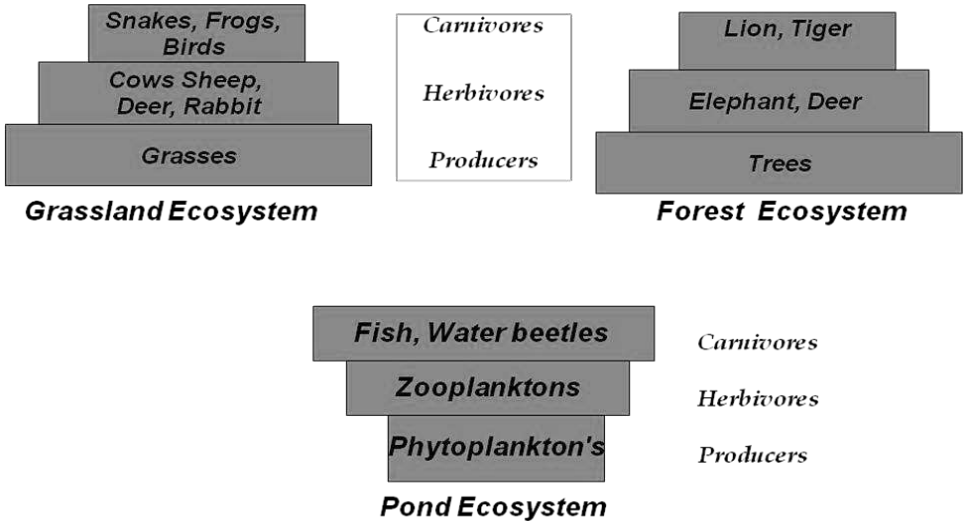


Figure 3.4: Pyramid of Biomass
(Grassland, Forest, Pond Ecosystem)

3.1.6 Pyramid of Energy

A pyramid of energy shows the flow of energy through each trophic level in an ecosystem. It represents the amount of energy available at each level, with producers at the base and consumers at higher levels.

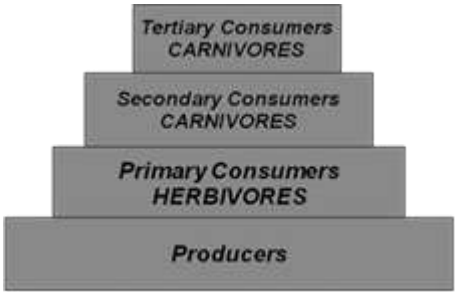


Figure 3.5: Pyramid of Energy

3.1.7 Energy Flow in an Ecosystem

Energy flow in an ecosystem refers to how energy moves through the various components of an ecosystem, from producers to consumers and eventually to decomposers. Here's a simple breakdown:

- **Solar Energy:** The primary source of energy for most ecosystems is sunlight. Producers, like plants and algae, capture this solar energy through photosynthesis to create food.
- **Producers:** Plants and algae (producers) convert solar energy into chemical energy, stored in the form of glucose. These organisms form the base of the food chain.
- **Primary Consumers:** Herbivores (like insects, small fish, or rabbits) eat the producers and gain the energy stored in them.
- **Secondary Consumers:** Carnivores (like small fish, frogs, or birds) feed on herbivores, transferring the energy they gained from the plants to higher trophic levels.
- **Tertiary Consumers:** These are top predators (like large carnivores or apex predators) that eat secondary consumers. At

each step, energy is transferred but also lost as heat (due to metabolism) and waste.

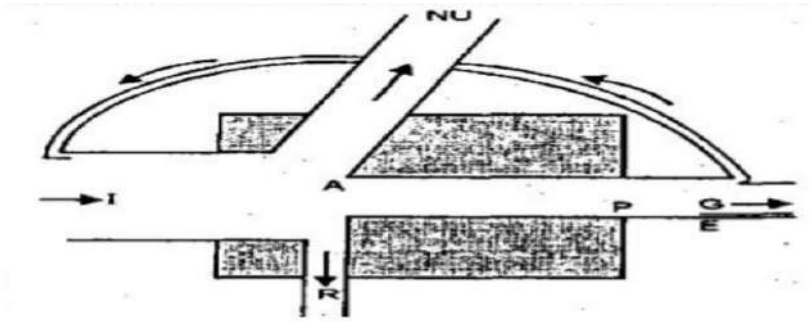
- **Decomposers:** Organisms like bacteria and fungi break down dead organisms, releasing energy and recycling nutrients back into the ecosystem.

Energy Transfer Efficiency

- Typically, only about 10% of the energy from one trophic level is passed on to the next, while the rest is lost as heat or used for metabolic activities.

Energy Flow Model: The flow of energy through various trophic level is given by energy flow models. Some of them are as follows:

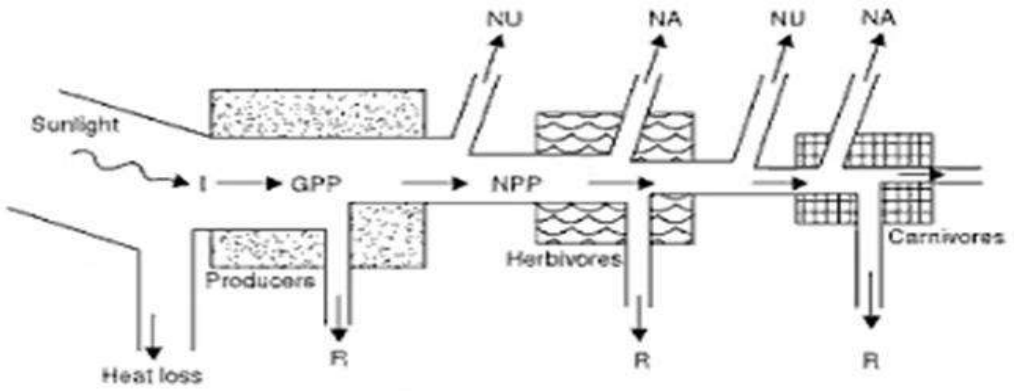
1. **Universal Energy Flow Model:** It is given by E. P. Odum. The **universal energy flow** model describes how energy moves through an ecosystem. Solar energy is captured by producers (plants), then transferred to consumers (herbivores and carnivores), and finally to decomposers. At each step, energy is lost as heat, with only about 10% passed to the next level.



I = Energy Input; A = Assimilated Energy; P = Production;
NU = Energy not used
Figure 3.6: Universal energy flow model.

2. **Single Channel Energy Flow Model:** The single channel energy flow model shows how energy flows in a straight line from producers to consumers and then to decomposers, with minimal energy loss at each step.

Example: single channel energy flow model for grazing food chain is given below.



I = solar energy input; GPP = Gross primary production; NPP = Net primary production; NU = Energy not used; NA = Energy not assimilated (Excretion); R = Respiratory loss

Figure 3.7: Unidirectional Energy Flow Model

3. **Double Channel or Y- Shaped Energy Flow Model: The Double Channel (Y-shaped) Energy Flow Model** shows two separate energy pathways: one where energy flows from producers to herbivores and another from producers to detritivores (organisms that feed on dead matter), with both paths eventually leading to decomposers.

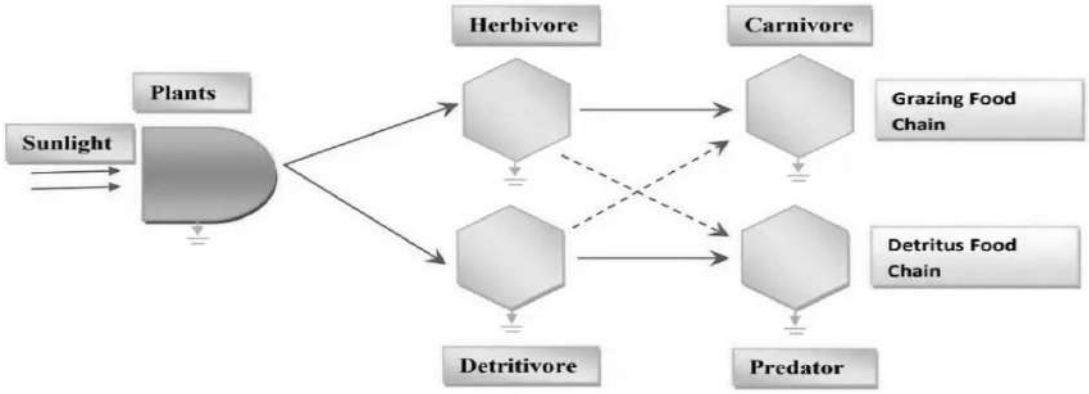


Figure 3.8: Double Channel Energy Flow Model

3.1.8 Ecological Succession

Ecological succession is the process by which an ecosystem changes and develops over time. It starts with simple organisms (like moss or small plants) and gradually evolves into a more complex community with larger plants and animals. There are two types:

1. **Primary Succession:** Happens in a place where no life existed before, like after a volcanic eruption or glacier retreat.

2. **Secondary Succession:** Occurs in an area where life once existed but was disturbed, like after a forest fire or farming.

There are two types of ecological succession:

1. **Primary Succession**
   - **Primary succession** is the process where life begins in a completely new or barren area, such as after a volcanic eruption, glacier retreat, or the creation of a new island.
   - It starts with simple organisms like lichens and mosses, which gradually break down the environment to make it suitable for plants, and over time, more complex life forms like trees and animals begin to appear.
   - Ends with a "climax community" – ecosystem stays constant, provided there are no changes in abiotic influences.

2. **Secondary Succession**
   - **Secondary succession** occurs in an area where life previously existed but was disturbed or destroyed, like after a forest fire, flood, or farming.
   - The soil is still present, so life can return more quickly.
   - It starts with small plants and grasses, and over time, it gradually progresses to more complex communities, eventually leading to a stable ecosystem.

Types of Succession on the Basis of the Moisture Contents

1. **Hydrach or Hydro Sere:** Hydrach or hydrosere refers to the type of ecological succession that begins in aquatic environments such as ponds, lakes, streams, swamps, and bogs.
2. **Mesarch:** Mesarch refers to the type of succession that occurs in areas with sufficient moisture.
3. **Xerach or Xerosere:** Xerach refers to succession that starts in dry, xeric environments with limited moisture, such as rocks or deserts. Xerosere is the term for a temporary community that develops during ecological succession in a barren, dry habitat. It can occur in one of the following three types:
   - **Lithosere:** Succession that begins on bare rock.
   - **Psammosere:** Succession that begins on sand.
   - **Halosere:** Succession that begins in saline water or soil.

Process of Ecological Succession

The process of ecological succession occurs in stages, where an ecosystem gradually changes over time. Here are the typical stages:

1. **Pioneer Stage:** The first organisms to colonize an area are called pioneers, such as lichens and mosses. These organisms can survive in harsh, bare conditions and start breaking down the environment to create soil.

2. **Establishment Stage:** Simple plants like grasses and small herbs begin to grow as soil develops. These plants help further enrich the soil, allowing more complex plants to establish themselves.

3. **Intermediate Stage:** As the environment becomes more stable, shrubs, bushes, and small trees start to grow. More species of plants and animals establish themselves, increasing biodiversity.

4. **Climax Stage:** The final stage of succession, where the ecosystem reaches a stable, mature state. Large trees and complex organisms

dominate, creating a balanced and diverse ecosystem that remains relatively unchanged unless disturbed.

The entire process can take years, decades, or even centuries, depending on the environment and conditions.

Example of Ecological Succession

1. **Hydrosere (Hydrarch):** This type of succession begins in aquatic environments, such as ponds, and eventually leads to a climax community, typically a forest. It follows these stages

2. **Phytoplankton Stage:** The pioneer community consists mainly of organisms like blue-green algae, green algae, diatoms, and bacteria.
   - **Rooted Submerged Stage:** As phytoplankton die and decompose, organic matter begins to accumulate on the pond's surface. This creates a new environment that supports the growth of rooted, submerged hydrophytes such as Hydrilla and Elodea.

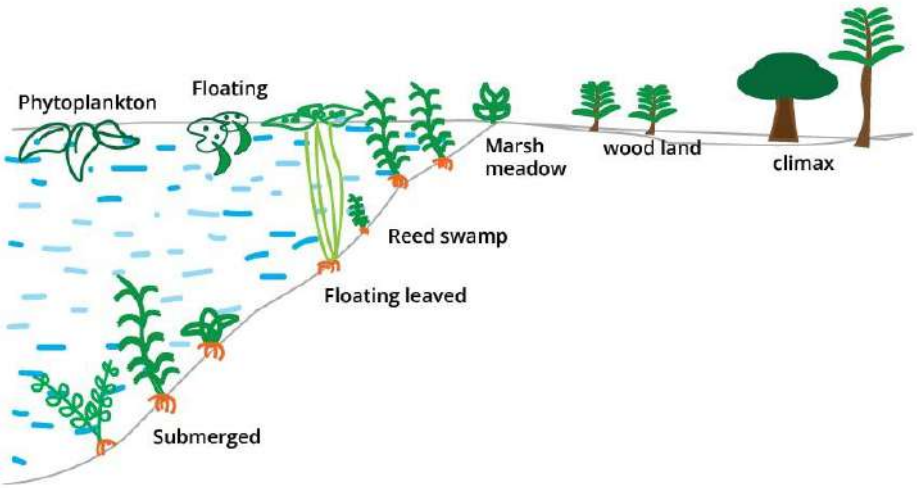


Figure 3.9: Ecological Succession (Image source: Vedantu.com)

3. **Rooted Submerged Stage:** As the lake's depth decreases to 2-5 feet, it creates favorable conditions for rooted hydrophytes with

large leaves that float on the water's surface. Examples include Nelumbo, Trapa, and Azolla.

4. **Reed-Swamp Stage:** This stage is also known as the amphibious stage, as the plants are rooted but most of their parts are above the water. Examples include Scirpus and Sagittaria.

5. **Sedge-Meadow Stage:** They form mat-like vegetation, leading to increased water loss through the process of evapo-transpiration. The marshy conditions of the previous stage are replaced by areas with remaining soil moisture. Examples include Carex, Cyperus, and others.

6. **Woodland Stage**: As the marshland disappears, the soil becomes drier, leading to the growth of vegetation such as shrubs (Salix, Cornus) and trees (Populus, Alnus).

7. **Forest Stage:** This is the climax community, which can develop into either a tropical rainforest or a mixed forest, depending on the region's climate. Examples include Ulmus, Acer, and Quercus.

3.2 Forest Ecosystem (Terrestrial Ecosystem)

A **forest ecosystem** is a complex community of plants, animals, and microorganisms interacting with each other and their environment in a forested area. Forests play a crucial role in maintaining ecological balance by supporting biodiversity, regulating the climate, and cycling nutrients.

Key Components of a Forest Ecosystem:

- **Producers (Plants):** Trees, shrubs, and grasses that use sunlight to produce food through photosynthesis. Examples include oak, pine, and fern.
- **Consumers (Animals)**

  Herbivores: Animals like deer and rabbits that feed on plants.

Carnivores: Animals like wolves and foxes that eat other animals.

Omnivores: Animals like bears that eat both plants and animals.

- **Decomposers:** Organisms like fungi, bacteria, and insects that break down dead plants and animals, returning nutrients to the soil.

Abiotic Factors: Non-living elements such as sunlight, temperature, rainfall, soil, and air, which influence the types of species that can thrive in a forest.

Types of Forest Ecosystems

- **Tropical Rainforest:** Found near the equator, these forests are characterized by high rainfall and biodiversity. Examples include the Amazon Rainforest.
- **Temperate Forest:** Found in regions with moderate climates, these forests have a mix of deciduous and coniferous trees. Examples include the forests in North America and Europe.
- **Boreal Forest (Taiga):** Found in cold, northern regions, dominated by coniferous trees like pines and firs.

Importance of Forest Ecosystems

- **Biodiversity:** Forests are home to a wide variety of species, supporting a rich ecosystem.
- **Climate Regulation:** Forests absorb carbon dioxide and release oxygen, helping to regulate the Earth's climate.
- **Water Cycle:** Forests help maintain the water cycle by absorbing and releasing water through transpiration.
- **Soil Protection:** The roots of forest plants prevent soil erosion and maintain soil fertility.

3.3 Grassland Ecosystem (Terrestrial Ecosystem)

A **grassland ecosystem** is a type of terrestrial ecosystem dominated by grasses and herbaceous plants, with few trees or shrubs. These ecosystems can be found in regions with moderate rainfall, often between forests and deserts. Grasslands play a crucial role in supporting a variety of herbivores, predators, and plant species, and are vital for agricultural production.

Key Components of a Grassland Ecosystem

- **Producers (Plants):** Grasses and other herbaceous plants are the primary producers, harnessing sunlight through photosynthesis to create energy. Common examples include buffalo grass, wheatgrass, and ryegrass.
- **Consumers (Animals)**

  Primary Consumers (Herbivores): Animals like zebras, bison, antelope, and rabbits feed on grasses.

  Secondary Consumers (Carnivores): Predators such as lions, wolves, and hawks feed on herbivores.
- **Omnivores:** Animals like raccoons and certain birds that eat both plants and animals.
- **Decomposers**

  Fungi, bacteria, and insects decompose dead plants and animals, returning nutrients to the soil, which helps maintain the productivity of the ecosystem.

Abiotic Factors

Non-living factors such as sunlight, temperature, soil type, and rainfall influence the types of plants and animals that can thrive in the grassland ecosystem. Grasslands typically experience seasonal droughts and periodic fires, which can shape the ecosystem.

Types of Grassland Ecosystems

- **Tropical Grasslands (Savannas):** Found in regions with distinct wet and dry seasons, these grasslands have scattered trees like acacias and baobabs. Examples include the African savannas and parts of Australia.
- **Temperate Grasslands (Prairies, Steppes):** Found in areas with hot summers and cold winters, these grasslands have a variety of grasses and few trees. Examples include the prairies of North America, the steppes of Eurasia, and the pampas of South America.
- **Cold Grasslands (Tundra):** Located in colder regions, these grasslands are often home to hardy grasses and mosses. Examples include the tundra regions of the Arctic and high mountain areas.

Importance of Grassland Ecosystems

- **Biodiversity:** Grasslands support a wide variety of herbivores, carnivores, and other species, making them ecologically rich.
- **Soil Formation:** The deep roots of grasses help prevent soil erosion, maintain soil fertility, and contribute to the formation of rich soils.
- **Carbon Sequestration:** Grasslands absorb and store carbon, helping mitigate climate change.
- **Human Benefits:** Grasslands provide resources for agriculture, grazing, and hunting. They are also home to numerous species used in food, medicine, and other industries.

3.4 Desert Ecosystem

A desert ecosystem is a type of terrestrial ecosystem characterized by low rainfall, extreme temperatures, and limited water availability. Deserts can be hot (like the Sahara) or cold (like the Gobi), but they

all share a harsh environment where only specially adapted plants and animals can survive.

Key Components of a Desert Ecosystem

- **Producers (Plants):** Desert plants are adapted to conserve water and survive in extreme conditions. These include cacti, succulents, and drought-resistant shrubs. Examples are the saguaro cactus and the creosote bush.
- **Consumers (Animals)**

  Primary Consumers (Herbivores): These are animals that feed on plants, such as camels, kangaroo rats, and desert tortoises.
- **Secondary Consumers (Carnivores):** Predators like snakes, foxes, and lizards feed on herbivores or smaller carnivores.
- **Omnivores:** Some animals, like ravens and certain rodents, eat both plants and animals.
- **Decomposers:** Microorganisms, fungi, and certain insects decompose dead organic matter, returning nutrients to the soil. In desert ecosystems, decomposition is slower due to dry conditions.

Abiotic Factors

The desert is characterized by extreme temperature variations (hot during the day, cold at night), very little rainfall, and poor soil quality. Sand, rocks, and occasional water sources like oases are typical features of desert landscapes.

Types of Desert Ecosystems

- **Hot Deserts:** These deserts experience high daytime temperatures and very little rainfall, often with dramatic temperature drops at night. Examples include the Sahara Desert, the Arabian Desert, and the Sonoran Desert.

- **Cold Deserts:** These deserts have cold winters and cooler temperatures overall. They receive low rainfall but can have snow in winter. Examples include the Gobi Desert in Asia and parts of the Great Basin Desert in the United States.

Importance of Desert Ecosystems

- **Biodiversity:** Despite harsh conditions, deserts support a unique range of plants and animals adapted to survive with minimal water.
- **Water Conservation:** Desert plants have evolved specialized adaptations like deep roots, water-storing tissues, and reduced leaf surfaces to conserve water.
- **Soil Formation:** While deserts have limited vegetation, they contribute to the formation of soils in dry regions, supporting specific plant life.
- **Cultural and Economic Value:** Deserts often contain valuable resources like minerals and oil, and many indigenous cultures have adapted to desert life in sustainable ways.

3.5 Aquatic Ecosystems

- An aquatic ecosystem is a water-based environment, including oceans, rivers, lakes, and ponds, where plants, animals, and microorganisms live.
- These ecosystems provide food and shelter to many species and play a key role in Earth's water cycle.
- Depending upon the quality and nature of water, the aquatic ecosystem is categorized into:
  - Freshwater Ecosystem
  - Marine Ecosystem.

3.5.1 Freshwater Ecosystems

A **freshwater ecosystem** is a water-based environment with low salt content, found in lakes, rivers, ponds, and streams. It supports a

variety of plant and animal life and plays an important role in the Earth's water cycle.

Biotic Components (Living organisms)

- **Producers** (Plants and Phytoplankton)

  Aquatic Plants: Water lilies, cattails, and reeds.

  Phytoplankton: Tiny algae and microorganisms that perform photosynthesis, forming the base of the food chain.

- **Consumers (Animals)**

  Primary Consumers: Herbivores like fish, insects, and small aquatic animals that feed on plants and algae.

  Secondary Consumers: Carnivores like larger fish, amphibians, and some insects that feed on herbivores.

  Tertiary Consumers: Top predators such as birds, large fish, and some reptiles that feed on smaller carnivores.

- **Decomposers**

  Bacteria, Fungi, and Detritivores: Organisms like decomposing bacteria, fungi, and insects that break down dead organic matter, recycling nutrients back into the ecosystem.

Abiotic Components (Non-Living Factors)

- **Water:** The main component of the ecosystem, affecting all aspects of life and nutrient cycling.
- **Temperature:** Influences the metabolism and behavior of aquatic organisms.
- **Light:** Essential for photosynthesis, impacting the growth of plants and algae.
- **Oxygen:** Crucial for respiration in animals and plants; levels vary in different parts of the water.

- **Nutrients:** Minerals and organic materials like nitrogen and phosphorus that support plant and algae growth.
- **Substrate:** The type of bottom surface (sand, mud, rocks) in aquatic environments, providing habitats for organisms.

3.5.3 Marine or Ocean Ecosystem

A marine or ocean ecosystem is a complex environment that includes both **biotic** (living) and **abiotic** (non-living) components. Here's a breakdown of its key features:

Biotic Components (Living Organisms)

- **Producers** (Autotrophs)

  Phytoplankton: Microscopic algae that perform photosynthesis, forming the base of the marine food web.

  Seaweed and Algae: Larger aquatic plants that provide energy and habitat.

- **Consumers**

  Primary Consumers: Herbivores like zooplankton, small crustaceans, and some fish species (e.g., sardines).

  Secondary Consumers: Carnivores or omnivores like larger fish (e.g., tuna), squid, and sea turtles.

  Tertiary Consumers: Apex predators like sharks, orcas, and large sea birds.

- **Decomposers**

  Bacteria and fungi that break down organic material, recycling nutrients back into the ecosystem.

Symbiotic Organisms

- Coral and zooxanthellae (algae) share a mutualistic relationship within coral reefs.

- Cleaner fish and host species (e.g., cleaner wrasse and larger fish).

Abiotic Components (Non-Living Factors)

- **Water**

  Salinity: Determines the types of organisms that can survive.

  Temperature: Influences metabolic rates and habitat suitability.

  Ocean Currents: Affect nutrient distribution and organism movement.

- **Light**

  Sunlight is crucial for photosynthesis in the photic zone (upper layer of the ocean).

  Deeper regions (aphotic zone) lack sunlight, relying on chemosynthesis or detritus for energy.

- **Nutrients**

  Nitrogen, phosphorus, and iron are critical for phytoplankton growth.

  Upwelling zones bring nutrient-rich waters to the surface.

- **Pressure**

  Increases with depth, influencing the types of organisms found at various levels.

- **Substrate**

  Includes sandy bottoms, rocky outcrops, coral reefs, and deep-sea sediment, providing habitats for different species.

- **Gases**

  Oxygen and carbon dioxide levels impact respiration and photosynthesis.

Interactions in the Marine Ecosystem

- **Food Webs:** Energy flows from producers (phytoplankton) to various consumer levels and decomposers.
- **Habitat Formation:** Coral reefs, mangroves, and seagrass beds create complex habitats.
- **Adaptations:** Biotic components adapt to abiotic factors, such as fish with streamlined bodies for fast swimming or bioluminescent organisms in the deep sea.

  This delicate balance between biotic and abiotic factors ensures the sustainability and productivity of the ocean ecosystem.

Estuarine Ecosystem

An **estuarine ecosystem** is a dynamic environment where freshwater from rivers and streams meets and mixes with saltwater from the ocean. This transition zone supports unique interactions between **biotic** (living) and **abiotic** (non-living) components.

Biotic Components (Living Organisms)

1. **Producers (Autotrophs)**
   - **Phytoplankton:** Primary producers that perform photosynthesis.
   - **Mangroves:** Trees adapted to saline water, common in tropical estuaries.
   - **Seagrass:** Submerged aquatic vegetation providing food and habitat.
   - **Algae:** Attached or free-floating species.
2. **Consumers**
   - **Primary Consumers:** Organisms like zooplankton, crabs, shrimp, and mollusks that feed on plants or phytoplankton.
   - **Secondary Consumers:** Small fish, larger crustaceans, and birds (e.g., herons, egrets).

- **Tertiary Consumers:** Top predators like larger fish (e.g., barracuda), birds of prey (e.g., ospreys), and estuarine crocodiles.

3. **Decomposers**
   - Bacteria and fungi that break down organic matter, recycling nutrients in the ecosystem.

4. **Symbiotic Relationships**
   - Mangrove roots provide shelter for juvenile fish and other marine organisms.
   - Mutualism between certain fish species and algae.

Abiotic Components (Non-Living Factors)

1. **Water**
   - **Salinity:** Varies between fresh and saltwater, creating a brackish environment.
   - **Tides:** Regular tidal movements bring in nutrients and flush out waste.
   - **Temperature:** Influences organism metabolism and distribution.

2. **Soil and Sediments**
   - Rich in organic matter due to decaying plants and animals.
   - Mudflats and sandbars provide habitats for burrowing organisms.

3. **Nutrients**
   - High nutrient levels from river runoffs support robust primary production.
   - Includes nitrogen, phosphorus, and organic carbon.

4. **Light**
    - Crucial for photosynthesis in shallow waters; varies with water turbidity.

5. **Oxygen Levels**
    - Affected by tides, temperature, and decomposition rates.
    - May vary between water surface and deeper layers.

6. **Flow and Mixing**
    - River inflow and tidal movements create a gradient of salinity and nutrient mixing.

Key Interactions

- **Food Web:** Primary producers (phytoplankton, algae) are consumed by primary consumers (zooplankton, shrimp), supporting higher trophic levels (fish, birds, and predators).
- **Nutrient Cycling:** Decomposers break down organic matter, releasing nutrients that fuel primary production.
- **Habitat Creation:** Mangroves and seagrass beds stabilize sediments and provide shelter for numerous organisms.

Questions

1. How do the structure and functions of an Ecosystem interact to sustain life?
2. Discuss about Food chain.
3. Discuss about Food Web.
4. Why are some ecological pyramids upright while others are inverted across ecosystems?
5. How do ecosystems develop and transform over time through ecological succession?
6. Explain the structure and function of Grassland Ecosystem.
7. What is the scope & importance of Environmental studies?
8. Explain different types of Aquatic Ecosystem.

9. How do the structure and functions of a forest ecosystem support biodiversity and ecological balance?
10. How is energy transferred through ecosystems according to various flow models?
11. Explain the structure and function of Desert Ecosystem.
12. Can you describe the structure and ecological roles within a pond ecosystem?
13. How does an estuarine ecosystem function, and what makes it unique?

Unit-4

Biodiversity & Its Conservation

Introduction

- Biodiversity refers to the variety of life forms on Earth, including plants, animals, fungi, microorganisms, and the ecosystems they inhabit.
- It encompasses the diversity of species, genetic diversity within species, and the variety of ecosystems.
- Biodiversity is essential for the stability of ecosystems, providing vital services such as pollination, water purification, climate regulation, and food production.
- It also contributes to cultural, economic, and recreational benefits.
- However, biodiversity is under threat from human activities like deforestation, pollution, climate change, and overexploitation, making its conservation crucial for the health and survival of the planet.

4.1 Levels of Biodiversity

Biodiversity is typically described in terms of three fundamental and interconnected levels of biological organisms. These are

Genetic diversity refers to the variety of genes within a species or population. It includes the differences in DNA sequences among individuals, which can lead to variations in traits such as appearance, behavior, and resistance to diseases.

Example 1: Genetic diversity within the human species can be seen in variations among different populations, such as those in India and

Africa. In simple terms, genetic diversity explains differences like height, hair color, or eye color. Various methods, including DNA analysis and others, can be used to identify and measure genetic diversity.

Examples 2: While all rice varieties belong to the Oryza sativa species, there are thousands of wild and cultivated types that differ genetically in characteristics such as grain color, size, shape, and nutrient content.

1. **Species diversity** refers to the variety of species within a particular ecosystem or on the entire planet.
    - It includes the number of different species (species richness) and the distribution or abundance of each species (species evenness).
    - High species diversity indicates a healthy, balanced ecosystem, where many species coexist and interact, contributing to ecological stability and resilience.
    - It is vital for ecosystem functions such as pollination, nutrient cycling, and maintaining food webs.
    - However, species diversity is threatened by factors like habitat loss, pollution, climate change, and overexploitation.

2. **Ecosystem diversity** refers to the variety of ecosystems in a given area or on Earth as a whole.
    - It includes different types of habitats, such as forests, grasslands, wetlands, oceans, and deserts, each with its own set of species and environmental conditions.
    - Ecosystem diversity is important because it supports the variety of life forms that depend on specific habitats for survival.
    - Healthy ecosystems provide essential services, like regulating climate, purifying water, and supporting biodiversity.
    - The loss of ecosystem diversity due to human activities like deforestation and urbanization can disrupt these vital processes.

Example: Godavari – Delta ecosystem which consists of grassland ecosystem, river ecosystem, estuarine ecosystem, fresh water aquatic ecosystem, marine water aquatic ecosystem.

4.2 Biodiversity at National Level

- **Biodiversity at the National Level** refers to the variety of species, ecosystems, and genetic resources found within a specific country. Each nation has its unique biodiversity, shaped by factors such as geography, climate, and ecosystems. National biodiversity includes a wide range of species, from plants and animals to microorganisms, and encompasses various ecosystems such as forests, grasslands, wetlands, and marine environments.
- At the national level, biodiversity plays a critical role in supporting livelihoods, food security, cultural heritage, and economic development. Many countries have established protected areas, national parks, and wildlife reserves to conserve biodiversity. Governments also implement policies and strategies for sustainable resource management and conservation.
- However, national biodiversity is often threatened by human activities like deforestation, pollution, over-exploitation of resources, and climate change. Efforts to protect and preserve biodiversity at the national level involve legal frameworks, conservation programs, and public awareness campaigns.

4.3 India as a Mega Diversity Nation

- India is one of the 12 mega diversity countries in the world.
- India is considered a **megadiverse nation** due to its vast variety of ecosystems, species, and genetic resources. It is home to about 7-8% of the world's recorded species, despite having only 2.4% of the world's land area. There are several reasons why India is classified as a megadiverse country:

- **Geographical Diversity:** India's diverse landscapes, ranging from the towering Himalayas to the coastal regions, deserts, and tropical rainforests, provide varied habitats for numerous species. This geographical diversity leads to high species richness and variety in ecosystems.
- **Climate Diversity:** The country experiences different climates, including tropical, temperate, arid, and alpine. This climatic variation supports a wide range of plant and animal species adapted to different environmental conditions.
- **Rich Flora and Fauna:** India is home to a wide array of species, including iconic animals like Bengal tigers, Indian elephants, and rhinoceroses. It has a large number of endemic species—plants and animals found only in this region—thanks to its diverse ecosystems.
- **Biodiversity Hotspots:** India contains four of the world's 36 recognized biodiversity hotspots—Western Ghats, Eastern Himalayas, Indo-Burma, and Sundaland. These areas are known for their rich and unique species but are also highly threatened.
- **Cultural and Traditional Knowledge:** India has a deep cultural connection with nature. Indigenous communities have maintained sustainable practices over centuries, contributing to the conservation of many species and ecosystems.

Due to these factors, India is recognized as one of the most biodiverse nations globally, and protecting its biodiversity is crucial for global environmental health.

4.4 Values of Biodiversity

In terms of its economic value, ecological and social benefits, and aesthetic value, biodiversity is extremely valuable.

Consumptive Use Value: It refers to the direct benefits or value derived from natural resources that are consumed or used up. These resources are typically utilized for personal or community needs and

are not intended for commercial sale. The consumptive use value includes things like:

1. **Food and Water:** The value of resources like fish, fruits, vegetables, and drinking water, which are directly consumed for sustenance.
2. **Fuel:** The use of firewood, biomass, and other natural materials for cooking, heating, and energy.
3. **Medicinal Plants:** Plants used for traditional medicines or remedies, consumed directly by communities.

   This value highlights the importance of natural resources in supporting daily life and livelihoods, particularly for local communities that rely heavily on these resources for survival.

Social Value: It refers to the benefits that natural resources, ecosystems, or environmental practices bring to society beyond their economic value. These benefits can improve the well-being, quality of life, and social fabric of communities. Social value encompasses various aspects such as:

1. **Cultural Significance:** Natural landscapes, species, and ecosystems often have cultural, spiritual, or religious importance for local communities or indigenous groups.
2. **Recreation and Tourism:** Natural areas such as parks, beaches, forests, and mountains provide spaces for recreation, leisure, and tourism, which enhance social life and promote mental health.
3. **Social Cohesion:** Environmental conservation and sustainable resource management can bring communities together, encouraging collective action and fostering a sense of shared responsibility and connection.
4. **Educational Value:** Natural environments serve as learning resources for education, research, and awareness, contributing to the development of environmental consciousness in society.

Social value is crucial for enhancing social equity, cultural heritage, and the overall quality of life, often leading to stronger communities and better environmental stewardship.

Productive Value: Productive value refers to the economic value generated by natural resources or ecosystems through their direct use in production activities. These resources contribute to the economy by supporting industries, agriculture, and other sectors. The productive value includes:

1. **Agriculture:** The value of soil, water, and biodiversity in producing crops, livestock, and other agricultural products.
2. **Forestry:** The value of timber, non-timber forest products (like fruits, medicinal plants, and resin), and the ecosystem services forests provide in terms of carbon sequestration, water regulation, and soil fertility.
3. **Fishing:** The value derived from fisheries, including fish and other aquatic species, which are vital for food production and economic activity in many regions.
4. **Energy:** The value of resources like coal, oil, natural gas, and renewable energy sources (such as solar, wind, and hydropower) used in power generation and industrial processes.
5. **Raw Materials:** The value of minerals, metals, and other raw materials used in manufacturing, construction, and technology.

Ethical Value: It refers to the moral importance of natural resources and ecosystems, emphasizing the intrinsic worth of nature beyond its utilitarian or economic benefits. This value is grounded in the belief that all living organisms and ecosystems have a right to exist and should be respected and protected. Ethical value includes:

- **Intrinsic Value of Nature:** Recognizing that nature has value in and of itself, regardless of its usefulness to humans. This viewpoint advocates for the preservation of species and ecosystems for their own sake, not just for human benefit.

- **Animal Rights:** The ethical responsibility to protect animals and ensure their well-being, preventing harm and exploitation, especially in the context of habitat destruction, poaching, and commercial exploitation.
- **Intergenerational Justice:** The ethical obligation to conserve natural resources so that future generations can enjoy a healthy environment and have access to the resources needed for their survival and well-being.
- **Respect for Indigenous Cultures:** Acknowledging the rights of indigenous communities to maintain their traditional lifestyles, cultures, and relationships with nature, and ensuring their role in conservation efforts.

The ethical value means that human beings may or may not use a certain species but knowing the very fact that this species exists in nature gives pleasure.

- **Example:** a peculiar species of Pigeon, grey / white bird with short legs is no more on this earth. Similarly, Dodo species is also no more. Human beings are not deriving anything direct from Kangaroo, giraffe but strongly feel that these species should exist in nature.

Aesthetic Value

It refers to the beauty and appeal that nature, such as landscapes, plants, and animals, provides. It is about enjoying the visual, sensory, and emotional experience that nature offers, like the beauty of a sunset, a forest, or a mountain view. Aesthetic value helps people feel relaxed, inspired, and connected to the world around them.

Values that are Optional: Among these values lies biodiversity's untapped potential, which warrants further exploration. For example, tropical rainforests or deep marine ecosystems may hold the key to treatments for diseases like cancer or AIDS. The value of recognizing that biological resources exist in our biosphere, potentially offering significant alternatives in the future, is known as option value.

4.5 Bio Diversity at Regional or Local Level

Biodiversity at the regional or local level refers to the variety of species, ecosystems, and genetic resources found within a specific geographic area, such as a region, state, or community. It focuses on the diversity of life forms and habitats that are unique to a particular area, shaped by local climate, geography, and human activity.

At the regional or local level, biodiversity is important because:

1. **Local Ecosystems:** Unique ecosystems, like wetlands, forests, or grasslands, provide essential services such as water filtration, soil protection, and habitat for local wildlife.
2. **Cultural Significance**: Local biodiversity often has cultural and traditional importance, with communities relying on certain species for food, medicine, and livelihoods.
3. **Conservation Focus:** Protecting local biodiversity helps ensure the sustainability of resources for communities and maintains ecosystem health on a smaller scale, often acting as a model for larger conservation efforts.
4. **Threats:** Local biodiversity can be threatened by urbanization, agriculture, deforestation, and pollution, making conservation efforts critical to preserve unique species and ecosystems.

4.6 Hot- Spots of Biodiversity

Biodiversity hotspots are regions that are both rich in species and have experienced significant habitat loss. These areas are considered critically important for conservation efforts due to their unique and diverse ecosystems, which support a large number of endemic species (species found nowhere else in the world). According to conservation scientists, there are 36 recognized biodiversity hotspots worldwide.

Key features of biodiversity hotspots include:

1. **High Species Richness:** These regions are home to a wide variety of species, many of which are endemic and found only in those areas.
2. **Significant Threats:** Hotspots are under severe threat due to habitat destruction, deforestation, climate change, and human activities like agriculture and urbanization.
3. **Critical for Global Conservation:** Protecting these hotspots is crucial for preserving global biodiversity. Conserving these areas can help save a significant portion of the world's species.
4. **Examples:** Some of the well-known biodiversity hotspots include:
    - **The Western Ghats** and **Sundaland** in Asia
    - **The Amazon Rainforest** in South America
    - **Madagascar** in Africa
    - **The Philippines** and **Indo-Burma** regions

Indo- Burma & Eastern Himalayas Hotspot

- **Indo-Burma Hotspot** and **Eastern Himalayas Hotspot** are two of the world's most important biodiversity hotspots, rich in unique species and ecosystems, but also heavily threatened by human activities.

Indo-Burma Hotspot

- **Location:** The Indo-Burma hotspot spans parts of Southeast Asia, covering countries like India (Northeastern states), Myanmar, Cambodia, Laos, Vietnam, Thailand, and southern China.
- **Biodiversity:** It is home to an extremely diverse range of species, including over 1,500 species of plants, more than 300 species of birds, and a wide variety of mammals, amphibians, and reptiles. Some iconic species found here include the **Indochinese tiger, Sunda pangolin, and Greater adjutant stork.**

- **Threats:** The Indo-Burma hotspot is under significant threat from deforestation, habitat loss, hunting, and agricultural expansion. This has led to the loss of many species and ecosystems.
- **Conservation Efforts**: The hotspot has protected areas like the Phong Nha-Kẻ Bàng National Park in Vietnam and efforts to protect river systems and tropical forests.

Eastern Himalayas Hotspot

- **Location:** The Eastern Himalayas hotspot covers the eastern part of the Himalayan Mountain range, spanning across India, Bhutan, Nepal, and parts of China and Myanmar.
- **Biodiversity:** Known for its diverse habitats, from lush forests to alpine meadows, it supports a rich variety of flora and fauna, including species like the red panda, Himalayan tahr, and snow leopard. It is also home to over 10,000 plant species and around 300 mammal species.
- **Threats:** The region faces threats from deforestation, mining, agriculture, and climate change. The fragile ecosystems are highly sensitive to environmental disturbances, especially due to the region's biodiversity and topography.
- **Conservation Efforts:** Conservation programs here focus on habitat restoration, creating protected areas like Kaziranga National Park in India, and implementing sustainable development practices that involve local communities in conservation.

Western Ghats and Sri Lanka

- The Western Ghats and Sri Lanka form another significant biodiversity hotspot, rich in unique species and ecosystems, but also facing numerous conservation challenges.

Western Ghats Hotspot

- **Location:** The Western Ghats are a mountain range that stretches along the western coast of India, covering the states of Gujarat, Maharashtra, Goa, Karnataka, Kerala, and Tamil Nadu. The region also extends into parts of Sri Lanka.
[3]
- **Biodiversity:** The Western Ghats are one of the world's most biologically diverse regions, home to more than 7,000 species of flowering plants, over 500 species of birds, 139 mammal species, and 290 amphibian species. Notable species include the Nilgiri tahr, Lion-tailed macaque, and Western Ghats shrub frog.
- **Threats:** The region is under threat from deforestation, habitat fragmentation, poaching, and human encroachment for agriculture, mining, and urban development. Climate change also poses a threat to many species due to the delicate balance of the ecosystem.
- **Conservation Efforts:** The Western Ghats have several protected areas, including Silent Valley National Park and Periyar Wildlife Sanctuary, as well as numerous conservation programs focused on preserving both flora and fauna. The area is also a UNESCO World Heritage Site, underscoring its global importance.

Sri Lanka Hotspot

- **Location:** Sri Lanka is an island nation in the Indian Ocean, just off the southern tip of India. Its diverse ecosystems, from tropical rainforests to coastal mangroves, are rich in endemic species.

- **Biodiversity:** Sri Lanka has a high level of endemism with many species found nowhere else in the world. Notable species include the Sri Lankan elephant, Sri Lankan leopard, and the purple-faced langur. The island also has a wide variety of plant species, including numerous orchids and medicinal plants.

- **Threats:** Major threats to Sri Lanka's biodiversity include habitat destruction from agricultural expansion, logging, urbanization, and human-wildlife conflicts. Additionally, climate change impacts such as rising temperatures and shifting rainfall patterns threaten the region’s ecosystems.

- **Conservation Efforts:** Sri Lanka has created several national parks and nature reserves like Yala National Park and Udawalawe National Park to protect its rich biodiversity. Conservation programs focus on sustainable practices, wildlife protection, and maintaining the integrity of the island's ecosystems.

Shared Significance

- The Western Ghats and Sri Lanka share many common species due to their proximity and similar ecological zones. The region is rich in endemic species, making it critical for global biodiversity conservation.
- Both regions are essential for the conservation of tropical rainforests, which provide crucial ecosystem services like carbon sequestration, water regulation, and habitat for a wide variety of species.
- The Western Ghats and Sri Lanka are both important areas for global conservation efforts, and preserving these areas is key to maintaining the health of the planet's biodiversity.

4.7 Endangered Species of India

- A plant, animal or microorganism that is in immediate danger of biological extinction is called endangered species or threatened species.

- The International Union of Conservation of Nature and Natural Resources (IUCN) publish the RED DATA BOOK which includes the list of endangered species of plants and animals.

- India is home to a wide variety of species, many of which are at risk of extinction due to habitat loss, poaching, and environmental changes. Some of the endangered species of India include:

1. **Bengal Tiger (Panthera Tigris Tigris)**
   - **Status:** Endangered
   - **Habitat:** Found in forests, grasslands, and wetlands across India, especially in national parks like **Jim Corbett** and **Kaziranga.**
   - **Threats:** Habitat loss, poaching for fur and bones, and human-wildlife conflict.

2. **Indian Rhinoceros (Rhinoceros Unicornis)**
   - **Status:** Vulnerable
   - **Habitat:** Grasslands and riverine areas, primarily in **Kaziranga National Park** in Assam.
   - **Threats:** Poaching for their horns, habitat destruction, and climate change.

3. **Asiatic Lion (Panthera leo persica)**
   - **Status:** Endangered
   - **Habitat:** Found only in the **Gir Forest National Park** in Gujarat.
   - **Threats:** Limited range, poaching, and human-wildlife conflict.

4. **Snow Leopard (Panthera uncia)**
   - **Status:** Vulnerable
   - **Habitat:** Cold, mountainous regions of the **Himalayas** and other northern regions.
   - **Threats:** Poaching, habitat fragmentation, and loss of prey species.

5. **Indian Elephant (Elephas Maximus Indicus)**
   - **Status:** Endangered
   - **Habitat:** Forests and grasslands across India, especially in **Karnataka, Kerala,** and **Assam.**
   - **Threats:** Habitat loss, human-wildlife conflict, and poaching for ivory and skin.

6. **Great Indian Bustard (Ardeotis Nigriceps)**
   - **Status:** Critically Endangered
   - **Habitat:** Grasslands and semi-arid regions, primarily in Rajasthan, Gujarat, and Maharashtra.
   - **Threats:** Habitat destruction, hunting, and collisions with power lines.

7. **Siberian Crane (Leucogeranus Leucogeranus)**
   - **Status:** Critically Endangered
   - **Habitat:** Wetlands in northern India, particularly in **Bharatpur Bird Sanctuary.**
   - **Threats:** Habitat loss, hunting, and climate change affecting migration patterns.

8. **Blackbuck (Antilope Cervicapra)**
   - **Status:** Near Threatened
   - **Habitat:** Grasslands and open forests, mainly in **Gujarat, Madhya Pradesh,** and **Rajasthan.**
   - **Threats:** Habitat loss, hunting, and competition with livestock.

9. **Green Sea Turtle (Chelonia Mydas)**
   - **Status:** Endangered

- **Habitat:** Coastal areas and beaches along the **Andaman and Nicobar Islands and Gujarat.**
- **Threats:** Poaching for eggs and meat, accidental capture in fishing nets, and habitat degradation.

10. Ganges River Dolphin (Platanista Gangetica Gangetica)

- **Status:** Endangered
- **Habitat:** Freshwater rivers, especially the **Ganges, Yamuna**, and **Brahmaputra rivers.**
- **Threats:** Water pollution, dam construction, and fishing nets.

Conservation Efforts

- India has implemented various conservation programs, such as the **Project Tiger** and **Project Elephant**, to protect these endangered species.
- National parks, wildlife sanctuaries, and biosphere reserves have been established to protect their habitats.
- Strict laws, like the **Wildlife Protection Act of 1972,** and active participation of local communities in conservation efforts also play a role in protecting these species.
- The conservation of these endangered species is crucial for ecological balance and crucial medicinals.

4.8 Endemic Species of India

- Endemic species are those species that are found only in a specific geographic area and nowhere else in the world. India is home to a rich variety of endemic species due to its diverse ecosystems and unique climatic conditions. Some of the key endemic species of India include:

1. **Indian Tiger (Panthera Tigris Tigris)**
   - **Habitat:** Found in the forests and grasslands of India, particularly in **Jim Corbett National Park**, **Kanha,** and **Bandhavgarh.**
   - **Description:** The Bengal tiger is India's national animal and a symbol of strength and grace. It is one of the most iconic and endangered species in India.

2. **Nilgiri Tahr (Nilgiritragus Hylocrius)**
   - **Habitat:** Endemic to the **Western Ghats** of India, particularly in areas like **Eravikulam National Park** in Kerala.
   - **Description:** A herbivorous mountain goat, this species is known for its distinct curved horns and is critically endangered due to habitat loss and poaching.

3. **Indian One-Horned Rhinoceros (Rhinoceros Unicornis)**
   - **Habitat:** Found mainly in **Kaziranga National Park** in Assam and some parts of **Uttarakhand.**
   - **Description:** This large herbivore, also known as the **Great Indian Rhinoceros,** is characterized by a single horn and thick armor-like skin. It is endangered due to poaching and habitat degradation.

4. **Lion-Tailed Macaque (Macaca Silenus)**
   - **Habitat:** Found in the **Western Ghats** of India, particularly in **Kerala, Tamil Nadu**, and **Karnataka.**
   - **Description:** Known for its unique silver mane, this primate is highly endangered and is mainly found in rainforests and highland areas.

5. **Andaman and Nicobar Nicobar Pigeon (Caloenas Nicobarica)**
   - **Habitat:** Endemic to the **Nicobar Islands** of India.

- **Description:** A large, brightly colored pigeon with a greenish-blue sheen, it is considered to be a vulnerable species due to habitat destruction and hunting.

6. **Himalayan Brown Bear (Ursus Arctos Isabellinus)**
   - **Habitat:** Found in the high-altitude regions of **Ladakh, Himachal Pradesh,** and **Uttarakhand.**
   - **Description:** A subspecies of the brown bear, it is critically endangered due to habitat loss and human-wildlife conflict.

7. **Kashmir Stag (Hangul or Cervus Hanglu)**
   - **Habitat:** Endemic to the **Himalayan region of Jammu and Kashmir.**
   - **Description:** The **Hangul** is a large deer species with antlers that grow in unique, twisted forms. It is critically endangered due to habitat loss and hunting.

8. **Great Indian Bustard (Ardeotis Nigriceps)**
   - **Habitat:** Found in the grasslands of **Rajasthan, Gujarat,** and **Madhya Pradesh.**
   - **Description:** Known for its size and striking appearance, this bird is critically endangered due to habitat loss, hunting, and threats from high-voltage power lines.

9. **Indian Star Tortoise (Geochelone Elegans)**
   - **Habitat:** Found in dry areas of **Tamil Nadu, Karnataka, Maharashtra,** and **Rajasthan.**
   - **Description:** This species of tortoise is named for the star-like patterns on its shell. It is threatened due to illegal pet trade and habitat destruction.

10. **Black-breasted Butterfly (Zographetus Bhagavanensis)**
    - **Habitat:** Found in the **Western Ghats,** especially in the **Sahyadri hills.**

- **Description:** This butterfly is known for its dark brown and orange markings and is endemic to the region. It is considered rare due to its specific habitat requirements.

Conservation Efforts

- India has several national parks, wildlife sanctuaries, and biosphere reserves to protect endemic species. Programs such as Project Tiger, Project Elephant, and Project Snow Leopard have been crucial in the conservation of these species.
- The species which are found only in a particular region are known as endemic species.
- In India of 47000 species of plants 7000 are endemic.
- Nearly 62% of our endemic species are found available in Himalayas and Western Ghats.
- The Western Ghats are particularly rich in Amphibians (frogs, toads) and reptiles (lizards, crocodiles). About 62% amphibians and 50% of lizards are endemic to India.
- Some of the endemic flora include orchid and species like sapria Himalayan, uvaria lurida, nepenthes khasiana.,etc
- Different species of monitor lizards, reticulated python, Indian salamander and viviparous toad are some important endemic species of our country.

4.9 Threats to Biodiversity

Threats to biodiversity are factors that cause harm to the variety of life on Earth, endangering species and ecosystems. These threats can lead to the loss of biodiversity, affecting the health of the planet. Some of the major threats to biodiversity include:

1. **Habitat Destruction**
   - **Deforestation,** urbanization, agriculture, and infrastructure development lead to the destruction of natural habitats, forcing species to migrate or face extinction.
   - For example, clearing forests for farming destroys the homes of numerous plant and animal species.

2. **Climate Change**
   - Changes in climate, such as rising temperatures and extreme weather patterns, can alter habitats, affect migration patterns, and disrupt ecosystems.
   - Species may struggle to adapt to these changes, leading to population declines or extinctions.

3. **Pollution**
   - **Air, water, and soil pollution** from industrial waste, chemicals, plastics, and pesticides harms ecosystems and species.
   - For instance, pollutants like oil spills can destroy marine ecosystems, while chemical fertilizers and pesticides harm aquatic life and pollinate species.

4. **Overexploitation of Resources**
   - Overfishing, hunting, logging, and harvesting of plants for trade depletes natural resources faster than they can regenerate.
   - This leads to the decline in species populations, such as the overfishing of marine species or the illegal poaching of tigers and elephants for their body parts.

5. **Invasive Species**
   - Non-native species introduced to new environments can outcompete, prey on, or spread diseases to native species, often disrupting the balance of ecosystems.

- For example, the introduction of the **cane toad** in Australia or the **zebra mussel** in North America has had destructive impacts on native wildlife.

6. Agricultural Expansion

- The conversion of forests and grasslands into agricultural fields and monoculture crops reduces biodiversity by eliminating natural habitats and introducing harmful pesticides and fertilizers.
- This reduces the variety of plants, animals, and microorganisms, threatening ecosystem stability.

7. Deforestation

- Large-scale logging, illegal tree cutting, and land clearance for agriculture lead to deforestation, removing vital forest habitats for wildlife.
- Forests, which are home to diverse species, play an essential role in maintaining global ecological balance.

8. Global Trade

- The movement of goods and species across borders can introduce invasive species, diseases, and pests to new areas, threatening native species.
- Trade also often leads to overharvesting of wildlife for trade, such as the illegal wildlife trade in elephants, rhinoceros, and exotic birds.

9. Land Use Change

- Urban sprawl, industrialization, and infrastructure development fragment natural habitats, isolating species and reducing their chances for survival.
- This leads to "habitat islands" where species may struggle to find food, mates, or shelter.

10. Disease

- Disease can spread quickly across populations, especially in vulnerable species. The introduction of pathogens from other regions can decimate native wildlife populations.

  For example, the **chytrid fungus** has severely impacted amphibian species worldwide.

11. Genetic Erosion

- The loss of genetic diversity within species due to overpopulation of a few individuals or inbreeding can make them more vulnerable to diseases, climate change, and other threats.
- This leads to the weakening of species over time and can eventually lead to extinction.

12. Agricultural Pollution

- Excessive use of fertilizers, herbicides, and pesticides in agriculture pollutes soil and water, harming organisms living in those ecosystems.
- For example, nutrient runoff from fertilizers can lead to **eutrophication,** which creates dead zones in oceans and freshwater systems, killing aquatic life.

13. Forest Fires

- While natural fires can be part of some ecosystems' regeneration, human-induced fires often destroy vast areas of habitat, contributing to species loss and altering ecosystems.
- Large-scale fires, often due to agricultural activities or climate change, can devastate plant and animal populations.

14. Fragmentation of Ecosystems

- The breaking up of large ecosystems into smaller, isolated patches makes it difficult for species to migrate, reproduce, and find food, ultimately threatening their survival.

- Habitat fragmentation also prevents the genetic exchange between populations, leading to inbreeding.

15. Landfills and Waste Disposal

- Landfills and improper waste disposal contaminate soils and water, affecting biodiversity. Animals may ingest or become trapped in human-made waste.
- Toxic waste and plastic pollution have a particularly severe impact on marine life, including birds, fish, and turtles.

Conclusion

- The threats to biodiversity are interlinked and often worsen each other. For instance, climate change exacerbates the effects of habitat destruction, while pollution accelerates the decline of species. To prevent further loss of biodiversity, concerted efforts are needed at local, national, and global levels, focusing on conservation, sustainable practices, and awareness.

4.10 Conservation of Biodiversity

- **Conservation of biodiversity** refers to the practice of protecting and preserving the variety of life on Earth, including plants, animals, ecosystems, and the genetic diversity within these species.
- Biodiversity is crucial for ecosystem stability, food security, medicine, and overall environmental health.
- Effective conservation helps maintain the balance of nature, ensures the survival of species, and contributes to the sustainability of ecosystems. Here's how biodiversity can be conserved
- The two basic approaches to wildlife conservation in protected habitats are:
  - In- situ conservation
  - Ex- situ conservation

- **In- situ Conservation**
  - **In-situ conservation** refers to the conservation of species in their natural habitats.
  - It involves protecting and managing ecosystems and species within their natural environment, allowing them to continue to evolve and thrive in the wild.
  - **Eg:** Biosphere reserves, National Parks, Sanctuaries, sacred forests &sacred lakes.

- **Biosphere Reserves**

Key Features of Biosphere Reserves

- **Conservation of Biodiversity:** They help protect the diversity of life, including both species and ecosystems, while maintaining ecological processes.

Multiple Zones: Biosphere reserves are usually divided into three main zones:

- **Core Zone:** This is the protected area where human activities are restricted to preserve the natural environment and wildlife.
- **Buffer Zone:** Surrounding the core zone, this area allows limited human activities, such as research or ecotourism, that do not harm the environment.
- **Transition Zone:** This is the outermost zone where sustainable human activities, such as agriculture, forestry, and community development, can take place.

Sustainable Development: Biosphere reserves encourage sustainable use of natural resources to benefit local communities without depleting the environment.

International Recognition: Many biosphere reserves are part of the **UNESCO Man and the Biosphere (MAB) Program,** which recognizes areas of global importance for conservation.

Examples of Biosphere Reserves

- **Sundarbans Biosphere Reserve (India):** Known for its vast mangrove forests and the Bengal tiger, it protects unique species and ecosystems in the delta region.
- **Nanda Devi and Valley of Flowers Biosphere Reserve (India):** A UNESCO-listed site that conserves the rich flora and fauna of the Himalayas.
- **Great Barrier Reef Biosphere Reserve (Australia):** A marine biosphere reserve that protects the world's largest coral reef ecosystem and promotes sustainable tourism.

Benefits of Biosphere Reserves

- **Conservation of Ecosystems:** They preserve biodiversity, habitats, and ecosystems for future generations.
- **Research Opportunities:** They provide spaces for scientists to study ecological processes and species in their natural environments.
- **Community Involvement:** Local communities are involved in conservation efforts and can benefit from sustainable practices like eco-friendly tourism and agriculture.
- **Global Recognition and Support:** Being part of the UNESCO MAB program can bring international attention and resources to conservation efforts.

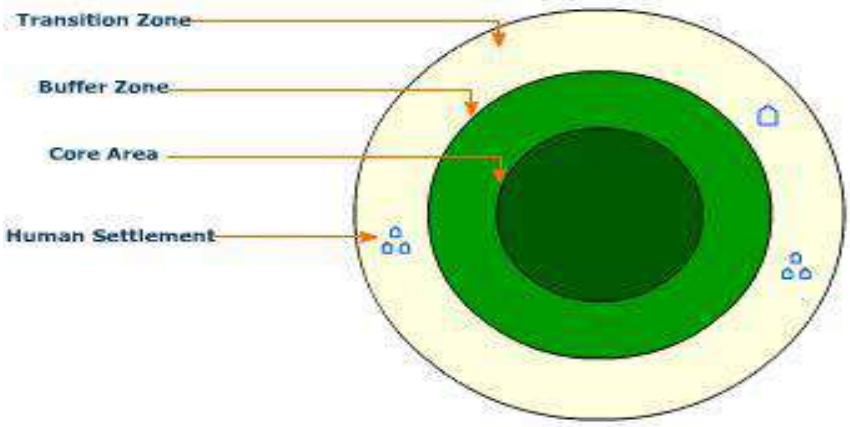


Figure 4: Zones of Biosphere Reserve

Examples: Nandadevi (UP), Manas (Assam), Sundarbans (West Bengal), Nilagiri (Karnataka, Kerala&Tamil Nadu).

- **National Parks**
  - A national park is a designated area aimed at conserving wildlife and the environment. It typically spans between 100 to 500 square kilometers within a biosphere reserve.
  - India has a total of 89 national parks.
  - These parks are also designed for recreational purposes, such as tourism, without causing harm to the environment.
  - However, in India we have 89 national parks.
  - It is also meant for enjoyment through tourism without effecting the Environment.

    Examples: Kazi Ranga NP–Assam;
    Gir NP- Gujrat;
    Bandipur NP- Karnataka;
    Khana NP- Madya Pradesh.

- **Sanctuaries**
  - A sanctuary is a protected area established for the conservation of wildlife and their natural habitats. In a sanctuary, hunting,

poaching, or any activities that could harm the ecosystem are strictly prohibited.

Eg: Ghana bird sanctuary (Rajasthan)

Madhuramalai wild life sanctuary (Tamil Nadu).

Sariska Wildlife Sanctuary (India): Known for the conservation of tigers and other wildlife species.

Keoladeo National Park (Bharatpur Bird Sanctuary, India): Famous for its migratory birds and a World Heritage Site.

Sacred Forests and Sacred Lakes

Sacred Forests: Sacred forests are areas of forest that are protected and revered by local communities due to cultural or religious beliefs. These forests are often considered sacred and are preserved to maintain spiritual connections and biodiversity.

Sacred Lakes: Sacred lakes are bodies of water that hold religious significance for certain communities. They are often protected due to cultural or spiritual importance and are seen as sacred places for rituals, worship, or pilgrimage.

Ex-Situ Conservation

- **Ex-situ conservation** is the protection of species by removing them from their natural habitats and placing them in controlled environments like zoos, botanical gardens, or seed banks.
- It helps conserve endangered species when in-situ conservation is not possible.
- In India, we have following important Gene bank/ Seed bank facilities.
  - **National Bureau of Plant Genetic Resources (NBPGR)** is situated in New Delhi, where agricultural and horticultural crops, along with their wild relatives, are preserved through cryopreservation techniques. This involves storing seeds,

eggs, semen, ova, cells, germplasm, and pollen using liquid nitrogen at temperatures as low as –196°C.

- **National Bureau of Animal Genetic Resources (NBAGR)** Located in Karnal, Haryana, this facility preserves the semen of domesticated bovine animals.
- **National Facility for Plant Tissue Culture Repository (NFPTCR)** for the development of a facility to conserve various plant and tree varieties through tissue culture.

Questions

1. Identify the different values associated with biodiversity.
2. Compare and contrast In-situ and Ex-situ conservation methods for biodiversity.
3. Write about the levels of Biodiversity.
4. Explain the Threats to Biodiversity.
5. India is considered as a mega diversity nation. Explain.
6. What are Hotspots of Biodiversity? Discuss their salient features?
7. Discuss about Endangered Species.
8. Explain briefly about Endemic Species.
9. Explain man wildlife conflicts and discuss how to overcome the conflicts.
10. Write a short note on i) Habitat Loss and ii) Poaching

Unit-5

Environmental Pollution Effects & Control Methods

Introduction

Environmental pollution is the contamination of air, water, soil, or other parts of the environment by harmful substances. These pollutants come from human activities, like industrial processes, waste disposal, and vehicle emissions, and can harm living organisms, including humans, animals, and plants.

Pollution is generally two types:

1. **Natural Pollution:** Natural pollution refers to the contamination of the environment caused by natural events or processes, such as volcanic eruptions, wildfires, or the release of gases from decaying plants.

2. **Man-Made Pollution:** Man-made pollution is the contamination of the environment caused by human activities, such as industrial waste, vehicle emissions, and the use of chemicals, that harm air, water, and land.

Pollutant: A pollutant is any substance that contaminates the environment and causes harm to living organisms.

5.1 Air Pollution

Definition

Air pollution is the presence of harmful substances in the air that can negatively affect health and the environment.

Classification Air Pollutants

- **Primary Pollutant:** A primary pollutant is a harmful substance directly released into the air from sources like vehicles or factories.
- **Secondary Pollutant:** A secondary pollutant is a harmful substance formed when primary pollutants react with other chemicals in the atmosphere. Eg: PAN, Ozone..

Sources/Causes of Air Pollution

Natural Sources: volcanic eruptions, wild fires, biological emissions, ocean spray, pollen grains of flowers, spores etc.,

Man – Made Activities: Thermal power plants, agricultural activities, industrial emissions, vehicle exhaust, fossil fuel burning, Deforestation, Construction, Waste disposal.

I. Effects on Human Health

- **Respiratory Problems:** It can cause or worsen conditions like asthma, bronchitis, and chronic obstructive pulmonary disease (COPD).
- **Heart Disease:** Long-term exposure to air pollution increases the risk of heart attacks, strokes, and other cardiovascular diseases.
- **Lung Cancer:** Prolonged exposure to pollutants like particulate matter and carcinogenic substances increases the risk of lung cancer.
- **Premature Death:** Air pollution is linked to higher rates of premature death, especially from heart and lung diseases.
- **Weakened Immune System:** Air pollutants can impair the immune system, making the body more vulnerable to infections.

- **Cognitive and Developmental Issues:** Children exposed to air pollution may experience developmental delays and cognitive impairments.
- **Eye and Throat Irritation:** Pollutants like ozone and sulfur dioxide can irritate the eyes, throat, and lungs, causing discomfort and breathing difficulties.
- **Pregnancy Risks:** Air pollution can lead to complications during pregnancy, such as low birth weight and preterm birth.

II. Effects on Plants

- Reduced Photosynthesis
- Leaf Damage
- Stunted Growth
- Decreased Crop Yields
- Increased Susceptibility to Disease
- Altered Reproduction
- Disrupted Water and Nutrient Uptake

III. Effect on Aquatic Life

When air pollutants mix with rain, they can increase the acidity (lower the pH) of freshwater lakes, ultimately leading to the death of fish.

IV. Effect on Materials

- Corrosion of Metals
- Degradation of Buildings and Monuments
- Deterioration of Paint and Coatings
- Fading of Fabrics
- Damage to Electronic Equipment
- Decay of Wood and Other Materials

Control Measures

Air pollution can be minimized by following methods

- **Use Cleaner Energy:** Shift to renewable energy sources like solar, wind, and hydroelectric power instead of burning fossil fuels.
- **Promote Public Transport:** Encourage the use of buses, trains, or cycling instead of private cars to reduce vehicle emissions.
- **Reduce Industrial Emissions:** Install filters and cleaner technologies in factories to limit the release of harmful gases.
- **Regulate Vehicle Emissions:** Implement stricter emission standards for vehicles and promote electric cars.
- **Plant Trees:** Trees absorb carbon dioxide and other pollutants, improving air quality.
- **Reduce Waste Burning:** Avoid burning waste, especially plastics, and promote recycling to reduce the release of harmful chemicals.
- **Switch to Eco-friendly Products:** Use products that are free from harmful chemicals, such as low-VOC paints and cleaning agents.
- **Energy Efficiency:** Use energy-efficient appliances and building designs to reduce overall pollution from energy use.

Removing particulate from stack exhaust gases by employing **electrostatic precipitators, bag-house filters, cyclone separators, scrubbers** etc.

Reduction of Air Pollution at Source

I. Gaseous Pollutants

Gaseous pollutants can be reduced through physical absorption onto porous solid materials such as activated charcoal, silica gel, and fuller's earth. They can also be removed by condensation.

Particulate Matter: Various devices are available, depending on factors such as the characteristics of the particulate, flow rate, collection efficiency, and cost.

A. Cyclone Separator

- A cyclone separator is a device used to remove particles from air or gas streams through centrifugal force.
- As the air or gas enters the cyclone, it is spun rapidly, causing heavier particles to be pushed toward the walls and collected at the bottom.
- It is commonly used in industries to separate dust and other particles from gases.

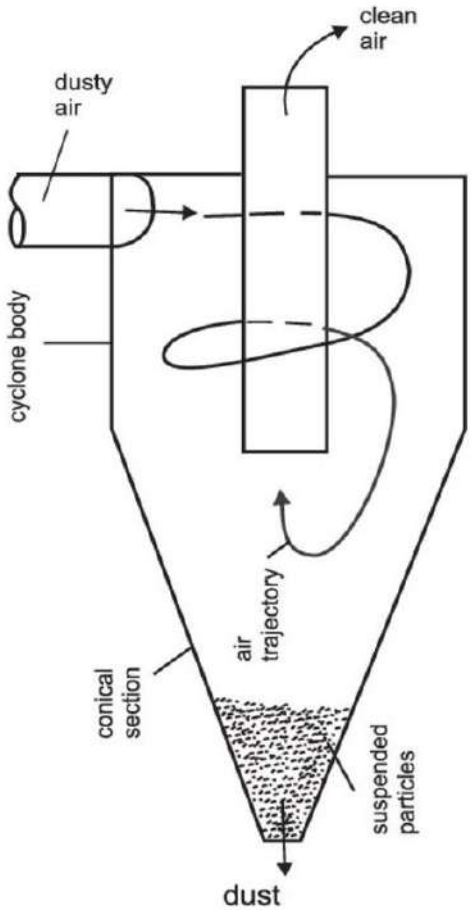


Figure 5.1 Cyclone separator
(Image source: https://www.researchgate.net/publication/258569257_Transversal_Pressure_Effect_in_Circulative_Separators/figures?lo=1)

B. Bag House Filters

- Baghouse filters are industrial air filtration systems that use fabric bags to capture dust and particulate matter from gases.
- The air or gas stream passes through the bags, trapping particles while allowing clean air to flow through.
- These filters are commonly used in industries like cement, mining, and power plants to reduce air pollution.

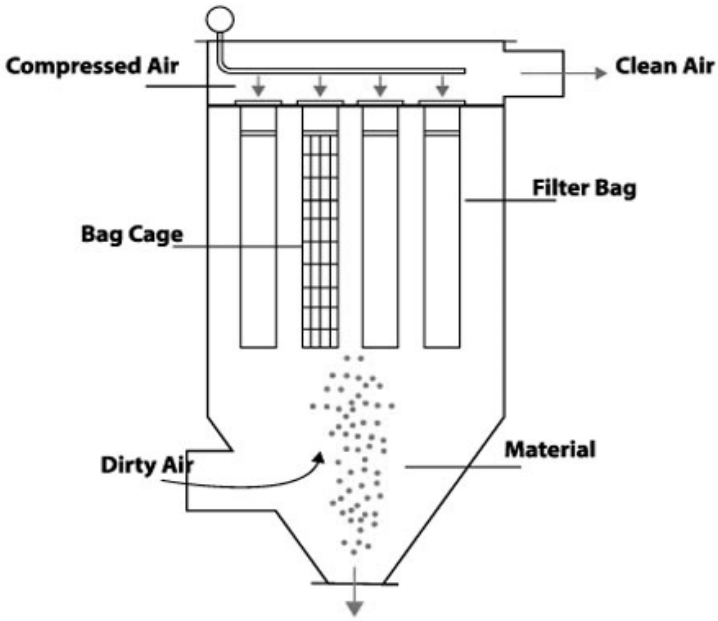


Figure 5.2: Bag House Filters

C. Wet Scrubbers

- Wet scrubbers are air pollution control devices that remove pollutants by spraying water or a liquid solution into the gas stream.
- The pollutants dissolve or react with the liquid, trapping them and preventing them from being released into the air.
- They are commonly used to remove gases like sulfur dioxide (SO_2) and particulate matter.

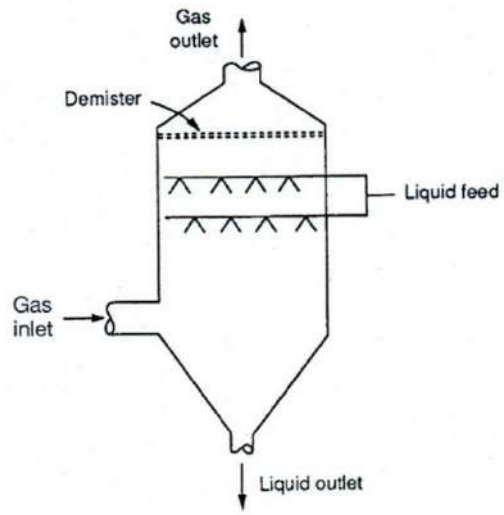


Figure 5.3: Wet Scrubber

5.2 Water Pollution

Water is an essential commodity for survival. We need water for drinking, cooking, bathing, washing, irrigation, and for industrial operations. Most of water for such uses comes from rivers, lakes or groundwater sources. Water has the property to dissolve many substances in it, therefore, it can easily get polluted.

Definition

Water pollution is the contamination of water bodies, such as rivers, lakes, and oceans, with harmful substances that degrade water quality and harm aquatic life.

Sources

- **Industrial Waste:** Chemicals, heavy metals, and toxic materials released by factories.
- **Agricultural Runoff:** Fertilizers, pesticides, and animal waste from farms.
- **Sewage and Wastewater:** Domestic and industrial waste released into water bodies.
- **Oil Spills:** Accidental release of oil into oceans or rivers.

- **Plastic Waste:** Litter and plastic products that enter water bodies.

Effects

- **Harm to Aquatic Life:** Pollutants can kill fish, plants, and other marine organisms.
- **Health Risks:** Contaminated water can cause diseases like cholera, dysentery, and typhoid.
- **Ecosystem Imbalance:** Disruption of ecosystems due to nutrient overload (eutrophication).
- **Decreased Water Quality:** Polluted water becomes unsuitable for drinking, agriculture, and recreational use.

Bio-magnification

- It is the process where toxic substances like pesticides, heavy metals, or industrial chemicals increase in concentration as they move up the food chain.
- Organisms at lower trophic levels, such as plants and herbivores, absorb these pollutants from the environment.
- As predators consume these organisms, the toxins accumulate and become more concentrated in their bodies.
- This is particularly harmful to top predators, including humans, as they may ingest high levels of toxins over time.
- Bio-magnification can lead to health problems, such as neurological damage and reproductive issues, and disrupt ecosystems. Reducing harmful chemical use is key to preventing it.

Following is the example of Bio-magnification of DDT in aquatic food chain:

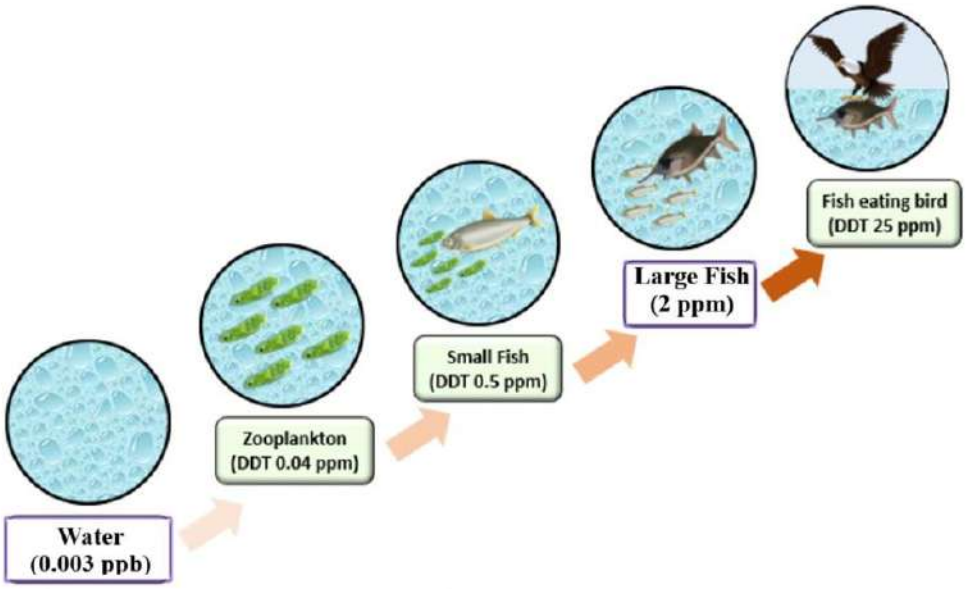


Figure 5.4: Bio magnification

Control Measures

- **Wastewater Treatment:** Treating water before releasing it into water bodies to remove pollutants.
- **Reduction of Pesticides and Fertilizers:** Using organic farming methods and reducing chemical runoff.
- **Proper Waste Disposal:** Ensuring proper treatment and disposal of industrial and domestic waste.
- **Pollution Prevention Regulations:** Implementing stricter laws and policies to reduce water pollution.
- **Public Awareness:** Educating communities on the importance of protecting water bodies and reducing pollution.

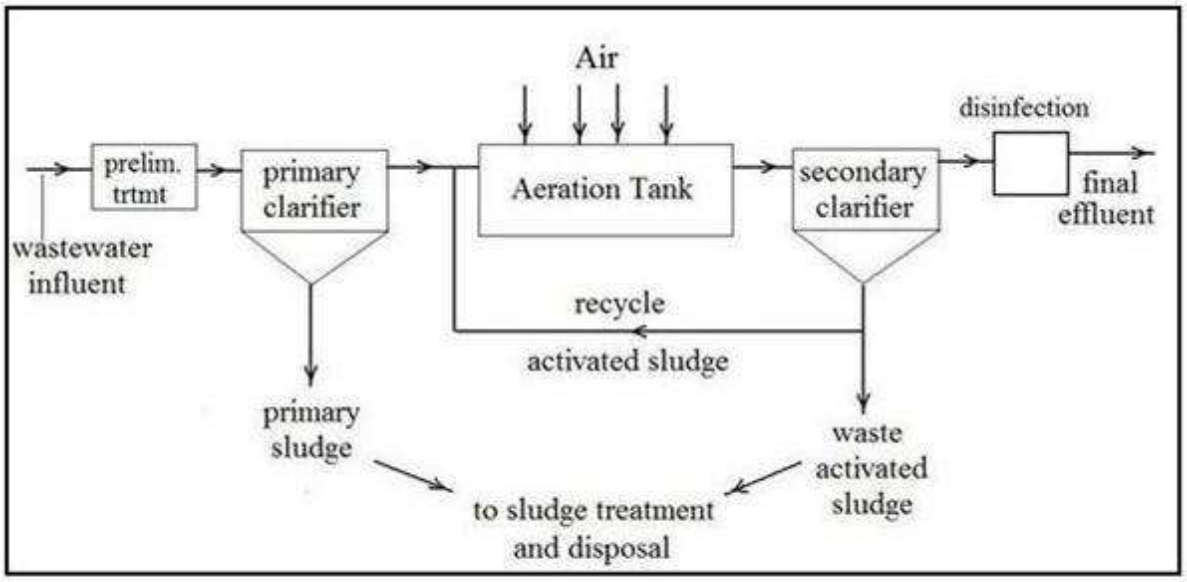
5.2.1 Waste Water Treatment

I. Preliminary Treatment

The removal of wastewater constituents like rags, sticks, floatable grit, and grease that could cause maintenance issues or operational problems in treatment processes and ancillary systems.

Preliminary treatment consists of following units

1. **Screening:** For the removal of floating debris
2. **Grit Chamber:** For the removal of sand and grit
3. **Comminuters:** For grinding large suspended solids
4. **Floatation Units:** For removal of oil and grease
5. Skimming Tank



Activated Sludge Wastewater Treatment Flow Diagram

Figure 5.5: Flow Chart of Waste Water Treatment Plant

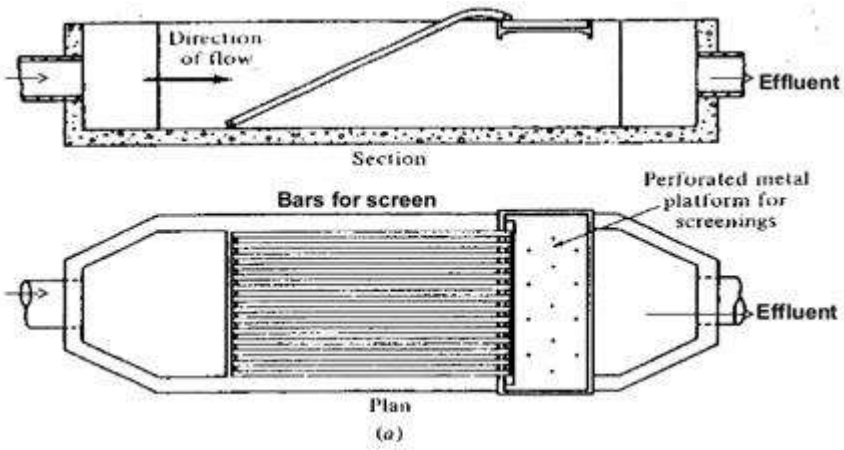


Figure 5.6: Screening

Screening: Screening in air pollution refers to the process of removing large particles and debris from the air using filters or screens. It helps to prevent larger pollutants from entering air treatment systems and reduces air pollution.

Grit Chamber: A grit chamber in air pollution control is used to remove heavy particulate matter like dust, sand, and larger debris from the air. It slows down the air flow, allowing these particles to settle, preventing damage to equipment and improving air quality.

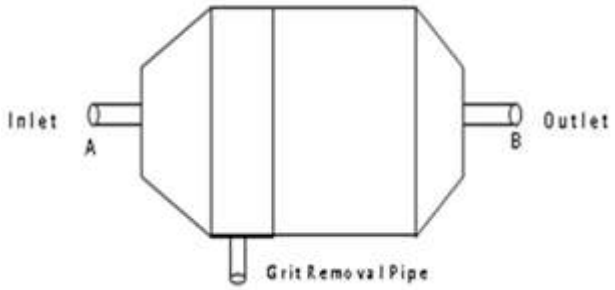


Figure 5.7: Grit Chamber

II. Primary Treatment

Primary treatment involves the removal of a portion of suspended solids and organic matter from wastewater and consists of the following processes/units.

Sedimentation: Primary Settling Tanks

Sedimentation is the process where solid particles in wastewater settle at the bottom of a tank due to gravity. It is used to remove suspended solids from the water before further treatment.

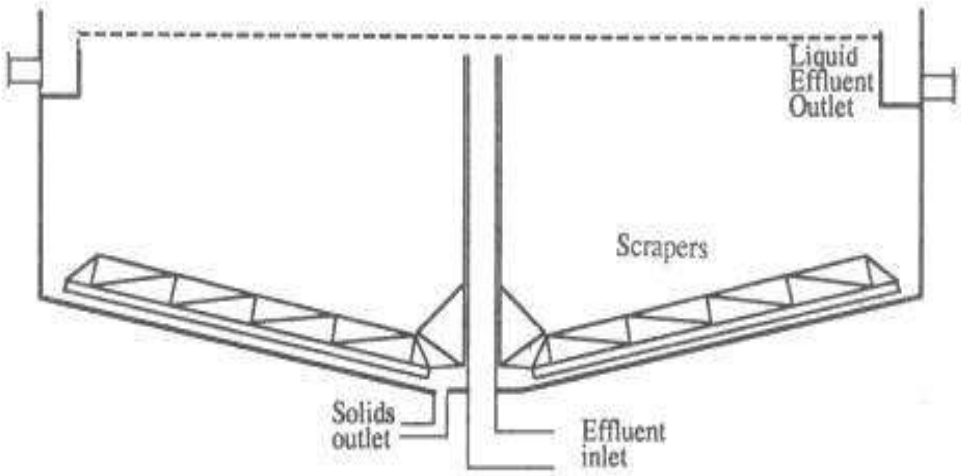


Figure 5.8: Primary Settling Tank

Coagulation

Secondary Settling Tank

- **Coagulation** is a water treatment process that involves adding chemicals, known as coagulants, to water to neutralize the charges on suspended particles.
- This causes the particles to clump together and form larger particles called flocs.
- Coagulants commonly used include aluminum sulfate (alum) and iron salts.
- The process begins with rapid mixing to distribute the coagulants evenly, followed by slower mixing (flocculation) to encourage the formation of flocs.
- These flocs then settle out during sedimentation, effectively removing suspended solids, organic matter, and some dissolved substances.
- Coagulation is crucial for improving water quality in both drinking water and wastewater treatment.

Flocculation

Flocculation is the process of gently mixing water to encourage the formation of flocs, which are clusters of coagulated particles.

It follows coagulation and helps to agglomerate small particles into larger ones that can easily settle out during sedimentation.

III. Secondary Treatment

Removal of biodegradable organic matter [insoluble or suspension state] and suspended solids.

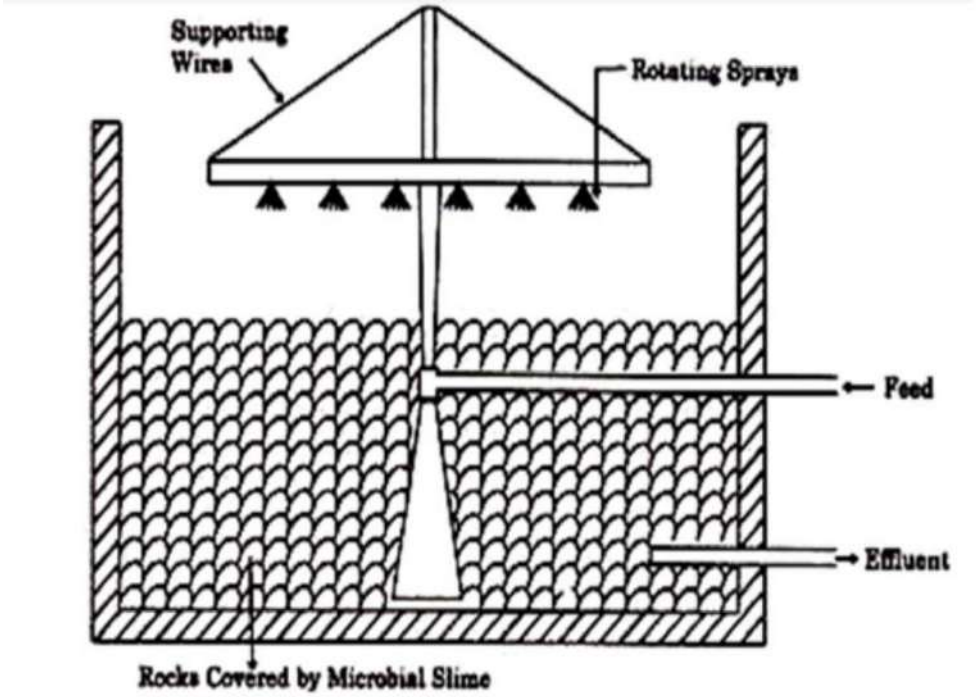


Figure 5.9: Conventional Trickling Filter

Secondary treatment consists of following processes.
1. Trickling Filter
2. Activated Sludge Process
3. Oxidation Ponds and lagoons

Trickling Filter

- **A trickling filter** is a biological wastewater treatment system where wastewater is spread over a bed of coarse material, such as stones or plastic media.
- As the water trickles down, microorganisms growing on the surface of the media break down organic pollutants.
- This process helps to reduce the organic load in wastewater before further treatment.

Activated Sludge Process (ASP)

- The activated sludge process is a widely used method for treating wastewater biologically.
- It involves aerating wastewater to promote the growth of microorganisms that break down organic matter.
- The wastewater is mixed with a population of activated sludge (microorganisms), which consumes the pollutants.
- After aeration, the mixture goes to a settling tank where the sludge is separated from the treated water.
- The separated sludge is either returned to the aeration tank to maintain the microbial population or removed for further processing.

IV. Tertiary Treatment

Tertiary Treatment (Advance waste water treatment) the main function of tertiary treatment is to decrease the load of nitrogen and phosphorous compound present in the effluent by the following process

Tertiary Treatment Consists of Following Processes and Units

- Membrane Filtration and Separation
- Dechlorination and Disinfection Systems

- Reverse Osmosis (RO) Systems
- Ion Exchange
- Activated Carbon Adsorption
- Physical/Chemical Treatment

5.3 Soil Pollution

Soil is a natural mixture of minerals, organic matter, water, and air that supports plant life. It forms the upper layer of the Earth's crust and provides nutrients and a habitat for various organisms.

Definition

Soil pollution is the contamination of soil with harmful substances, leading to a decline in soil quality, fertility, and health.

Sources

- **Industrial Waste:** Chemicals, heavy metals, and hazardous materials from factories.
- **Agricultural Practices:** Overuse of pesticides, fertilizers, and herbicides.
- **Urbanization:** Dumping of solid waste, including plastics and chemicals.
- **Mining Activities:** Release of toxic substances like mercury and arsenic.
- **Waste Disposal:** Improper disposal of hazardous waste and sewage.

Effects

- **Reduced Soil Fertility:** Toxic chemicals can kill beneficial microorganisms and plants.
- **Contamination of Food Crops:** Pollutants can be absorbed by plants, affecting food safety.

- **Loss of Biodiversity:** Harm to soil-dwelling organisms such as earthworms and insects.
- **Water Pollution:** Pollutants in the soil can leach into groundwater, affecting water quality.

Control Measures

1. **Sustainable Agriculture:** Reducing chemical use and adopting organic farming practices.
2. **Proper Waste Disposal:** Ensuring proper treatment and disposal of industrial and household waste.
3. **Reforestation and Soil Conservation:** Planting trees and maintaining soil health through erosion control.
4. **Remediation Techniques:** Using methods like bioremediation and phytoremediation to restore contaminated soil.
5. **Legislation:** Implementing laws to regulate waste disposal and pollution control in industries.

5.4 Noise Pollution

Definition

Noise pollution is the presence of harmful or disturbing levels of sound in the environment, often caused by human activities.

It is measured in units of decibels and is denoted by dB

The human ear can perceive sounds up to around 85 decibels, but beyond this limit, the eardrum can no longer withstand the sound pressure.

Noise standards recommended by CPCB

Table 5.1: Noise Standards recommended by CPCB committee

Area code	Category of Area	Noise level in dB (A) L eq	
		Day	Night
A	Industrial area	75	70
B	Commercial	65	55
C	Residential	55	45
D	Silence zone	50	48

Table 5.2: Different Sounds and their Sound levels on decibel Scale

	Sound level (dB)	Source of sound
	180	Rocket engine
	150	Jet plane take off
Threshold of pain	140	
	130	Maximum recorded rock music
	120	Thunder cap
	110	Auto horn
	100	Construction work
	90	Motor cycle
	70	Vacuum cleaner
	60	Air condition unit
	50	Average living room
	30	Library
	20	Broadcasting studio
	10	Rustling leaf
Threshold limit of hearing	0	

Sources of Noise Pollution

The main sources of noise are

- **Transportation:** Traffic noise from cars, trucks, trains, and airplanes.
- **Industrial Activities:** Machinery, equipment, and operations in factories and construction sites.
- **Urbanization:** Noise from crowded streets, shopping malls, and other urban areas.

- **Public Events:** Noise from concerts, sports events, and festivals.
- **Household Appliances:** Sounds from lawnmowers, vacuum cleaners, air conditioners, and other household devices.
- **Electric Generators:** Noise produced by backup generators and other heavy machinery.
- **Agricultural Activities:** Noise from tractors, harvesters, and other farm equipment.
- **Entertainment:** Noise from loudspeakers, clubs, and bars.

Effects of Noise Pollution

Noise Causes the Following Effects

- **Hearing Loss:** Prolonged exposure to loud noise can cause permanent hearing damage or loss.
- **Stress and Anxiety:** Continuous noise can lead to increased stress levels, anxiety, and irritation.
- **Sleep Disturbances:** Noise pollution can disrupt sleep patterns, leading to insomnia and fatigue.
- **Heart Problems:** Long-term exposure to noise has been linked to increased blood pressure, heart disease, and stroke risk.
- **Decreased Productivity:** Noise can interfere with concentration, leading to reduced work efficiency and performance.
- **Negative Impact on Wildlife:** Animals depend on sound for communication and navigation, and excessive noise can disrupt their behaviour, reproduction, and survival.
- **Reduced Quality of Life:** Persistent noise can lower the overall quality of life, causing frustration and decreasing overall well-being.

Control and Preventing Measures

- **Use of Noise Barriers:** Installing soundproof barriers along highways, railways, and around industrial areas to block noise from reaching residential areas.
- **Regulating Noise Emissions:** Implementing laws to limit noise levels from vehicles, machinery, and industrial activities.
- **Urban Planning:** Designing cities with noise-reducing features, such as separating residential areas from high-traffic zones and industrial sites.
- **Quiet Equipment and Vehicles:** Promoting the use of quieter machines, vehicles, and appliances with low noise emissions.
- **Building Sound Insulation:** Installing soundproof materials in buildings, particularly in homes, offices, and schools, to reduce indoor noise.
- **Traffic Management:** Implementing measures like speed limits, road surface improvements, and low-noise tires to reduce traffic noise.
- **Public Awareness:** Educating the public about the harmful effects of noise pollution and encouraging quieter lifestyles.
- **Landscaping:** Planting trees, shrubs, or grass along roads and around buildings to act as natural sound barriers.
- **Noise Monitoring:** Establishing systems to monitor and control noise pollution levels in urban and industrial areas. There are four fundamental ways in which noise can be controlled.

5.5 Nuclear Hazards

- **Nuclear pollution** refers to the contamination of the environment by radioactive substances, which can result from nuclear accidents, improper disposal of radioactive waste, or the testing of nuclear weapons.

- This type of pollution poses significant risks to human health and the environment due to the harmful effects of ionizing radiation.
- In general, the more massive and charged the particle, the less its penetration power. For example, alpha particles are easily stopped by barriers, while gamma rays can penetrate deeply into materials and require heavier shielding.

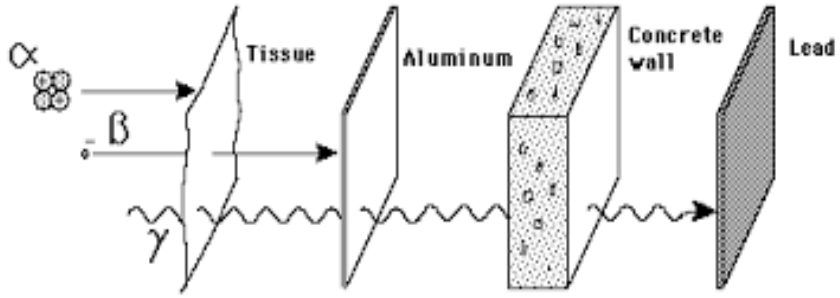


Figure 5.10: Penetration power of alpha particles, beta particles and gamma rays.

Sources

Various sources of radioactivity can be grouped into
1. Natural sources
2. Anthropogenic (man-made) sources.

Natural Sources: Sources of natural radioactivity include cosmic rays from outer space, radioactive radon-222, soil, rocks, air, water and food, which contain one or more radioactive substances.

Anthropogenic Sources: These sources are nuclear power plants, nuclear accidents, X-rays, diagnostic kits, test laboratories etc. where radioactive substances are used.

Effects of Radiations

- Ionization radiations can affect living organisms by causing harmful changes in the body cells and also changes at genetic level.

- **Genetic damage** is caused by radiations, which induce mutations in the DNA, thereby affecting genes and chromosomes. The damage is often seen in the off springs and may be transmitted up to several generations.
- Somatic damage includes burns, miscarriages, eye cataract and cancer of bone, thyroid, breast, lungs and skin.
- Radioisotopes enter the environment during mining of uranium. The radioactivity in the earth's crust enters the crops grown there and ultimately in human beings.
- Radioactive iodine (I - 131) accumulates in thyroid gland and causes cancer. Similarly, strontium-90 accumulates in the bones and causes.
- Exposure of the brain and central nervous system to high doses of radiation causes delirium, convulsions and death within hours or days.
- Acute radiation sickness is marked by vomiting; bleeding of gums and in severe cases mouth ulcers.
- Nausea and vomiting often begin a few hours after the gastrointestinal tract is exposed. Infection of the intestinal wall can kill weeks afterwards.
- Unborn children are vulnerable to brain damage or mental retardation, especially if irradiation occurs during formation of the central nervous system in early pregnancy.

Control of Nuclear Pollution

- Siting of nuclear power plants should be carefully done after studying long term and short-term effects.
- 'Proper disposal of wastes from laboratory involving the use of radioisotopes should be done.
- Nuclear devices should never be exploded in air.

5.6 Solid Waste Management

Solid waste (waste other than liquid or gaseous) can be classified as municipal, industrial, agricultural, medical, mining waste and sewage sludge.

Types/Sources of Solid Wastes

Urban wastes Sources

*Domestic wastes – Food waste, Cloth, Waste paper.
*Commercial wastes – Packing material, cans, bottles, polythene.
*Construction Wastes – Wood, concrete debris.
*Bio medical wastes – Anatomical wastes, infectious wastes.

Industrial wastes Sources

*Nuclear power plants – generates radioactive wastes
*Thermal power plants – produces fly ash in large quantities

Management of solid waste is very important in order to minimize the adverse effects of solid wastes.

Effects of Solid Wastes

- Municipal solid wastes heap up on the roads due to improper disposal system.
- Waste dumping allows biodegradable materials to decompose under uncontrolled and unhygienic conditions produces foul smell and breeds various types of insects and infectious organisms besides spoiling the aesthetics of the site.
- Industrial solid wastes are sources of toxic metals and hazardous wastes, which may spread on land and can cause changes in physico- chemical and biological characteristics thereby affecting productivity of soils.

- Toxic substances may leach or percolate to contaminate the ground water.
- Various types of wastes like cans, pesticides, cleaning solvents, batteries (zinc, lead or mercury) radioactive materials, plastics are mixed up with paper, scraps and other non-toxic materials which could be recycled.
- Burning of some of these materials produce dioxins, furans and polychlorinated biphenyls, which have the potential to cause various types of ailments including cancer.

Steps Involved in Solid Waste Management

For waste management we stress on three R's. - Reduce, reuse and recycle before destruction and safe storage of wastes.

Reduction in use of Raw Materials: Reduction in the use of raw materials will correspondingly decrease the production of waste. Reduced demand for any metallic product will decrease the mining of their metal and cause less production of waste.

Reuse of Waste Materials

The refillable containers which are discarded after use can be reused.

Villagers make casseroles and silos from waste paper and other waste materials.

Making rubber rings from the discarded cycle tubes which are used by the newspaper vendors, instead of rubber bands, reduces the waste generation during manufacturing of rubber bands.

Recycling of Materials: Recycling is the reprocessing of discarded materials into new useful products.

Formation of some old type products e.g. old aluminium cans and glass bottles are melted and recast into new cans and bottles.

Formation of New Products: Preparation of cellulose insulation from paper, preparation of fuel pellets from kitchen waste. Preparation of automobiles and construction materials from steel cans.

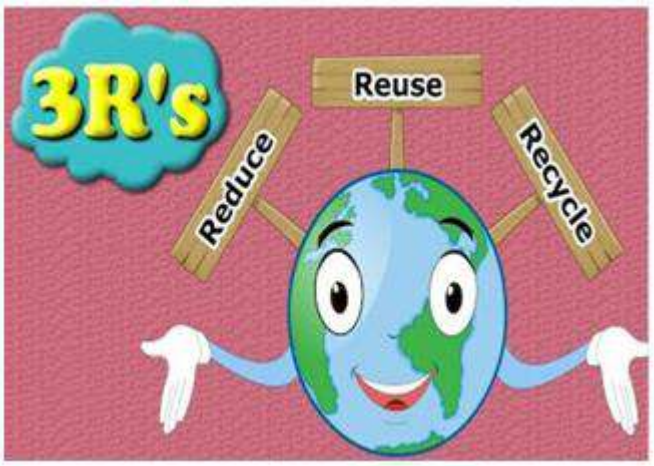


Figure 5.10: Principles of Solid Waste Management

For discarding wastes the following methods can be adopted:

Sanitary Landfill

Garbage is compacted, spread out in thin layers, and covered with clay or plastic foam in a sanitary landfill. An impermeable liner, typically consisting of multiple layers of clay, thick plastic, and sand, covers the bottom of contemporary landfills. The liner prevents leachate from seeping into the ground water and contaminating it. The bottom leachate is pumped and sent to be treated. To stop water seepage, the landfill is covered with top soil, sand, clay, and gravel when it is full. A number of wells are dug close to the landfill to check for leaks that could contaminate the groundwater. Anaerobic decomposition produces methane, which is gathered and burned to generate heat or electricity.

Composting: Due to shortage of space for landfill in bigger cities, the biodegradable yard waste is allowed to degrade or decompose in an oxygen rich medium. A good quality nutrient rich and environmentally friendly manure is formed which improves the soil conditions and fertility.

Incineration: Incinerators are burning plants capable of burning a large amount of materials at high temperature. The initial cost is very high. During incineration high levels of dioxins, furans, lead and cadmium may be emitted with the fly ash of incinerator. Dioxin level may reach many times more than in the ambient environment. For incineration of materials, it is better to remove batteries containing heavy metals and plastic containing chlorine before burning the material. Prior removal of plastics will reduce emissions of dioxins and polychlorinated biphenyls (PCBs).

5.7 Pollution Case Studies

5.7.1 Delhi’s Winter Pollution

Delhi's winter pollution is a major concern, and there have been several studies on the issue. One such study by the Council on Energy, Environment and Water (CEEW) analyzed Delhi's air pollution scenario during the winterof2021.Thestudy found that about 75%of the days during the winter of 2021 had 'very poor' to 'severe' air quality .

Key Findings

- Transport, dust, and domestic biomass burning are the largest local contributors to air pollution, accounting for around 12%, 7%, and 6% of the pollution load, respectively.
- Regional influence plays a significant role, with about 64% of Delhi's winter pollution load coming from outside its boundaries.
- Forecasts can predict pollution trends, but their accuracy decreases when predicting high pollution episodes.

Recommendations

- Implementing the Graded Response Action Plan (GRAP) based on modelled source contributions obtained from forecasts.

- Conducting surveys to explore the prevalence of biomass usage for heating and cooking purposes.
- Promoting alternatives for space heating and encouraging households to use clean fuels.
- Relaying air quality forecasts to the public via social media platforms to reduce individual exposure and activity levels.

5.7.2 Bhopal gas Tragedy

The Bhopal Gas Tragedy, one of the world's worst industrial disasters, occurred on December 3, 1984, at the Union Carbide pesticide plant in Bhopal, Madhya Pradesh, India.

Key Facts

- Date: December 3, 1984
- Time: 12:30 AM
- Location: Union Carbide pesticide plant, Bhopal
- Gas leaked: Methyl Isocyanate (MIC)
- Deaths: Estimated 3,787 immediate deaths; 16,000+ long-term deaths
- Injured: Over 500,000 people
- Affected area: 40 square kilometers

Causes

- Safety measures negligence
- Poor maintenance
- Lack of emergency preparedness
- Inadequate training

Consequences

- Environmental damage
- Health effects: respiratory issues, birth defects, cancer
- Economic impact: significant losses for Union Carbide
- Social impact: displacement, psychological trauma

Investigations and Legal Action

- Government inquiry: headed by Justice N. M. Ahuja
- Criminal charges against Union Carbide CEO, Warren Anderson
- Civil lawsuit: settled for $470 million in 1989
- Ongoing controversies and activism

Lessons Learned

- Importance of safety protocols
- Corporate accountability
- Government regulation and oversight
- Community engagement and preparedness

5.7.3 Yamuna River Case Study

The Yamuna River case study is a classic example of how pollution can devastate a vital water source. Once considered sacred and the lifeline of the people in its surrounding areas, the Yamuna River has become one of India's most polluted river.

Causes of Pollution

- **Industrial Wastes:** The River receives untreated wastewater from industrial units in Haryana, Delhi, and Uttar Pradesh.
- **Domestic Sewage:** Nineteen drains from Delhi, carrying sewage and waste, flow into the Yamuna.

- **Agricultural Runoff:** Pesticides and fertilizers from agricultural lands in Haryana and Uttar Pradesh contaminate the river.
- **Garbage Dumping:** The River is used as a dumping ground for organic, metallic waste, and polythene bags.
- **Ash Scattering and Murti Immersion:** Rituals like ash scattering and idol immersion contribute to the pollution.

Effects of Pollution

- **Water Quality:** The river's water quality is severely compromised, making it unfit for drinking or human use.
- **Health Risks:** Exposure to polluted water poses serious health risks to humans and wildlife.
- **Environmental Damage:** The pollution harms aquatic life and disrupts the ecosystem.

Efforts to Restore the Yamuna

- **Yamuna Action Plan (YAP):** Launched in 1993, YAP aims to improve water quality and increase public awareness.
- **Sewage Treatment Plants:** Several treatment plants have been built to treat wastewater before releasing it into the river.
- **Community Participation:** Initiatives encourage public involvement in cleaning and maintaining the river.

5.7.4 Love Canal Tragedy

The Love Canal tragedy is a landmark environmental disaster in the United States.

Location: Niagara Falls, New York

Timeline

- **1890s:** Love Canal constructed as a hydroelectric canal.
- **1940s-1950s:** Hooker Chemical Company (now Occidental Petroleum) dumped approximately 21,800 tons of toxic waste, including dioxin, furans, and other hazardous chemicals.
- **1953:** Hooker Chemical sold the canal to the Niagara Falls Board of Education for $1.
- **1954-1978:** Homes and schools built on top of the canal, unaware of the toxic waste.
- **1976-1978:** Residents began reporting health issues and contaminated soil and water.
- **1978:** Federal and state authorities intervened, declaring Love Canal a hazardous waste site.
- **1980:** Love Canal Emergency Declaration signed by President Jimmy Carter.

Health Consequences

- High rates of birth defects, miscarriages, and still births.
- Increased incidence of cancer, neurological disorders, and respiratory problems.
- Contaminated soil, water, and air.

Environmental Impact

- **Soil Contamination:** Dioxin levels 100 times higher than acceptable limits.
- **Groundwater Contamination:** Chemicals leaked into nearby streams and rivers.
- **Air Pollution:** Toxic vapors released into the atmosphere.

Government Response

- EPA Superfund program established (1980) to clean up hazardous waste sites.
- Love Canal declared a Superfund site (1983).
- Occidental Petroleum held accountable for cleanup costs.

Lessons Learned

- Importance of environmental regulations and enforcement.
- Corporate accountability for environmental damage.
- Community involvement in environmental decision-making.

Legacy

- Love Canal became a symbol of environmental activism.
- Inspired passage of the Superfund law (CERCLA).
- Continues to influence environmental policy and toxic waste management.

5.7.5 Fukushima Daiichi Nuclear Disaster

Fukushima Daiichi Nuclear Disaster (2011)

Location: Okuma, Fukushima Prefecture, Japan

Background

- Earthquake (9.0 magnitudes) and tsunami hit Fukushima Daiichi Nuclear Power Plant.
- Cooling systems failed, causing radioactive material release.

Pollution Impact

- Radioactive iodine-131, cesium-137, and strontium-90 contaminated:

  - **Air:** 10% of Japan's land area affected.
  - **Water:** Ocean contamination, affecting marine life.
  - **Soil:** 100,000 hectares contaminated.
- Health Effects
  - Increased cancer risk (thyroid, leukemia).
  - Radiation exposure (workers, residents).
- Environmental Damage
  - Contaminated forests, rivers, and wildlife habitats.
  - Long-term ecosystem disruption.

Cleanup Efforts

- Radioactive material removal.
- Containment structure construction.
- Water treatment and discharge.
- Decontamination of surrounding areas.

Challenges

- Radioactive waste management.
- Long-term health monitoring.
- Community displacement and psychological trauma.

Lessons Learned

- Nuclear safety and regulation.
- Emergency preparedness and response.
- Transparency and communication.

Similar Case Studies

1. Chernobyl Nuclear Disaster (1986)
2. Three Mile Island Accident (1979)
3. Wind scale Fire (1957)

5.7.6 Chernobyl Nuclear Disaster

The Chernobyl Nuclear Disaster was a catastrophic event that occurred on April 26, 1986, at the Chernobyl Nuclear Power Plant, located in Ukraine, Soviet Union (now Ukraine).

Background

- Chernobyl Nuclear Power Plant, constructed in 1970, was one of the largest nuclear power plants in the Soviet Union.
- **Safety Test Gone Wrong:** Operators attempted to conduct a safety test during routine maintenance, which led to an uncontrolled power surge.

Disaster

Explosion: At 1:23 AM, a massive explosion occurred, releasing radioactive materials into the atmosphere.

Fire: The graphite moderator caught fire, spreading radioactive debris.

Radioactive Release: 50 tons of radioactive material, including iodine-131, cesium-137, and strontium-90, contaminated the environment.

Consequences

- **Immediate Effects**
  - 28 immediate deaths.
  - 200,000 evacuations from surrounding areas.
  - Radioactive contamination of 50,000 km².

- **Long-Term Effects**
  - Increased cancer risk (thyroid, leukemia, etc.).
  - Birth defects and genetic mutations.
  - Environmental damage (forests, water sources, wildlife).
- **Economic Impact**
  - Estimated $235 billion in damages.
  - Contamination of agricultural land and water sources.

Cleanup and Containment

- **Liquidators:** 500,000 workers involved in cleanup efforts.
- **Sarcophagus:** A concrete structure built to encase the damaged reactor.
- **New Confinement:** A replacement confinement structure, the New Safe Confinement, was completed in 2017.

Investigations and Lessons Learned

- Human error and design flaws.
- Lack of safety culture and regulation.
- Importance of transparency and international cooperation.

Legacy

- Nuclear safety reforms.
- Increased public awareness of nuclear risks.
- Ongoing research on nuclear accidents and environmental impact.

5.8 Disaster Management

5.8.1 Cyclone

- A cyclone is essentially a massive air mass that rotates around a low-pressure area, with winds spiraling inward.
- Imagine a giant coil of air that can bring severe weather, heavy rain, and strong winds. Cyclones can form over any body of warm air or water, and they're classified into different types based on their location and characteristics.

Types of Cyclones

- **Tropical Cyclones:** These form over warm ocean waters in the tropics and are fueled by heat released from condensation. Examples include hurricanes and typhoons.
- **Extra Tropical Cyclones:** These form outside the tropics and are associated with weather fronts. They're often weaker than tropical cyclones.
- **Subtropical Cyclones:** These have characteristics of both tropical and extra tropical cyclones.

Cyclones can have devastating effects on coastal regions, causing storm surges, flooding, and damage to infrastructure [1]. However, they also play a crucial role in maintaining global atmospheric circulation and can even relieve drought conditions.

Earthquakes are a fascinating and complex phenomenon that shape our planet's surface. Here's an overview:

5.8.2 Earthquake

An earthquake is a sudden release of energy in the Earth's lithosphere, causing seismic waves to propagate through the Earth's crust. This energy release creates ground shaking, sometimes violent enough to cause damage, tsunamis, and landslides.

Types of Earthquakes

- **Tectonic Earthquakes:** Caused by plate movement and stress buildup.
- **Volcanic Earthquakes:** Related to volcanic activity.
- **Collapse Earthquakes:** Result from underground caverns or voids collapsing.
- **Human-induced Earthquakes:** Caused by human activities like injection of fluids into the ground.

Earthquake Characteristics

- **Magnitude:** Measures earthquake size (Richter scale).
- **Intensity:** Measures earthquake effects on the Earth's surface (Mercalli scale).
- **Epicenter:** The point on the Earth's surface directly above the earthquake's focus.
- **Fault:** A fracture in the Earth's crust where earthquakes occur.

Earthquake Effects

- **Ground Shaking:** Causes damage to structures and infrastructure.
- **Tsunamis:** Ocean waves triggered by earthquakes.
- **Landslides and Soil Liquefaction:** Earthquake-induced ground instability.
- **Fire and Gas Leaks:** Secondary hazards.

Famous Earthquakes

- San Francisco (1906)
- Chile (1960)
- Alaska (1964)

- Japan (2011)
- Nepal (2015)

Earthquake Safety

- Prepare emergency kits.
- Practice earthquake drills.
- Secure heavy furniture and objects.
- Know evacuation routes.

Questions

1. What are causes, effects and control measures of air pollution?
2. What are causes, effects and control measures of water pollution?
3. Explain the role of an individual to prevent pollution?
4. Explain the effects and control measures of urban and industrial solid waste.
5. Write briefly about the unequal consumption patterns of the developed and developing countries?
6. Describe the sources, effects and methods to control Soil pollution?
7. What are the causes, effects and control measures of nuclear pollution?
8. What are the causes, effects and control measures of Noise pollution?
9. Explain the causes, effects and control measures of Thermal Pollution.
10. Explain the sources, effects and control measures of Marine Pollution.
11. Write a short note on i) Bhopal Gas Tragedy
    ii) Fukushima Daiichi Disaster.

Unit-6

Social Issues and the Environment

6.1 Urban Problems Related to Energy

Cities are the primary hubs for employment, trade, education, innovation, and economic growth. The vast majority of people lived in rural areas until recently, and their primary sources of income were farming, raising cattle, fishing, hunting, and other cottage industries. About 200 years ago, when the Industrial Era began, the cities began to rapidly develop. Approximately half of the world's population now resides in cities, and more and more people from rural areas are moving to cities in pursuit of work. It is getting harder to fit all of the commercial, residential, and industrial facilities inside a small municipal boundary due to the rapid urban growth. Urban sprawl is the term used to describe the resultant expansion of cities into rural or sub-urban areas. People in cities have far greater energy needs than people in rural areas. This is due to the higher standard of living enjoyed by urban dwellers, whose lifestyle necessitates greater energy inputs in all areas of life.

The energy demanding activities include:

- Residential and commercial lighting.
- Transportation means including automobiles and public transport for moving from residence to workplace.
- Modern life-style using a large number of electrical gadgets in everyday life.
- Industrial plants using a big proportion of energy.
- A large amount of waste generation which has to be disposed of properly using energy-based techniques.

- Control and prevention of air and water pollution which need energy dependent technologies.

Due to high population density and high energy demanding activities, the urban problems related to energy are much more magnified as compared to the rural population.

6.2 Water Conservation Techniques

- Drip Irrigation
- Decreasing Run Off Losses
- Contour Cultivation
- Bench Terracing
- Water Spreading
- Chemical Wetting agents
- Chemical Conditioners
- Reducing Evaporation Loss
- Reducing irrigation losses
- Increasing block pricing

6.3 Rainwater Harvesting

"Rainwater harvesting is a technique of increasing the recharge of ground water by capturing and storing rainwater".

This is done by constructing special water-harvesting structures like dug wells, percolation pits, lagoons, check dams etc.

Rainwater harvesting has the following objectives:

- To reduce run off loss.
- To avoid flooding of roads.
- To meet the increasing demands of water.
- To raise the water table by recharging ground water.
- To reduce groundwater contamination.
- To supplement groundwater supplies during lean season.

Traditional Rain Water Harvesting

- In India, it is an old practice in high rainfall areas to collect rainwater from roof-tops into storage tanks.
- In foot hills, water flowing from springs is collected by embankment type water storage.
- In Himalayan foot-hills people use the hollow bamboos as pipelines to transport the water of natural springs.
- Rajasthan is known for its tankas (under- ground tanks) and khadins (embankments) for harvesting rainwater.
- In our ancient times we had adequate Talaabs, Baawaris, Johars, Hauz etc. in every city, and village and capital cities of our kings and lords, which were used to collect rain-water and ensured adequate water supply in dry periods.

Modern Techniques of Rain Water Harvesting

- In arid and semi-arid regions artificial ground water recharging is done by constructing shallow percolation tanks.
- Check-dams are constructed for harvesting runoff from large catchment areas.
- Groundwater flow can be intercepted by building groundwater dams for storing water underground. As compared to surface dams, groundwater dams have several advantages like minimum evaporation loss, reduced chances of contamination etc.,
- In roof top rainwater harvesting, which is a low cost and effective technique for urban houses and buildings, the rain-water from the top of the roofs is diverted to some surface tank or pit through a delivery system which can be later used for several purposes. Also, it can be used to recharge underground aquifers by diverting the stored water to some abandoned dug-well or by using a hand pump (Fig.).

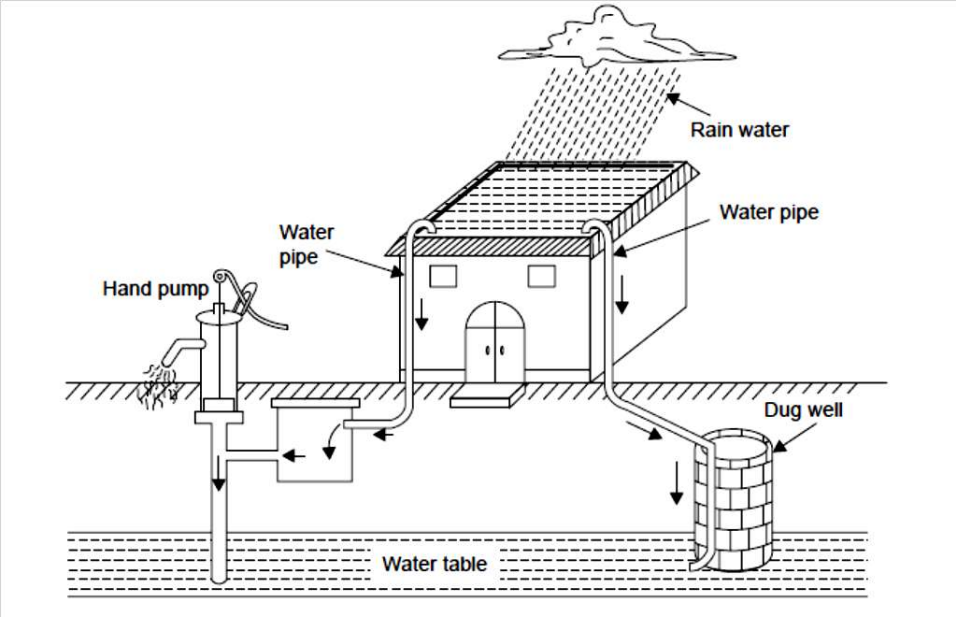


Figure 6.1: Systematic representation of rain water harvesting

6.4 Resettlement and Rehabilitation

Resettlement – Rehabilitation

A process by which people in any area are subjected to some natural disaster, they should be provided with temporary help, until they are settled in their environment.

Reasons for Displacement

- Natural disasters like earth quakes, cyclones, tsunamis, volcanic eruptions, prolonged drought conditions, floods etc.,
- Man-made disasters like industrial accidents, nuclear accidents, oil spills, toxic contamination of cities.
- Developmental projects like construction of dams, irrigation canals, reservoirs etc.
- Infrastructural projects like flyovers, bridges, roads etc.
- Energy related projects like power plants, oil exploration, mining activities etc.
- Projects related with the conservation of wildlife like national parks, sanctuaries and biosphere reserves etc.

(a) Displacement Due to Dams

- The Hirakund Dam has displaced more than 20,000 people residing in about 250 villages.
- The Bhakra Nangal Dam was constructed during 1950's and till now it has not been possible to rehabilitate even half of the displaced persons.
- Tehri Dam on the river Bhagirathi 10,000 residents of the Tehri town were to be displaced.

(b) Displacement due to Mining

- Mining is another developmental activity, which causes displacement of the native people. Several thousands of hectares of land area is covered in mining operation and the native people are displaced.
- Due to Jharia coal fields 0.3 million population has to be displaced.

(c) Displacement due to Creation of National Parks

- When some forest area is covered under a National Park, it is a welcome step for conservation of the natural resources. Where the entry of local dwellers or tribal's is prohibited.
- The wayanand wild life sanctuary in Kerala caused displacement of 53,472 tribal families.

Rehabilitation Issues

The major issues related to displacement rehabilitation as follows:

- **Little or no Support:** Tribal's are most affected among the displaced who are already poor. There is no support to affected people.
- **Meager Compensation:** The compensation for the land lost is often not paid or it is delayed or even if paid it is too small.

- **Loss of Livelihood:** The displaced people have to leave their ancestral land and forest on which they depend for their livelihood.
- **Lack of Facilities:** When the people are resettled in a new area, basic infrastructural and amenities are not provided. Very often temporary camps became permanent settlements.
- **Increase in Stress:** Resettlement disrupts the entire life style of the people and drives them towards stress.
- **Increase in Health Problems:** Lack of nutrition, due to the loss of agriculture and forest based livelihood lead to general decline in the health of people.
- Loss of identity and loss of intimate link between the people and the environment.
- Kinship systems, marriages, social and cultural functions, their folk-songs, dances and activities vanish with their displacement.
- Break up of families is an important social issue arising due to displacement in which the women are the worst affected.
- Resettlement increases the poverty of tribal due to loss of land, jobs, skills and food insecurity.

Rehabilitation Policy

There is a need for a comprehensive National Rehabilitation Policy. Different states are following different practices in this regard.

Case study

6.4.1 Sardar Sarovar Project

The Gujarat government is developing its policy regarding the Sardar Sarovar Project through a number of government resolutions. It has been determined that each landed outsee will have the right to be resettled on irrigable land in the state of his choice. The minimum amount of land granted to an outsee would be two hectares, and the

area of the land would be equivalent to that which he previously owned. However, there are issues with native people who were cultivating forest land and landless outsiders. There is no set deadline for determining whether a family has an adult son. Given that the adult son will be regarded as a distinct family, it is crucial.

The people of 20 submerged villages in Gujarat have been resettled at different locations leading to disintegration of joint families.

6.4.2 Pong Dam

Pong Dam's situation is distinct. In 1960, when Punjab was still a part of Himachal Pradesh, the dam was built on the Beas River. Rajasthan is irrigated by using the water. As a result, Rajasthan consented to give the displaced people land in the Indira Gandhi Canal's command area. However, another dam had to be constructed in order to transport Beas Water to Rajasthan, which resulted in the displacement of 20,722 additional families who Rajasthan had to relocate. Only 16,000 of the 30,000 families that were uprooted by the Pong dam were deemed eligible for allocation because they were the only legitimate farmers for whom 2.25 lakh acres of land had been set aside.It is unknown what became of the remaining 14,000 families. One of the dam's beneficiaries, Punjab, is completely unrelated to the rehabilitation problem. The only states attempting to resolve the issue are Rajasthan and Himachal Pradesh. Even those who have found homes are located in resettlement areas in Pakistan's desert, thousands of kilometres away from their hometown, severing their familial ties.

6.5 Environmental Ethics

Environmental ethics refers to the issues, principles and guidelines relating to human interactions with their environment.

It is rightly said, *The environmental crisis is an outward manifestation of the crisis of mind and spirit.*

It all depends on how do we think and act.

- If we think Man is all powerful and the supreme creature on this earth and man is the master of nature and can harness it at his will, it reflects our **human-centric thinking.**
- On the other hand, if we think, Nature has provided us with all the resources for leading a beautiful life and she nourishes us like a mother, we should respect her and nurture her., this is an **earth-centric thinking.**

Without much concern for the harm done to the planet Earth, the first viewpoint exhorts us to march forward triumphantly in order to subdue nature and establish our dominance over it through technological advancements, economic expansion, and development. According to the second point of view, we should live sustainably and as part of the planet as any other creation of nature. Thus, it is evident that our actions will reflect our thoughts.

When faced with important decisions, environmental ethics can help us make decisions and give us the rules for acting on our beliefs.

Some important ethical guidelines known as Earth ethics or Environmental Ethics are as follows:

- You should maintain standard environmental values.
- You should avoid deforestation.
- Non vegetarianism should be discouraged.
- Whenever it is possible people should grow plants.
- You should not kill animals for food.
- You should love and honour the earth since it has blessed you with life and governs your survival.
- You should keep each day sacred to earth and celebrate the turning of its seasons.
- You should not hold yourself above other living things and have no right to drive them to extinction.

- You should be grateful to the plants and animals which nourish you by giving you food.
- You should limit your offsprings because too many people will overburden the earth.
- You should not waste your resources on destructive weapons.
- You should educate the public about the maintenance of environment.

6.6 Environmental Legislations

India is the first nation in the world to include environmental conservation and protection clauses in its constitution. The U.N. Conference on the Human Environment in Stockholm, Sweden, first discussed the environment as a topic on the international agenda on June 5, 1972. Since then, June 5th has been observed globally as World Environment Day. The Water (Prevention and Control of Pollution) Act of 1974, the Forest (Conservation) Act of 1980, the Air (Prevention and Control of Pollution) Act of 1981, and finally the Environment (Protection) Act of 1986 were all passed after the Wildlife (Protection) Act of 1972.

Provisions of the Constitution

The following environmental protection provisions were added to the constitution in 1976, four years after the Stockholm Conference, through the 42nd amendment:

Article 48-A: The state shall Endeavour to protect and improve the environment and to safeguard forests and wildlife of the country.

Article 51A (g): It shall be the duty of every citizen of India to protect and improve the natural environment including forests, lakes, rivers and wildlife and to have compassion for living creatures.

Thus, our constitution includes environmental protection and conservation as one of our fundamental duties.

Some of the important Acts passed by the Government of India are discussed here.

6.6.1 The Air (Prevention and Control of Pollution) act, 1981

Salient features of the act are as follows:

- The Act provides for prevention, control and abatement of air pollution.
- In the Act, air pollution has been defined as the presence of any solid, liquid or gaseous substance (including noise) in the atmosphere in such concentration as may be or tend to be harmful to human beings or any other living creatures or plants or property or environment.
- Noise pollution has been inserted as pollution in the Act in 1987.
- Pollution control boards at the central or state level have the regulatory authority to implement the Air Act.
- Under section 17, the boards have to check whether the industry strictly follows the norms or not standards laid down by the Board, regarding the discharge of emission of any air pollutant. Based upon analysis report consent is granted or refused to the industry.
- Air Act has provisions for defining the constitution, powers and function of Pollution Control Boards, funds, accounts, audit, penalties and procedures.
- Section 20 of the Act has provision for ensuring emission standards from automobiles
- As per Section 19, in consultation with the State Pollution Control Board, the state government may declare an area within the state as air pollution control area and can prohibit the use of any fuel other than approved fuel in the area causing air pollution.

6.6.2 Water (Prevention and Control of Pollution) Act, 1974

Salient features of act:

- This act came into existence in 1974.
- It provides for maintaining and restoring the wholesomeness of water by preventing and controlling its pollution.
- This act gives definition to water pollution.
- According to act, water Pollution is defined as "such contamination of water, or such alteration of the physical, chemical or biological properties of water, or such discharge as is likely to cause a nuisance or render the water harmful or injurious to public health and safety or harmful for any other use or to aquatic plants and other organisms or animal life".
- It provides for maintenance and restoration of quality of all types of surface and ground water.
- It provides for the establishment of Central and State Boards for pollution control.
- It confers them with powers and functions to control pollution. The Central and State Pollution Control Boards are given comprehensive powers to advise, coordinate and provide technical assistance for prevention and control of pollution of water.
- The Act has provisions for funds, budgets, accounts and audit of the Central and State Pollution Control Boards.
- The Act makes provisions for various penalties for the defaulters and procedure for the same.

Central Pollution Control Board (CPCB)

- It advises the central govt. in matters related to prevention and control of water pollution.
- Coordinates the activities of State Pollution Control Boards and provides them technical assistance and guidance.

- Organizes training programs for prevention and control of pollution.
- Organizes comprehensive programs on pollution related issues through mass media.
- Collects, compiles and publishes technical and statistical data related to pollution.
- Prepares manuals for treatment and disposal of sewage and trade effluents.
- Lays down standards for water quality parameters.
- Plans nation-wide programs for prevention, control or abatement of pollution.
- Establishes and recognizes laboratories for analysis of water, sewage or trade effluent sample.

The **State Pollution Control Boards** also have similar functions to be executed at state level and are governed by the directions of CPCB.

- The Board advises the state govt. with respect to the location of any industry that might pollute a stream or a well.
- It lays down standards for effluents and is empowered to take samples from any stream, well or trade effluent or sewage passing through an industry.
- The State Board is empowered to take legal samples of trade effluent in accordance with the procedure laid down in the Act. If the samples do not conform to the prescribed water quality standards (crossing maximum permissible limits), then consent is refused to the unit.
- Every industry has to obtain consent from the Board (granted for a fixed duration) by applying on a prescribed Proforma providing all technical details, along with a prescribed fee following which analysis of the effluent is carried out.

- The Board suggests efficient methods for utilization, treatment and disposal of trade effluents.

6.6.3 Wildlife (Protection) Act, 1972

The act, a landmark in the history of wildlife legislation in our country, came into existence in 1972. Wildlife was transferred from State list to concurrent list in 1976, thus giving power to the Central Govt. to enact the legislation.

The Indian Board of Wildlife (IBWL) was took up the task of setting up wildlife National Parks and sanctuaries. The major activities and provisions in the act can be summed up as follows:

- It defines the wild-life related terminology.
- It provides for the appointment of wildlife advisory Board, Wildlife warden, their powers, duties etc.
- Under the Act, comprehensive listing of endangered wild life species was done for the first time and prohibition of hunting of the endangered species was mentioned.
- Protection to some endangered plants like Beddome cycad, Blue Vanda, Ladies Slipper Orchid, Pitcher plant etc. is also provided under the Act.
- The Act provides for setting up of National Parks, Wildlife Sanctuaries etc.
- The Act provides for the constitution of Central Zoo Authority.
- The Act imposes a ban on the trade or commerce in scheduled animals.
- It provides for legal powers to officers and punishment to offenders.
- It provides for captive breeding programme for endangered species.

- Several Conservation Projects for individual endangered species like lion (1972) Tiger (1973), Crocodile (1974) and Brown antlered Deer (1981) were started under this Act.
- The Act is adopted by all states in India except J & K, which has it own Act.

Some of the major drawbacks of the Act include mild penalty to offenders, illegal wild life trade in J & K, personal ownership certificate for animal articles like tiger and leopard skins, no coverage of foreign endangered wildlife, pitiable condition of wildlife in mobile zoos and little emphasis on protection of plant genetic resources.

6.6.4 Forest (Conservation) Act, 1980

The salient features of the Act are as follows:

- This act came in to existence in 1980.
- This act deals with the conservation of forests and related aspects.
- Except J & K, the act is adopted all over India.
- The Act covers under it all types of forests including reserved forests, protected forests or any forested land irrespective of its ownership.
- The State Govt. has been empowered under this Act to use the forests only for forestry purposes. If at all it wants to use it in any other way, it has to take prior approval of central Government.
- It makes provision for conservation of all types of forests and for this purpose there is an Advisory committee which recommends funding for it to the Central Government.
- Any illegal non-forest activity within a forest area can be immediately stopped under this Act. Non-forest activities include clearing of forest land for cultivation of any type of plants/crops or any other purpose (except re-afforestation).

Amendment in the Forest Act

In 1992, some amendment was made in the Act which made provisions for allowing some non-forest activities in forests, without cutting trees or limited cutting with prior approval of Central Govt.

- Wildlife sanctuaries, National Parks etc. are totally prohibited for any exploration or survey under this Act without prior approval of Central Govt. even if no tree-felling is involved.
- Cultivation of tea, coffee, spices, rubber and plants which are cash-crops, are included under non-forestry activity and not allowed in reserve forests.
- Even cultivation of fruit-bearing trees, oil-yielding plants or plants of medicinal value in forest area need to be first approved by the Central Govt.
- Plantation of mulberry for rearing silkworm is considered a non-forest activity.
- Mining is a non-forestry activity and prior approval of Central Govt. is mandatory.
- Removal of stones, bajri, boulder etc from river-beds located within the forest area fall under non-forest activity.

6.7 Public Environmental Awareness

Public awareness about environment is at a stage of infancy. Of late, some awareness has taken place related to environmental degradation, pollution etc. but incomplete knowledge and information and ignorance about many aspects has often led to misconceptions.

Some of the main reasons responsible for widespread environmental ignorance can be summed up as follows:

- Our courses in Science, technology, economics etc. have so far failed to integrate the knowledge in environmental aspects as an essential component of the curriculum.

- Our planners, decision-makers, politicians and administrators have not been trained so as to consider the environmental aspects associated with their plans.
- In zeal to go ahead with some ambitious development projects, quite often there is purposeful concealment of information about environmental aspects.
- There is greater consideration of economic gains and issues related to eliminating poverty by providing employment that overshadows the basic environmental issues.

6.7.1 Methods to Propagate Environmental Awareness

Environmental awareness needs to be created through formal and informal education to all sections of the society. Everyone needs to understand it because environment belongs to all and every individual matters. When it comes to conservation and protection of environment.

Various stages and methods that can be useful for raising environmental awareness in different sections of the society are as follows:

- **Among Students through Education:** Environmental education must be imparted to the student's right from the childhood stage. It is a welcome step that now all over the country we are introducing environmental studies as a subject at all stages including school and college level, following the directives of the Supreme Court.
- **Among the Masses through Mass-Media:** Media can play an important role to educate the masses on environmental issues through articles, environmental rallies, plantation campaigns, street plays, real eco-disaster stories and success stories of conservation efforts. TV serials like Virasat, Race to save the Planet, Heads and Tails, Terra-view, Captain Planet etc.

- **Among the Planners, Decision-Makers and Leaders:** Since this elite section of the society plays the most important role in shaping the future of the society, it is very important to give them the necessary orientation and training through specially organized workshops and training programmes.

Some other approaches are:

- Door to door contact with concerned parties.
- Community group meetings
- Distribution of pamphlets and flyers
- Talk shows through media
- Publicity through radio, TV and newspaper
- Short movies
- Local plays etc.,

6.8 Global Environmental Challenges

6.8.1 Climate Change

Climate is an area's average weather. It refers to a region's overall weather, seasonal variations, and weather extremes. The climate has changed in the last few years. The delicate balance that has developed between the different elements of the environment is being upset by man-made activities. The average global temperature is rising as a result of an increase in greenhouse gases in the atmosphere. This could disrupt the hydrological cycle, lead to droughts and floods in various parts of the world, raise sea levels, and alter agricultural productivity. Agriculture may be disrupted by even minor changes in the climate, which could cause animal migration, including human migration.

6.8.2 Global Warming

The earth's surface absorbs about 75% of the solar energy that reaches it, raising the planet's temperature. The remaining heat returns to the atmosphere through radiation. Gases like carbon dioxide, methane, and ozone, as well as water vapour and chlorofluorocarbon compounds (CFCs), trap part of the heat. As a result, they contribute to atmospheric heating. Global warming results from this. The main cause of global warming is carbon dioxide. Methane, water vapour, nitrous oxide, CFCs, and ozone are additional greenhouse gases in addition to carbon dioxide. When plants burn, digest, or decompose without oxygen, methane is naturally produced. Paddy fields, coal mines, decaying landfills, and fossil fuels all emit significant amounts of methane. Man-made industrial chemicals called chlorofluorocarbons (CFCs) are used in air conditioners and other devices. The environment naturally contains nitrous oxide. Because of the use of chemical fertilisers and the burning of fossil fuels, their numbers have dramatically increased in recent years.

Consequences

- Temperature will increase to a level which may lead to melting of polar ice caps and flooding of low-lying areas all over the earth.
- Increasing the global temperature increases the incidence of infectious diseases like dengue, malaria, yellow fever, sleeping sickness etc.
- Soil moisture will decrease and evaporation will increase, which may drastically affect wheat and maize production.

Measures to Check

- Reduce the use of CFCs and fossil fuel.
- Use energy more efficiently.
- Plant more trees.
- Stabilize population growth.
- Adopt sustainable agriculture.

6.8.3 Ozone Layer Depletion

The ozone layer is a thin layer in the atmosphere made up of oxygen atoms O3 that absorbs harmful UV-radiation from reaching the earth surface. The ozone is being depleted by chemicals released in to the atmosphere like chlorofluorocarbons (CFCs), carbon tetra fluoride, methyl chloroforms and chlorofluoromethanes. Several pollutants attack the ozone layer. However, chlorofluorocarbons (CFCs) are known to be chief among them. CFC molecules are stable, virtually indestructible until they reach the stratosphere. Here, the CFC molecules are broken apart by intense ultraviolet radiation and release the chlorine atoms they contain. These chlorine atoms react with ozone and break it down into ordinary oxygen molecules that do not absorb UV-radiation.

$CF_2Cl_2 + hv \rightarrow Cl^* + CF_2Cl^*$

$Cl^* + O_3 \rightarrow ClO^* + O_2$

$ClO^* + O \rightarrow Cl^* + O_2$

Net reaction: $2O_3 \rightarrow 3O_2$

Effects of Ozone Depletion

- Ozone depletion will result in more UV radiation reaching the earth especially UV-B (290-320nm) and these radiations affect DNA and the photo synthetic chemicals. Any change in DNA can result in mutation and cancer.
- Easy absorption of UV rays by the lens and cornea of eye will cause cataract.
- Phytoplanktons are sensitive to UV exposure.
- Yield of vital crops like corn, rice, soybean, cotton, bean, pea and wheat will decrease.
- Degradation of paints, plastics and other polymer material will result.

6.8.4 Acid Rains

- Oxides of sulfur and nitrogen originating from the industrial operations and fossil fuel combustion are the major sources of acid forming gases.
- Acid forming gases are oxidized by which time they travel thousand kilometers. In the atmosphere these gases are ultimately converted in to sulfuric and nitric acids. These acids cause acid rain.
- Acid rain is only one component of acidic deposition. Acidic deposition is the total of wet acidic deposition (acid rain) and dry acidic deposition.

  $SO_2 + H_2O \rightarrow H_2SO_4$

  $NO_2 + H_2O \rightarrow HNO_3$

Impact of Acid Rain on Aquatic Environment

- Direct exposure to acid may damage some fish species: e.g. solmarid group species.
- Acid rains leached down the toxic metals in to the water bodies. These toxic metals break down the metabolic system in the fishes.
- These toxic metals are also deposited in the gills and effect respiratory system and then entering into food chain and food webs.
- Normally phytoplankton's disappears at the pH between 5.8-6.0.

Effect on Human Beings

- Human nervous system, respiratory system and digestive system are affected by acid rains.
- It also causes the premature death from heart and lung disorders such as asthma and bronchitis.

Effect on Buildings

- Acid rain corrodes houses, monuments, statues, bridges and fences.
- Taj-Mahal in Agra suffering at present due to SO_2 and H_2SO_4 acid fumes released from Mathura refinery.

6.9 Consumerism and Waste Products

Consumerism refers to the consumption of resources by the people. While early human societies used to consume much less resources, with the dawn of industrial era, consumerism has shown an exponential rise. It has been related both to the increase in the population size as well as increase in our demands due to change in life-style.

Two types of conditions of population and consumerism exist.

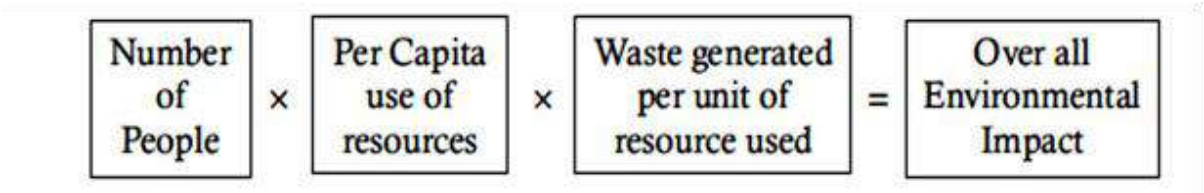
(i) People Over-Population

It happens when the population of an area exceeds the supply of food, water, and other essential resources. The limited resources are deteriorated by excessive population pressure, leading to extreme poverty, malnutrition, and early mortality. Less developed nations (LDCs) are where this happens. Because there are so many people here, not everyone has access to enough resources. Therefore, even though overall consumption is high, per capita consumption is lower.

(ii) Consumption Over-Population

In the more developed nations (MDCs), this takes place. Although there are more resources available, the population is smaller, and because of the opulent lifestyle, per capita resource consumption is very high. The more resources are used, the more waste is produced, and the worse the environment gets.

This concept can be explained by using the model of
Paul Ehrlich and John Hodlren (1972):



In LDC.s - No. of people is very high, but per capita use of resources and waste generated are less.

In MDC.s - No. of people is low, but per capita use of resources and wastes generated are very high.

The overall environmental impact of these two types of consumerism may be same or even greater in case of MDC.s.

Thus, consumerism varies with the country and USA is known for maximum consumerism.

6.10 Wasteland Reclamation

Wasteland reclamation, also known as land reclamation or land rehabilitation, is the process of restoring degraded or contaminated land to a usable or productive state.

Types of Wasteland Reclamation

1. **Physical Reclamation:** Involves physical removal of debris, filling of low-lying areas, and grading of land.
2. **Biological Reclamation:** Uses plants and microorganisms to stabilize and improve soil quality.
3. **Chemical Reclamation:** Involves application of chemicals to neutralize or remove pollutants.

Methods of Wasteland Reclamation

1. Excavation and removal of contaminants
2. Soil amendment and fertilization
3. Vegetation establishment (e.g., afforestation, grassland restoration)
4. Hydrological restoration (e.g., wetland creation)
5. Engineering solutions (e.g., containment walls, drainage systems)

Benefits of Wasteland Reclamation

1. Improved environmental quality
2. Increased land value
3. Enhanced biodiversity
4. Reduced pollution
5. Creation of recreational or agricultural land

Examples of Successful Wasteland Reclamation Projects

1. Chicago's Lake Calumet Area (USA) - transformed industrial wasteland into wildlife habitat
2. London's Olympic Park (UK) - reclaimed contaminated industrial site for recreational use
3. Singapore's Jurong Island (Singapore) - transformed mangrove swamp into industrial hub
4. India's Navi Mumbai (India) - reclaimed coastal wetlands for urban development

Challenges and Limitations

1. High costs
2. Technical complexities
3. Regulatory frameworks
4. Community engagement and participation
5. Long-term monitoring and maintenance

Technologies Used in Wasteland Reclamation

1. Geographic Information Systems (GIS)
2. Remote Sensing (RS)
3. Drone technology
4. Bioremediation
5. Phytoremediation

Best Practices

1. Community involvement and engagement
2. Integrated planning and management
3. Monitoring and evaluation
4. Use of local and native species
5. Adaptation to climate change

6.11 Major Nuclear Accidents

Nuclear accidents can have devastating consequences for human health, the environment, and the economy.

1. **Chernobyl (1986, Ukraine):** Worst nuclear power plant accident in history, releasing radioactive materials into the environment.
2. **Fukushima Daiichi (2011, Japan):** Tsunami-caused meltdown, releasing radioactive materials into the ocean.
3. **Three Mile Island (1979, USA):** Partial meltdown, minimal radioactive release.
4. **Windscale (1957, UK):** Fire at a nuclear power plant, releasing radioactive material.
5. **SL-1 (1961, USA):** Experimental nuclear reactor accident, killing three operators.

Types of Nuclear Accidents

1. Core meltdown
2. Radiation exposure

3. Radioactive release
4. Fuel cycle accidents
5. Waste management accidents

Causes of Nuclear Accidents

1. Human error
2. Design flaws
3. Equipment failure
4. Natural disasters (e.g., earthquakes, tsunamis)
5. Sabotage or terrorism

Consequences of Nuclear Accidents

1. Radiation exposure to humans and environment
2. Health effects (cancer, genetic damage)
3. Environmental contamination
4. Economic costs (cleanup, compensation)
5. Psychological impact on affected communities

Safety Measures

1. Robust design and construction
2. Regular maintenance and inspection
3. Operator training and safety culture
4. Emergency preparedness and response planning
5. International cooperation and regulation

International Response

1. International Atomic Energy Agency (IAEA)
2. World Association of Nuclear Operators (WANO)
3. Nuclear Energy Agency (NEA)
4. Convention on Nuclear Safety (CNS)

Recent Incidents

1. **2022:** Radiation leak at Taishan Nuclear Power Plant, China
2. **2020:** Fire at Ukraine's Zaporizhzhia Nuclear Power Plant
3. **2019:** Radiation release at Russia's Kuznetsov Nuclear Power Plant

6.12 Holocaust

The Environmental Holocaust, also known as Ecocide or Environmental Genocide, refers to the destructive impact of human activities on the natural world, leading to the degradation or destruction of ecosystems, biodiversity loss, and harm to human health.

Types of Environmental Holocaust

1. Deforestation and land degradation
2. Climate change and global warming
3. Pollution (air, water, soil)
4. Overexploitation of resources (water, minerals, fossil fuels)
5. Loss of biodiversity and species extinction
6. Chemical contamination and toxic waste
7. Nuclear accidents and radioactive pollution

Examples

1. Chernobyl Nuclear Disaster (1986)
2. Exxon Valdez Oil Spill (1989)
3. Bhopal Gas Tragedy (1984)
4. Amazon Rainforest Deforestation
5. Great Barrier Reef Bleaching
6. Flint Water Crisis (2014)
7. Ok Tedi Environmental Disaster (Papua New Guinea)

Causes

1. Human population growth and consumption
2. Industrialization and technological advancements
3. Corporate greed and profit-driven policies
4. Lack of regulations and enforcement
2. Climate change denial and skepticism

Consequences

1. Human health impacts (respiratory diseases, cancer)
2. Ecosystem disruption and collapse
3. Loss of ecosystem services (clean air, water, food)
4. Economic costs (healthcare, environmental remediation)
5. Social and cultural impacts (displacement, community disruption)

Solutions

1. Sustainable development and resource management
2. Renewable energy and energy efficiency
3. Waste reduction and circular economy
4. Conservation and restoration efforts
5. Climate change mitigation and adaptation
6. Environmental education and awareness
7. Policy reforms and international cooperation

Questions

1. What are the major issues associated with resettlement and rehabilitation?
2. Illustrate the benefits of using rainwater harvesting for irrigation.
3. Examine the relationship between water conservation and energy savings.
4. Discuss the importance of ethics in decision making.
5. Explain the salient features of Wild life act.
6. Explain the role of state and central pollution control boards according to the EPA act?
7. Write about forest (prevention and control of pollution) act.

8. Write about water (prevention and control of pollution) act.
9. Write about air (prevention and control of pollution) act.
10. Write about Global Environmental Challenges.
11. Explain the primary causes and consequences of Holocaust.
12. How do nuclear accidents impact the environment.

Unit-7

Human Population and the Environment

7.1 Population Growth

In year 1800 the earth was home to about 1 billion people. There after a drastic change in human population growth. With scientific and technological advancement, life expectancy of humans improved. People started living in definite settlements leading a more stable life with better sanitation, food and medical facilities. It is about 90-95% of total population growth of the world in the last 50 years. The population growth takes the exponential increase.

Population Explosion

The enormous increase in population due to low death rate and high birth rate.

Causes of Population Explosion

- The rapid population growth is due to decrease in death rate and increase in birth rate.
- Availability of antibiotics, immunization, increased food production, clean water and air decreases the famine-related deaths.
- In agriculturally based countries; children are required to help parents in the field, that is why population increases in the developing countries.
- Modern medical facilities, life expectancy, illiteracy.

Effects of Population Explosion

- It can lead to depletion of resources.
- Severe competition for food and space.
- Increase in psychological stress and strain.
- Rapid pollution of environment.
- Large scale unemployment.
- Poverty, Environmental degradation, over –exploitation of natural resources, threats.
- Communal war.

Remedy

- Through birth control programs

7.2 Role of Information Technology in Environment

1. **Remote Sensing:** Remote sensing is a science of obtaining information about objects or areas from distance typically from aircraft or satellite.

   Components - A platform, aircraft, a balloon, rocket and satellite.

   Applications
   In agriculture, forestry, land cover, water resources.

2. **Data Base:** Collection of inter related data on various subjects.
   Applications
   - Ministry of environment and forest
   - National management information system
   - Environmental information system

3. **Geographical Information System (GIS):**It is a technique of superimposing various thematic maps using digital data on a large number of inter-related aspects.

Application

- Thematic maps are super imposed using soft ware's.
- Interpretation of polluted zones.
- To check unplanned growth and related environmental problems.

4. **Satellite data**
   - Helps in providing reliable information and data about forest cover.
   - Provide information about forecasting weather.
   - Reserves of oil, minerals can be discovered.

5. **World Wide Web:** It provides Current data.

Applications

- Online learning.
- Digital files or photos, animations on environmental studies.

7.3 Role of Information Technology in Human Health

The health service technology involves three systems

Finance and accounting

- Pathology.
- Patient Administration – clinical system.

Applications

- Data regarding birth and death rates.
- To monitor the health of the people effectively.
- The information regarding the outbreak of epidemic diseases.
- Online Consultation.
- Drugs and its replacement.

7.4 Family Welfare Program

- The family welfare program (FWP) is a national health program in India that aims to improve the health of mothers and children, reduce fertility, and stabilize population growth.

History of Family Welfare Program

- It was started in the year 1951.
- India is the first country in the world that implemented the Family Welfare Program at government level.
- India was the first country in the world to have launched a National Program for Family Planning in 1952.
- The FWP is a voluntary program that is implemented through a Community Needs Assessment Approach (C.N.A.A.).
- Government introduced this program in various five year plans.
- It was decided in national health policy 1983, that Net Reproductive Rate (NRR) should be 1 by the year 2000.

The Family Welfare Program (FWP) in India has several aims and objectives, including:

- **Population Stabilization:** The FWP aims to reduce the birth rate, avoid unwanted births, and regulate the interval between pregnancies.
- **Maternal and Child Health:** The FWP aims to improve maternal and child health services, including education on safe delivery and post-delivery care.
- **Disease Elimination:** The FWP aims to eliminate diseases like polio and leprosy.
- **Healthcare:** The FWP aims to improve hospital services and train doctors, nurses, and other paramedical staff.
- **Immunization:** The FWP aims to provide healthcare for infants' immunization against preventable diseases.

- **Contraceptive Use:** The FWP aims to increase the use of contraceptives among eligible couples.

The family welfare program in India faces a number of challenges, including:

- **Early Marriage and Childbearing:** Many women in India marry before the legal minimum age of 18, and many adolescents become mothers.
- **Limited Progress on Reproductive Health:** India's infant mortality rate is high, and maternal mortality is also a concern.
- **Mismatch between Infrastructure and Resource Flow:** Some populous states fail to utilize resources, while others have higher allocations than they need.
- **Awareness (Poor):** Inadequate awareness about family welfare program.
- **Infanticide (Girl Child):** Due to Female Infanticide, male – female sex ratio has reached an alarming stage at the national as well as state levels.

The environment can have a significant impact on human health, both positively and negatively:

- **Negative Impacts:** Environmental hazards like air and water pollution, exposure to toxic by-products, and climate change can increase the risk of chronic diseases, disability, and death. For example, air pollution can cause lung and heart disease, while noise pollution can lead to hearing loss and cardiovascular disease. Climate change can cause heatwaves, extreme weather events, and prolonged pollen seasons, which can increase the spread of infectious diseases.
- **Positive Impacts:** Spending time in nature can have health benefits, such as increased self-awareness, feelings of awe and wonder, and a sense of connection to nature. The relationship between human health and the environment is complex and dynamic. Some activities that promote human health, like

agriculture and energy production, can also have negative impacts on the environment.

Some protective actions that can be taken to improve environmental health include:

- Isolation, such as prohibiting people from entering certain areas
- Shielding, such as using physical shields or protective clothing
- Limiting the time people spend in hazardous areas
- Treatment, such as treating environmental media to reduce or eliminate toxic agents
- Prevention strategies, such as vaccination or eating antioxidants.

7.5 Human Rights

Human rights are a set of principles that establish the standards for how people should be treated and what they are entitled to as human beings:

- They are inherent to all people, regardless of their race, sex, nationality, ethnicity, language, religion, or any other status.
- They include the right to life, liberty, freedom from slavery and torture, freedom of opinion and expression, the right to work and education, and more.
- They are about living a life free from fear, harassment, or discrimination.
- They govern how people live in society and with each other, as well as their relationship with the state.
- They oblige governments to do some things, and prevent them from doing others.
- Individuals also have responsibilities to respect the rights of others.
- The Universal Declaration of Human Rights (UDHR) was adopted by the UN General Assembly in 1948 and is the

foundation of all international human rights law. The UDHR, along with the International Covenant for Civil and Political Rights, and the International Covenant for Economic, Social and Cultural Rights, make up the International Bill of Rights.

Some of the important articles of the universal Declaration of Human Rights are

Article no	**Article**
1	Right to Equality
2	Freedom from Discrimination
3	Right to Life, Liberty, Personal Security
4	Freedom from Slavery
5	Freedom from Torture and Degrading Treatment
6	Right to Recognition as a Person before the Law
7	Right to Equality before the Law
8	Right to Remedy by Competent Tribunal
9	Freedom from Arbitrary Arrest and Exile
10	Right to Fair Public Hearing
11	Right to be Considered Innocent until Proven Guilty
12	Freedom from Interference with Privacy, Family, Home and Correspondence
13	Right to Free Movement in and out of the Country
14	Right to Asylum in other Countries from Persecution
15	Right to a Nationality and the Freedom to Change It
16	Right to Marriage and Family
17	Right to Own Property
18	Freedom of Belief and Religion
19	Freedom of Opinion and Information
20	Right of Peaceful Assembly and Association
21	Right to Participate in Government and in Free Elections
22	Right to Social Security
23	Right to Desirable Work and to Join Trade Unions
24	Right to Rest and Leisure
25	Right to Adequate Living Standard
26	Right to Education

27	Right to Participate in the Cultural Life of Community
28	Right to a Social Order that Articulates this Document
29	Community Duties Essential to Free and Full Development
30	Freedom from State or Personal Interference in the above Rights

7.6 Value Education

Value education is the process of teaching moral and political values, as well as the skills and dispositions that come with them, to students. It's a key part of education that helps students make informed choices and develop the values of equality, democracy, freedom, and human rights.

Value education can take place in many different settings, including schools, families, universities, businesses, and sports.

Needs of Value Education

- Moral Development
- Culture Development
- Social Development
- Cooperative living
- Development of democratic qualities

Some ways to teach values include:

Activities: Organizing tours, visits, and activities that help students develop self-confidence, self-respect, and team spirit.

Social Service: Organizing social service camps where students can help clean, distribute food, and help accident victims.

Games: Playing sports, Scouting and Guiding, NSS, NCC, and Yoga to help students develop discipline, punctuality, and cooperation.

Advantages of Value Education

- Healthy and balanced Personality
- Ideal citizens
- Reconstruction of Experiences
- Adjustable Environment
- Promotes Social efficiency
- Develops cultural values

Value education is important because it helps students develop a value system that they can use to make choices and lead a full life. It also helps students understand their needs and visualize their goals, and can help remove confusions and contradictions.

7.7 HIV / AIDS

Human Immunodeficiency Virus (HIV): This is the virus that causes AIDS. HIV attacks the immune system, which fights all infections, making it unable to fend off other illnesses.

Acquired Immune Deficiency Syndrome (AIDS): Is a deadly disease that interferes with the body's natural ability to fight infection.

Steps of Disease

1. **HIV:** From infected person. May not show signs and if they do it will be flu-like (night sweats, weight loss). Unless tested, person probably will not know they are HIV positive. You can still infect another person.
2. **AIDS:** T-cell (fighter cells) have a count of 200 or less so immune system cannot fight off infections so now you can get an one and die from it.
3. **Death:** Due to opportunistic infections (pneumonia).

Tests for HIV

1. Elisa Test
2. Western Blot

CURE FOR HIV/AIDS: There is NOT a cure for this disease. Once a person has HIV, they will have it FOREVER.

HOW HIV IS SPREAD: Can be passed from one person to another ONLY by fluids.

HIV cannot survive outside the human body.

Fluids that pass HIV include:

- Blood (using the same drug needle as infected person, transfusions from infected person)
- Sexual Fluids (Semen and Vaginal Secretions from sexual relations with an infected person).
- Breast Milk (Mother with HIV breastfeeding her child)
- Prenatally (Mother to Fetus during Pregnancy at times can pass disease).

HIV is Not Spread through

- Through air.
- By kissing.
- Casual contact (hugging)
- Mosquitoes.
- Sharing eating utensils.
- Donating blood.

Prevention of HIV from Spreading

- Avoid all sexual contact!!! ABSTINENCE!
- Avoid illegal drugs (especially those that need needles)
- Universal Precautions (gloves when touching blood)

7.8 Women and Child Welfare

Women and child are usually soft, who suffer in a number of ways mainly because they are weak, helpless and economically dependent.

Women Welfare: The main aim of women welfare is to improve the status of the women by providing opportunities in education, employment and economic independence.

Need of Women Welfare

- In general, women suffer gender discrimination and devaluation at home, at work place, in matrimony, in public life.
- High number of cases of dowry deaths, rape, domestic violence.
- The Human rights of women are violated, in the male dominated society.
- Generally, in policy making and decision-making process, women are neglected.

Objectives of Women Welfare

- To provide education.
- To impart Vocational training.
- To generate awareness about the environment.
- To improve the employment opportunities.
- To aware problems of Population.
- To restore the dignity, status, equality and respect for women.

Schemes of various organizations towards women welfare

The National Network for Women and Mining (NNWM): It is fighting for a "gender Audit" of India's mining Company.

United Nations Decade for Women: It is witnessed inclusion of several women welfare related issues on international agenda.

Ministry of Women And Child Development: It aims to work for the upliftment of women by family planning, health care, education and awareness.

Child Welfare

Children occupy nearly 40% of total population. 20 Million children in our country are estimated to be working as child labors in various industries like match industry…

Reasons for Child Labours

- Poverty
- Low Family Income
- Schemes of various organizations towards women welfare

Questions

1. Give reasons for overpopulation and mention problems created by Overpopulation.
2. Discuss the impact of population, affluence and technology on environment with suitable examples.
3. Discuss briefly about Population Explosion.
4. "Environment Education can play an important role in environmental protection". Explain it.
5. Briefly discuss the various schemes launched for Women's education in India.
6. What is meant by value education? Discuss their concepts with the help of suitable illustrations.
7. Discuss the Role of IT on human health and Environment.
8. Analyze the effectiveness of existing Women's welfare policies.
9. Evaluate the impact of early childhood education policies.
10. Explain the primary objectives of Family welfare programmes.
11. Discuss the relationship between HIV/AIDS and socioeconomic factors.

Unit-8

Environmental Management

The importance of impact assessment different phases of environmental audit, ecotourism, EMP and EIS preparation, and EIA.

Definition of Impact: An impact can be defined as any change in physical, chemical, biological, cultural or socio-economic environmental system as a result of developmental activities

8.1 Environmental Impact Assessment (EIA)

EIA , it is a procedure to plan some developmental activity with well-defined environmental goals so that damage due to the activity both during developmental stage and production stage have minimum impacts on the natural system and the population in the area.

Goals of EIA

- To fulfill the responsibilities towards the coming generation as trustees of environment.
- To assure safe, healthy, productive, aesthetically as well as culturally pleasing surroundings.
- To provide wide range of beneficial use of environment without degradation or risk to health.
- To preserve historical, cultural and natural heritage.
- To achieve a balance between population and resource use for good standard of living.
- To ensure sustainable development with minimal environmental degradation.

EIA Methodology

Basic Steps in EIA: screening, scoping, baseline data, impact identification, evaluation, mitigation, EIS preparation, review and environmental audit, involving public participation at various stages.

1. **Screening:** It is done to see whether the project needs an EIA for clearance or not.
2. **Scoping:** It involves determination of extent of EIA required for the project. Depending on project it is two types.
   - **Rapid EIA:** when the EIA report is based on a single seasonal data (other than monsoon period) it is called rapid EIA.
   - **Comprehensive EIA:** When the EIA report is based on detailed seasonal data, it is called comprehensive EIA.
3. **Baseline Data:** It is an overall study of environmental setting of project location.

   The following environmental parameters are usually considered while preparing the baseline data.

   - Site location and topography
   - Regional demography, land use and water use pattern
   - Regional landmarks like historical and cultural heritage in the area
   - Geographical parameters like land form, soil type, land scape etc.,
   - Hydrological parameters like ground water and surface water resources, water quality, pollution sources etc.,
   - Meteorological parameters like temperature extremes, wind speed and direction, dew point, rain fall etc.,
   - Ecology-the flora, fauna, endangered species, successional stage etc.,

Baseline data is collected by various methods. They are,

i. **Primary Data Collection:** This can be done by actual visit to the area and collection of data by survey method. Questionnaire method can be used to collect data on socio-economic aspects, occurrence of some diseases, information on natural, cultural or historical heritage etc.,

ii. **Secondary Data Collection:** Various govt. agencies and boards have the relevant data available with them

- Meteorological stations provide data on annual rainfall, humidity, temperature, evaporation, soil temperature, wind speed and wind direction.
- Agricultural department can make available the data on cropping patterns and irrigation
- Archaeological departments can provide us the required data on any historical or cultural heritage of area.
- Water board can provide data on ground water level and surface water resources
- Senses department provides information on population.

iii. **Use of Remote Sensing and GIS for Data Acquiring:** Satellite data available through remote sensing and use of geographical information system are very useful modern techniques for acquisition of baseline data on vegetation, land use, land cover, water bodies, deforestation etc.,

4. **Impact Identification:** It includes the details of project characters and baseline data to ensure the identification of full range impacts. During the identification process, the positive and negative, direct and indirect, significant and insignificant impacts are considered.
5. **Impact Prediction:** Here magnitude of changes going to occur due to the project is predicted by using mathematical and mass balance models.
6. **Impact Evaluation:** Impact evaluation is done by considering cost and benefits of the project.

7. **Mitigation Measures:** Once the impacts are predicted and evaluated, mitigation measures are to be suggested to avoid, reduce or rectify the adverse changes due to project.

   Review and Draft Impact Statement is Prepared at This Stage

8. **Decision Analysis (Review):** Public participation is involved by arranging group discussions or by adopting questionnaire method to arrive at a decision about the project.

9. **Environmental Impact Statement (EIS):** Based on the data obtained and review suggestions a final EIS is prepared as per the format provided by the ministry of environment and forests in our country.

   The EIS clearly mentions the objectives of the project, its environmental impacts, impacts that are in avoidable, mitigation measures to minimize the impacts, alternatives to the proposed action etc.,

10. **Environmental Audit:** It compares the impacts predicted in EIS before the project was started and actual impacts after implementation of the project.

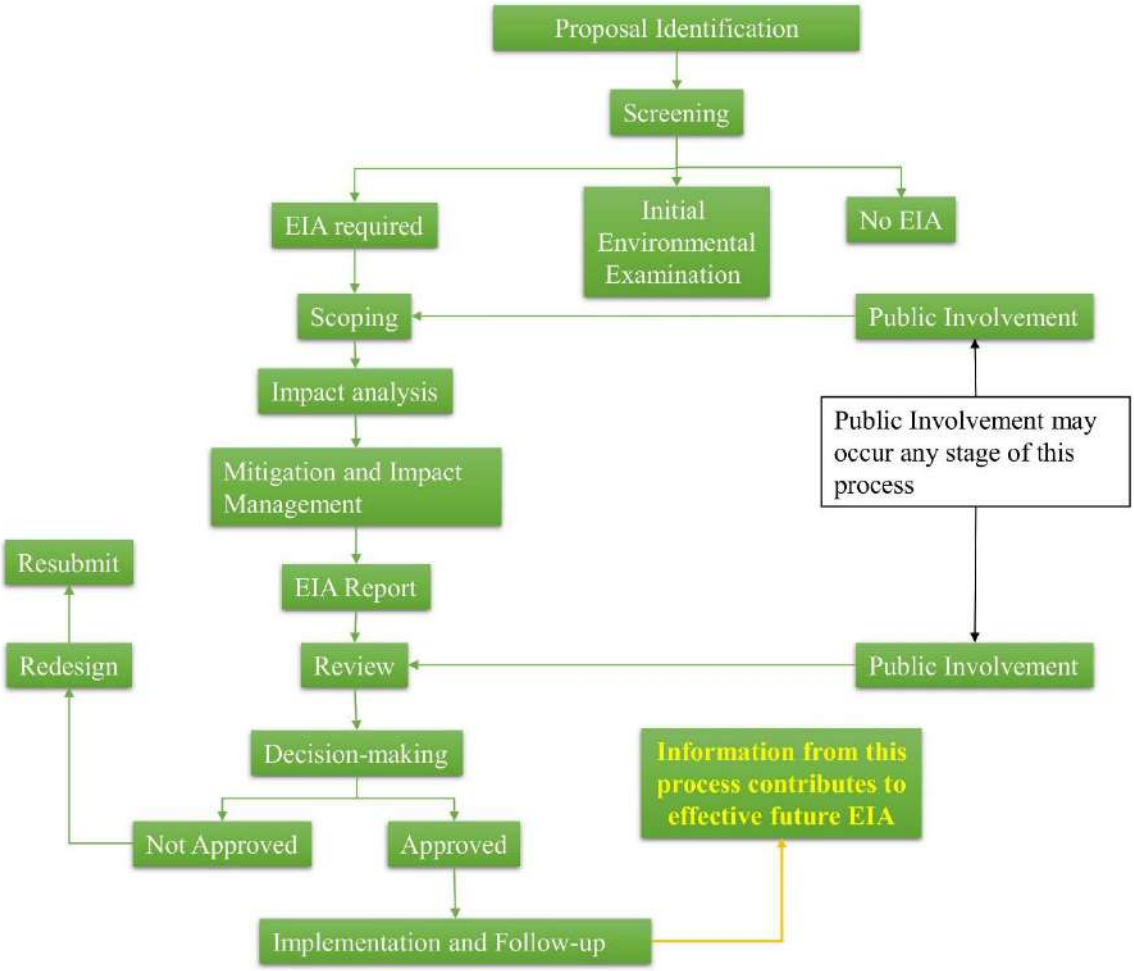


Figure 8.1: EIA Methodology flow-chart

Why EIA important?

Countries can better accomplish sustainable development goals, prevent negative effects on the environment, society, and culture, lower costs, improve infrastructure planning, and more by identifying potential alternatives and negative effects.

8.2 Environmental Impact Statement

The EIS is prepared by the project proponents at the time of submission of the proposal which is known as draft EIS. After evaluation and review by the impact assessment agency, the final EIS is prepared.

EIS Includes Following Aspects

- Effect on land including land degradation & subsistence
- Deforestation and compensatory afforestation
- Air pollution along with possible health effects
- Water pollution including surface and ground water pollution
- Noise pollution due to project
- Loss of flora and fauna
- Socio-economic impacts including displacement of people, cultural loss and health aspects.
- Risk analysis and disaster management
- Recycling and reduction of waste
- Efficient use of energy and matter.



Figure 8.2: EIS flow-chart

Hence an Environmental Impact Statement (EIS) which is a summary of the project is kept for the public to read.

8.3 Environmental Management Plan

Environmental Management Plan (EMP) is aimed to maintain the existing environmental quality.

Management plan is prepared so as to minimize the negative impacts, increase the positive impacts and restore the damage caused by the project.

The main objective of EMP is to investigate specific activities which are related to adverse impacts. The impacts can be first minimized by various planning activities.

Some more measures can be practiced to minimize the impacts on environment are as follows:

- **Technological Solutions:** there are technological solutions for the many adverse impacts caused by the various developmental activities.

  Examples: cleaner technologies are emerging that produce much less pollutants in the environment.

- **Preventive Methods:** in order to prevent or avoid damage to the environment by a developmental activity, the harmful hazardous substances produced should be handled properly.

  Examples: To prevent the loss of habitat of some wildlife due to a project, we have to migrate them to another nearby forest

  Examples: To prevent the loss of top fertile layer of soil during mining, layer by layer dumping of the soil is done during digging which are later restored.

- **Control Methods:** to control the damage due to various emissions coming from developmental projects like power plants, industries, stone-crushing units various control method are available.

  Examples: particulate matter can be controlled by using cyclone separators, bag house filters, wet scrubbers etc.,

Examples: Boundary walls around the stone crushing units and use of frequent water spray prevent the particulate emissions moving to nearby area.

- **Treatment Methods:** to convert the harmful substances produced by various industrial projects or to reduce their concentrations to safe permissible limits various treatment methods are available.
- **Green Belt Development:** Tree and shrubs are known to act as sinks of many toxic gases and particulates. They also absorb CO_2 and release O_2 thus purifies the air. Therefore, development of green belt around development site has been mandatory. Green belt development also has a special importance in hydroelectric projects as the project construction process produces lot of dust due to excavation works, crushing of material and batching of aggregates.

8.4 Environmental Auditing

A management tool comprising a systematic, documented, periodic and objective evaluation of how well environmental organization, management and equipment are performing in the aim of helping to safeguard the environment by: Facilitating management and control of environmental practices, assessing compliance with company policies, which would include meeting regulatory requirements".

Steps of an EA as Follows

- **Pre-Audit Activities:** Which include: – Selection and scheduling of facility to audit, – Selection of audit team, – Contact with facility and planning of the audit.
- **Site Activities:** which are divided into 5 steps: – First understanding of internal controls, – Assessment of internal controls, – Gathering of audit evidence, – Evaluation of audit findings, – Report of findings to facility.

- **Post Audit Activities:** which include: – Production of a draft report, – Production of a final report, – Preparation and implementation of an action plan, – Monitoring of action plan.

Scope of EA

- **Material Audit:** The thrust areas of material audit are the effective utilization of materials, conservation of materials and minimizing wastages in different processes.
- **Energy Audit:** It deals with overall energy consumption, minimizing pollution etc.
- **Water Audit:** Evaluation of raw water intake facilities and evaluation of water treatment facilities.
- **Health and Safety Audit:** Health and safety measures are examined e.g. occupational diseases, compliance with labour laws, adequacy of safety measures etc.
- **Environmental Quality Audit:** It examines noise level, status of existing air quality, review of greening program undertaken by the enterprise, Installation and functioning of anti pollutant equipment.
- **Water Disposal Audit:** Review of waste disposal measures relating to solid and hazardous wastes, recycling etc.
- **Engineering Audit:** It analyses the aspect relating to efficiency of plant and machinery, engineering designs, cost effective methods of production etc.

Benefits

- Facilitating comparison and interchange of information between operation or plants,
- Increasing employee awareness of environmental policies and responsibilities,
- Identifying cost-savings including those resulting from waste minimization,

- Evaluating training programs and providing data to assist in training personnel,
- Providing an information base for use in emergency response arrangements,
- Assuring an adequate, up-to-date environmental database for internal management awareness and decision making in relation to plant modifications, new plans, etc.,
- Helping to assist relations with authorities by convincing them that complete and effective audits are being undertaken, by informing them of the type of procedure adopted.

8.5 Ecotourism

The term ecotourism was coined by 'Ceballos-Lascurain' in 1983.

It is a visit of tourist to unpolluted areas rich in biodiversity for recreation and boosting the economy of the region.

Definition

According to the **international ecotourism society** ecotourism is defined as "responsible travel to natural areas that conserves the environment and improves the wellbeing of local people"

Principles of Ecotourism

- Conservation
- Community involvement
- Sustainable development

Conservation: the areas of tourist interest are attractive environments rich in biodiversity. In general, these areas are protected areas where the native species of flora and fauna are allowed to exist and propagate in their natural habitat without human intervention. Therefore, tourist should make minimal impact on the local environment.

Community Involvement: Tourists should treat local people as guides. They should be made part of ecotourism project and their social and cultural traditions should be respected. They should not be exploited by engaging them as cheap labor. Therefore, we can empower the local people to overcome poverty.

Sustainable Development: Sustainable development means development without degrading the environment and which can be sustained in future without any compromise. Ecotourism should not disturb the delicate ecological balance of natural environment. Ecotourism should not add unwanted material, in this way ecotourism provides the sustainable development.

Merits of Ecotourism

- Ecotourism has become an international trade which generates revenue and can empower local people to overcome poverty.
- It helps the tourists and local communities to share their experiences for environmental understanding through education and interpretation.
- It involves the recreation of economy, adventure and enjoyment.
- Ecotourism helps to spread environmental awareness for protection natural, social and cultural heritage and sustainable development.

Disadvantages

- Forests are cleared to build hotels and resorts.
- Agricultural practices are disturbed
- To make rooms for Eco tourists a lot of natural resources have to be sacrificed.
- Communities are displaced and there is a violation of their fundamental rights.

- Frequent moment of vehicles pollutes the environment by emission of air pollutants and noise.

Ecotourism and Indian Scenario

Ecotourism in India is in nascent stage. In India the ministry of tourism is promoting ecotourism in country. To encourage ecotourism practices the ministry has include categories of awards **'Best ecofriendly Hotel', best responsible tourism project, 'Best ecofriendly practices by** tour operators' to various segments of travel, tourism and hospitality sector.

- Corbett national park has highest density of tigers in India
- Kaziranga national park (Assam)
- Kanha national park (MP)
- Bandavgarh national park (MP)
- Pench tiger reserve (MP)
- Maduramalai wild life sanctuaries are some of the other important ecotourism destinations.

Questions

1. Explain the case study on EIA in industries?
2. What is the significance of EIA and various stages of EIA?
3. What is the significance of EMP and various stages of EMP?
4. What is EIA? Discuss its methodology.
5. Write short notes on Environmental Audit.
6. Discuss the concept of Ecotourism, its principles and merits.
7. What are the different types of impacts caused due to development activities?
8. Write a short note on mitigation measures to protect Environment.
9. Discuss briefly about Scope of Environmental Audit.
10. Describe the key components of an EIS.

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